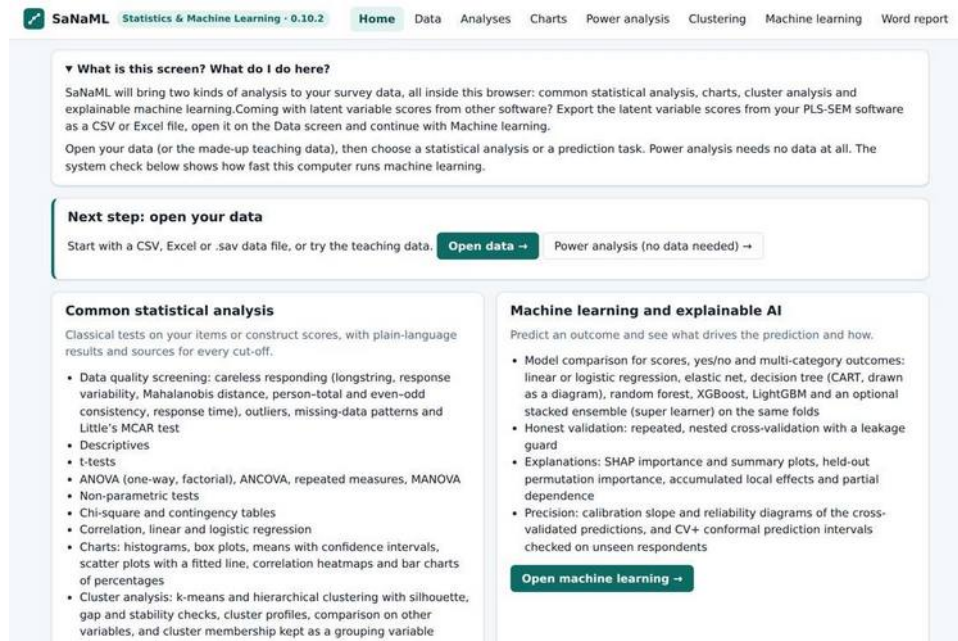


SaNaML

HR Analytics: a step-by-step manual with practice data

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Worked examples: 1,200 simulated employees and a 600-person simulated engagement survey
(*practice_hr.csv* and *practice_engagement.csv*)

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Contents

About this manual.....	4
Who this manual is for.....	4
What SaNaML can and cannot do for HR today.....	4
The practice data are simulated.....	5
What you need.....	5
How to read the instructions.....	5
Responsible people analytics.....	7
1. Getting started with HR data.....	9
1.1 What the screens are for.....	9
1.2 The worked examples.....	10
1.3 Open the file.....	10
1.4 Check the variable types.....	10
1.5 Missing values.....	12
1.6 Files that are not ready to use.....	12
1.7 Anonymisation and small groups.....	12
1.8 Making new variables.....	13
2. Describing the workforce.....	16
2.1 Descriptive statistics.....	16
2.2 Describing groups.....	16
2.3 Charts.....	18
2.4 Salary is skewed: when to transform.....	20
3. Survey scales: the engagement survey.....	23
3.1 Open the file.....	23
3.2 Data quality screening.....	23
3.3 Reliability of the scales.....	25
3.4 Scale scores.....	26
3.5 Comparing departments: ANOVA and Kruskal-Wallis.....	27
3.6 Comparing gender and remote work.....	30
4. Pay equity: the gender pay gap.....	32
4.1 Two questions, not one.....	32
4.2 The raw gap: t-test and Mann-Whitney test.....	32
4.3 Do women and men do different jobs?.....	34
4.4 The adjusted gap: regression of log salary.....	35
4.5 Is the result robust?.....	38
4.6 Does the gap differ by department?.....	38
4.7 What this analysis cannot show.....	40
4.8 Promotion.....	40
5. Training evaluation.....	41
5.1 The data.....	41
5.2 Did scores change? Paired t-test and Wilcoxon test.....	41
5.3 Does more training go with more gain?.....	43
5.4 Regression.....	45
5.5 What the design cannot show.....	47

5.6 Planning the next evaluation: power analysis.....	47
6. Engagement and job satisfaction: what goes with them?.....	49
6.1 The correlation matrix.....	49
6.2 Multiple regression of engagement.....	50
6.3 Hierarchical regression: background first, working conditions second.....	52
6.4 Job satisfaction.....	54
6.5 Engagement over the year: repeated-measures ANOVA.....	54
7. Absence and department differences.....	57
7.1 Absence by department.....	57
7.2 What goes with absence?.....	59
7.3 Leaving by department: the chi-square test.....	59
7.4 Leavers and stayers compared.....	63
8. Predicting attrition.....	64
8.1 What is being predicted.....	64
8.2 Logistic regression first.....	64
8.3 Machine learning on unseen employees.....	67
8.4 Results · Model comparison.....	69
8.5 The imbalance problem.....	70
8.6 Results · Explanations: what drives leaving?.....	71
8.7 Results · Calibration.....	75
8.8 Results · Cut-offs, precision and recall.....	76
8.9 Ethical use of attrition predictions.....	79
9. Employee types: clustering.....	81
9.1 Choosing the variables.....	81
9.2 The default solution: two clusters.....	82
9.3 Comparing solutions and choosing k.....	83
9.4 Reading the three-cluster solution.....	84
9.5 Describing the clusters with other variables.....	86
9.6 Keeping the clusters as a variable.....	87
9.7 Do the employee types differ in leaving?.....	87
10. Reporting and the Word report.....	90
10.1 Adding results.....	90
10.2 The Word report screen.....	90
10.3 What is inside the downloaded document.....	91
10.4 Using the report well.....	91
10.5 A checklist for an HR analysis.....	92
Appendix A. Codebooks.....	93
A.1 practice_hr.csv (1,200 employees).....	93
A.2 practice_engagement.csv (600 respondents).....	94
Appendix B. Which tool answers which HR question?.....	95
Appendix C. Glossary.....	97
References.....	98

About this manual

This manual shows how to use SaNaML 0.10.2 to answer common questions of human resource (HR) management with employee data: How do employees describe their work? Do women and men earn the same after allowing for the job they do? Did a training programme work? What goes with engagement? Who is likely to leave? Every chapter says what to click, what you will see, how to read it, and what the method cannot tell you. All screenshots and numbers come from real runs of SaNaML 0.10.2 on the two practice files described below, so you can follow along and obtain the same results.

Who this manual is for

Students of human resource management and organisational behaviour, researchers who analyse employee or survey data, and HR practitioners who want to understand their own data without writing code. You do not need to know any statistics software. Each statistical idea is explained in plain language in a "Concept" box when it first appears.

What SaNaML can and cannot do for HR today

Please read this section before you start, so that you know what to expect.

SaNaML is a general program for statistics and machine learning on tables of data. **It has no dedicated HR menu.** It does not know what "turnover", "pay gap" or "span of control" means. What it offers are general tools that, used with care, answer many HR questions: descriptive statistics and charts; making new variables from old ones (logarithms, groups, indicators, merged categories, differences and ratios of two variables); data quality screening of survey responses; reliability and scale scores; t-tests, analysis of variance and their rank-based alternatives; chi-square tests; correlation; linear and logistic regression; repeated-measures analysis; power analysis; clustering; and machine learning with explanations and calibration. Appendix B shows which tool answers which HR question.

The following are **not available in SaNaML 0.10.2**, and this manual does not pretend otherwise:

- survival analysis for time to leaving (Kaplan–Meier curves, log-rank test, Cox regression). Leaving is analysed here as a yes/no outcome within a fixed year, which ignores when in the year people left;
- fairness and adverse-impact tools (selection-rate ratios, error rates by group, and similar);
- automatic suppression of small groups. SaNaML prints the number of people in every group, but it will not hide a group of three people for you. Chapter 1 explains what to do;
- ordinal and multinomial regression (you can use linear regression on a rating with care, or collapse the outcome to yes/no and use logistic regression);
- mediation and moderation analysis with bootstrap tests;
- imbalance remedies for rare outcomes (class weights or resampling). SaNaML flags an unbalanced outcome, reports measures suited to it and shows precision and recall at any cut-off you choose, but it does not rebalance the data (sections 8.5 and 8.8);
- an Oaxaca–Blinder decomposition of pay gaps (section 4.7 shows what the regression can and cannot say).

Some of these are planned for later versions, but a manual should describe the program you have, not the one you may get.

The practice data are simulated

Both data files were generated by computer for teaching. **No real employee appears in them**, and no company is described. Effects were planted on purpose, so you can check that an analysis finds what was built in. Appendix A gives the codebooks. Real data are messier than these files: the effects will be weaker, the missing data less random and the questions less clean.

What you need

- A computer with an up-to-date browser. SaNaML runs entirely inside the browser; your data are not uploaded.
- The two practice files, `practice_hr.csv` and `practice_engagement.csv`, which are shipped with SaNaML.

Feedback on this manual is welcome at sabraz@seu.ac.lk.

How to read the instructions

Table 1

Conventions used in this manual

Convention	Meaning	Example
Bold green text	A button, tab, menu item, box or list on the screen. Click or choose exactly this.	Run the comparison
→	A sequence of clicks.	Data → Choose a file...
Numbered steps	Actions to carry out in this order.	1. Open ... 2. Click ...
<i>Italics</i>	A technical term explained nearby, or text exactly as it appears in the results.	<i>nested cross-validation</i>

Concept: what a term means

Explains a statistical idea in plain language.

Cut-off values: the rule of thumb

The thresholds SaNaML applies, with their sources. SaNaML prints the same rule under every check ("Rule: ...").

What it means for our example

What the numbers of the worked example mean.

Tip

A shortcut or good practice.

Note

A warning or a point that needs judgement.

How to report it

A model sentence for your report. SaNaML writes one for every result.

Exercise

A task for you to try, usually with the answer that was planted in the data.

Responsible people analytics

Employee data are about people who depend on the organisation that holds the data. Before any analysis, decide how the results will be used. These principles apply to every chapter.

Note: ten rules for working with employee data

- **Have a purpose and a legal basis.** Collect and analyse only what you need for a stated purpose, and follow the data protection law and the policies that apply to your organisation and country. This manual is not legal advice.
- **Tell people and, where required, ask them.** Employees should know what is collected and why. A survey that promises confidentiality must keep the promise: do not use it to identify individuals.
- **Remove names and numbers that identify people.** Analyse a file in which the employee number is the only identifier, keep the key that links numbers to names somewhere else, and delete the identifier too if you do not need it (section 1.7).
- **Report patterns, not people.** Statistics describe groups. A result such as "attrition is higher among people with low engagement" is a statement about a group; it does not say that this particular employee will leave.
- **Protect small groups.** A table of salary by department and gender may contain a cell with two people, and everybody knows who they are. Do not report a group smaller than a minimum you set in advance (many organisations use five to ten people), and merge categories instead. SaNaML will not do this for you (section 1.7).
- **Be careful with sensitive variables.** Gender, age, ethnicity, religion, disability and health are protected or sensitive in many places. Use them to check for unfair treatment (chapter 4), not to predict or to rank individuals. A model that never receives gender can still treat women differently if other variables stand in for it.
- **Do not decide about a person from a model.** A prediction of leaving is a prompt for a supportive conversation, never a reason to withhold a promotion or training. Chapter 8 returns to this.
- **Correlation is not causation.** Engagement and satisfaction go together; that does not show that raising one raises the other. Only design (for example a randomised trial) supports causal claims, and even then with limits.
- **Past data carry past bias.** If past promotions favoured one group, a model trained on them learns that pattern and repeats it. A well-known case outside HR is an algorithm that used health costs as a proxy for health needs and so under-served Black patients (Obermeyer et al., 2019).
- **Keep the humans in charge.** Document what was done, report its limits, and let people who understand the work review conclusions before anyone acts on them.

Concept: anonymous, pseudonymous, aggregated

Data are *anonymous* when nobody can work out who a row belongs to, even by combining columns. Data with an employee number are only *pseudonymous*: whoever holds the key can identify people. Combinations of ordinary columns (department, level, age, gender) can single out one person even without a name; the idea of *k-anonymity* asks that every combination occur at least k times (Sweeney, 2002). *Aggregated* results, such as group means, are safest, provided the groups are large enough.

1. Getting started with HR data

1.1 What the screens are for

SaNaML has these screens, shown in the menu at the top: **Home**, **Data**, **Analyses**, **Charts**, **Power analysis**, **Clustering**, **Machine learning** and **Word report**. Figure 1 shows the Home screen. Every screen starts with a box *What is this screen? What do I do here?* that explains it in two or three sentences, and a screen that needs data shows "No data yet" until you open a file.

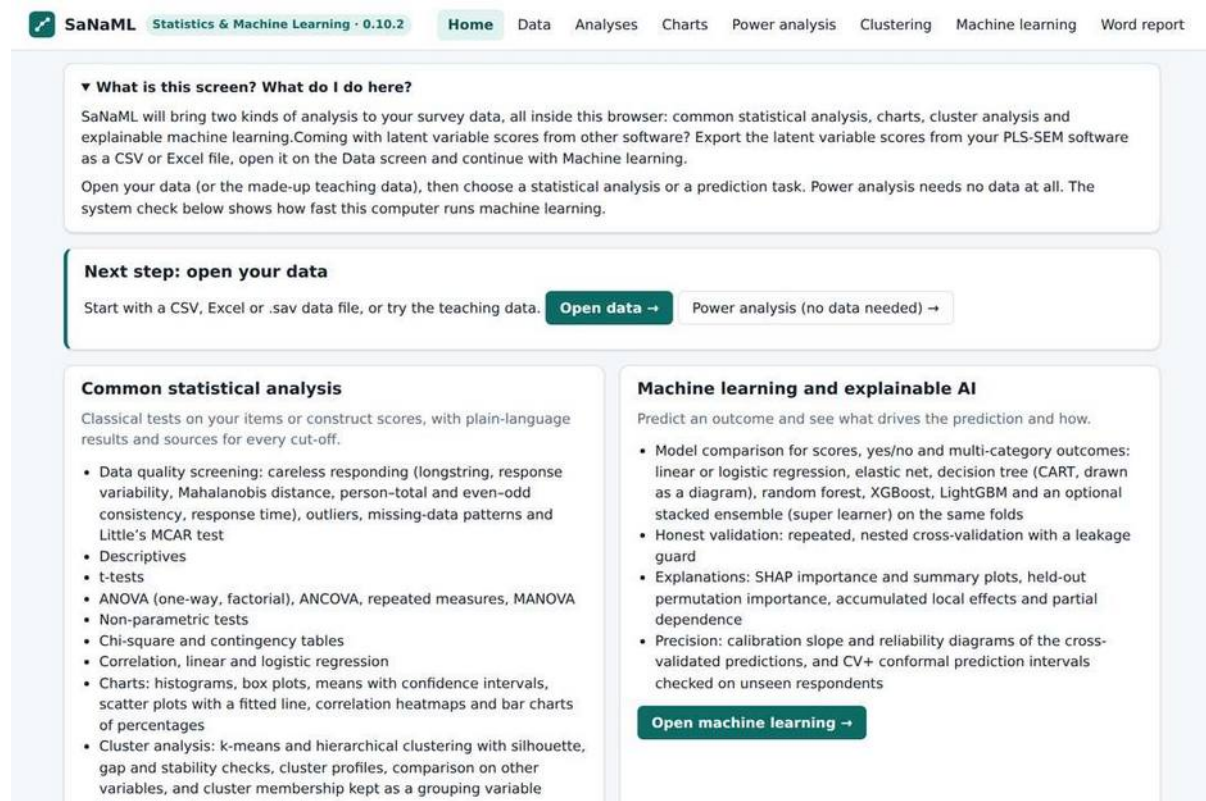


Figure 1. The Home screen of SaNaML.

Three principles apply throughout:

- **Your data stay on your computer.** Every calculation, machine learning included, runs inside your browser. Nothing is uploaded. This matters for employee data, but it does not replace the rules of the previous chapter: a file on a shared laptop is still a file.
- **Plain language with sources.** Every result starts with *What it means*, lists its checks with a verdict (OK, Note, Check, Info) and the rule behind it, and ends with *How to report it* (APA style).
- **Honest testing.** Prediction models are judged on employees they have not seen, and every step that learns from the data is repeated inside each training set.

SaNaML keeps your data and results in the browser between visits. They are lost if you clear the browser's site data, so download the Word report (chapter 10) at the end of a session.

1.2 The worked examples

Table 2

The datasets used in this manual

File	What it holds	Used in
practice_hr.csv	1,200 simulated employees of one company: department, level, gender, age, education, tenure, salary, performance, training, work conditions, survey scores, absence, promotion, a pre/post skills test, engagement in four quarters, and whether the person left within the year	Chapters 1, 2 and 4 to 9
practice_engagement.csv	600 simulated survey respondents: Vigor, Dedication and Absorption items (three each, rated 1 to 7), with department, gender, tenure band and whether the person works remotely	Chapter 3

Tip: follow along

Open SaNaML next to this manual and repeat each step. Random parts of the analysis (cross-validation splits, clusters) use fixed seeds, so you get the same numbers as the screenshots. Machine learning takes longer on a slow computer, but the results do not change.

1.3 Open the file

1. Click **Data** in the menu.
2. Click **Choose a file...** and pick practice_hr.csv. SaNaML reads CSV, Excel (.xlsx) and SPSS (.sav / .zsav) files with one row per person and one column per variable, with the column names in the first row.
3. SaNaML confirms: *Open practice_hr.csv: 1200 rows, 26 variables.*

Note: the data are simulated

practice_hr.csv is a simulated company. No real employee appears in it. The effects in it were planted so that you can check your results against the answers in the exercises.

1.4 Check the variable types

Step 2 lists every variable with its type, the number of valid values and the number missing (Figure 2). SaNaML guesses the type from the file. **Scale** means numbers you can average; **Categorical** means groups.

2 · Check the variable types

Scale = numbers you can average (ratings, scores, age). **Categorical** = groups or answers without a meaningful average (gender, education, yes/no). SaNaML guessed from your file; change any that are wrong.

Variable	Type	Valid	Missing	Values
EmployeeID	Categorical	1200	0	E0001, E0002, E0003, E0004, E0005, E0006, ...
Department	Categorical	1200	0	Finance, HR, IT, Operations, Sales
Job_level	Scale	1200	0	
Gender	Categorical	1200	0	female, male
Age	Scale	1200	0	
Education	Categorical	1200	0	Bachelor, Diploma, Doctorate, Master
Tenure_years	Scale	1200	0	
Annual_salary	Scale	1200	0	
Performance_rating	Scale	1200	0	
Training_hours	Scale	1200	0	
Overtime_hours_month	Scale	1200	0	
Commute_km	Scale	1165	35	
Remote_days_week	Scale	1200	0	
Manager_support	Scale	1160	40	
Work_life_balance	Scale	1200	0	
Engagement	Scale	1200	0	
Job_satisfaction	Scale	1148	52	
Absence_days	Scale	1200	0	
Promoted_last_3y	Categorical	1200	0	No, Yes
Skills_test_before	Scale	1200	0	
Skills_test_after	Scale	1200	0	
Engagement_Q1	Scale	1200	0	
Engagement_Q2	Scale	1200	0	
Engagement_Q3	Scale	1200	0	
Engagement_Q4	Scale	1200	0	
Left_company	Categorical	1200	0	No, Yes

Figure 2. Step 2 · Check the variable types for *practice_hr.csv*.

Read the list with HR eyes:

- **EmployeeID** was read as Categorical, which is right: it is a label with 1,200 different values, not a number to average. Never use an ID as a predictor or a grouping variable.
- **Department, Gender, Education, Promoted_last_3y and Left_company** are Categorical. Department has the groups Finance, HR, IT, Operations and Sales; Education has Bachelor, Diploma, Doctorate and Master; Gender has the two groups female and male (as coded in the file).
- **Job_level** (1 to 5) was read as Scale because it is a number. This is a judgement call. Treated as Scale, it enters a regression as one straight-line effect per level. Changing its type to Categorical (the **Type of Job_level** list) turns it into a grouping variable with the groups 1, 2, 3, 4 and 5 that can be used in tables and tests, and gives each level its own coefficient in a regression. Use whichever your question needs; section 4.5 shows both.

- The survey scores (Manager_support, Work_life_balance, Engagement, Job_satisfaction) are averages on a 1 to 5 scale and are correctly Scale.

Note: the order of categories

SaNaML lists categories alphabetically, and a regression uses the first one as the comparison group. So female is the reference group for Gender, Finance for Department and Bachelor for Education. This is only a naming convention, but you must remember it when you read coefficients.

1.5 Missing values

The columns Valid and Missing in Figure 2 show the missing data: Commute_km has 35 missing values, Manager_support 40 and Job_satisfaction 52. All other columns are complete. In the data file these are empty cells.

How SaNaML treats missing values: an analysis uses every person who has the values it needs. A t-test of Job_satisfaction uses the 1,148 people who answered it; a regression uses only people with complete data on every variable in the model and says how many were left out (for example "n = 1160 (40 respondents with missing values left out)"). Machine learning fills in missing values inside each training set (section 8.3).

Concept: why missing data matter

If values are missing at random, dropping those people loses precision but not accuracy. If people are missing *because of* what they would have answered (for example unhappy employees skipping the satisfaction question), the results are biased, and no program can see this from the data alone. Ask why the values are missing before you decide that it does not matter. The data quality screen (section 3.2) tests one simple version, Little's test of missing completely at random.

1.6 Files that are not ready to use

A few things to fix in your own files before opening them:

- one row per person, no merged cells, no totals row, no notes above the header;
- consistent spelling of categories ("Finance", not "Finance " and "finance");
- numbers stored as numbers, not as text such as "52,000 USD";
- yes/no outcomes coded in the same way throughout;
- dates converted into numbers that mean something, such as tenure in years.

1.7 Anonymisation and small groups

Before analysis:

1. **Replace names by numbers.** practice_hr.csv has an employee number and no name. Keep the list that links numbers to names outside the analysis file.

2. **Drop columns you do not need.** Date of birth is not needed when Age is; a home address is not needed when Commute_km is.
3. **Coarsen identifying detail.** Age in five-year bands is safer than the exact age; exact hire dates are safer as tenure bands.
4. **Count the people in each group before you report.** SaNaML shows the group sizes everywhere (the column n, or the labels n = 46 under a chart). Check them against your minimum group size. In practice_hr.csv the smallest education group is Doctorate with 46 people; the smallest Department, HR, has 113; but the HR department split by gender and education would produce cells with very few people.
5. **Merge or leave out small groups** instead of reporting them. SaNaML does not do this automatically.

How to report it

"Employee numbers were retained for linkage only and were removed before analysis. Groups with fewer than [minimum] employees are not reported."

1.8 Making new variables

Step 4 of the Data screen, *Compute and recode variables (optional)*, makes a new variable from an existing one. The original is never changed; the new variable is added to the list of variables and can be used on every other screen (Figure 3). Choose what you want to make:

- **Transform a scale variable.** Natural logarithm, $\ln(1 + x)$, square root, square, standardise (z-score), centre (subtract the mean) and reverse code (lowest + highest - x). A logarithm needs $x > 0$; other values become missing.
- **Make groups from a scale variable.** Two groups at the median, three equal-sized groups (tertiles), four equal-sized groups (quartiles), or your own cut points. With your own cut points the program notes that a value equal to a cut point goes to the lower group.
- **Make a 0/1 indicator of a category.** The indicator is 1 for the categories you tick.
- **Combine two scale variables.** Choose variable A, how to combine (difference $A - B$, sum, product, ratio $A \div B$, or percentage change $(A - B) \div B \times 100$) and variable B. For the ratio the program notes "Needs B different from 0; other respondents become missing"; the percentage change needs the same.
- **Merge categories into one.** Tick two or more categories and name the merged one.

4 • Compute and recode variables (optional)

Make a new variable from an existing one: a logarithm for a skewed variable, a standardised score, low and high groups, a 0/1 indicator for one category, or merged categories. The original variable is never changed.

What do you want to make?

Variable

Transformation

Pulls in a long right tail (income, salary, spend). Needs $x > 0$; other values become missing.

Name of the new variable

Letters, numbers, dots and underscores; must not be used by another variable yet.

Add the new variable

Figure 3. Step 4 of the Data screen: a natural logarithm of Annual_salary is ready to be added.

Pick the variable, pick the operation, and optionally type a name; if you leave the name empty the program supplies one. Select **Add the new variable**. The program confirms with, for example, *Added Annual_salary_ln: 1200 of 1200 respondents have a value..* A new name cannot repeat the name of another variable.

We make five variables, which are used in later chapters:

- **Annual_salary_ln** (natural logarithm of Annual_salary). Used for pay equity in chapter 4 and for clustering in chapter 9. Section 2.4 shows why.
- **Tenure_group** from Tenure_years, with three equal-sized groups. The program shows the cut points it found, 3.1 and 6.5 years, and names the groups Low third, Middle third and High third. A median split would have used the single cut point 4.7 years. The three groups hold 401, 407 and 392 employees.
- **Skills_gain**, the difference Skills_test_after – Skills_test_before (Figure 4). The program describes this choice as "A change or gain score: after minus before." It is used in chapter 5.
- **Left_yes**, a 0/1 indicator that is 1 when Left_company is Yes. Its mean is the proportion who left.
- **Dept_merged**, in which Finance and HR are merged into one category called Corporate (Figure 5). The groups of Dept_merged are Corporate (292 employees), IT (195), Operations (363) and Sales (350).

4 • Compute and recode variables (optional)

Make a new variable from an existing one: a logarithm for a skewed variable, a standardised score, low and high groups, a 0/1 indicator for one category, or merged categories. The original variable is never changed.

What do you want to make?

Variable A

How to combine A (the variable above) and B

Variable B

A change or gain score: after minus before.

Name of the new variable

Letters, numbers, dots and underscores; must not be used by another variable yet.

Add the new variable

Figure 4. Combining two variables: Skills_test_after – Skills_test_before.

4 • Compute and recode variables (optional)

Make a new variable from an existing one: a logarithm for a skewed variable, a standardised score, low and high groups, a 0/1 indicator for one category, or merged categories. The original variable is never changed.

What do you want to make?

Variable

Categories to merge (tick two or more)
☒ Finance
☒ HR
☐ IT
☐ Operations
☐ Sales

Name of the merged category

Name of the new variable

Letters, numbers, dots and underscores; must not be used by another variable yet.

Add the new variable

Figure 5. Merging Finance and HR into Corporate.

An interaction term, for example between tenure and overtime, can be made by centring each variable with the transformation **Centre (subtract the mean)** and then combining the two centred variables with the product $A \times B$. The Variable types step now lists the new variables with a short description of how each was made, for example "Tenure_years in groups" and "1 if Left_company is Yes".

Tip: dividing a scale into groups loses information

Groups of tertiles or a median split make results easy to show to managers, but they throw away the detail in the scale and can hide or create effects, especially a split at the median of a variable with no natural cut. Use groups for describing and presenting; for testing, prefer the scale itself (for instance, correlate or regress on Tenure_years instead of comparing tenure groups). In this manual Tenure_group is used only to describe and to show a cross-table with leaving (section 7.3).

Note: new variables are kept in the browser session

The new variables are part of the data in the browser. They are not written into your original file. If you reopen the file, make them again, or keep a note of the recipe (variable, operation, name) with your analysis plan.

2. Describing the workforce

Every HR analysis should start with a description of who is in the data. In this chapter we describe the 1,200 employees of practice_hr.csv with numbers (**Analyses**) and with pictures (**Charts**), and deal with the first practical problem of pay data: salaries are skewed.

2.1 Descriptive statistics

1. Open practice_hr.csv (section 1.3).
2. Click **Analyses** → **Descriptive statistics**.
3. Under *Variables to describe*, tick Age, Tenure_years, Annual_salary, Engagement and Absence_days.
4. Leave **Split by** at "none". The result appears below.

Concept: mean, median, skewness

The *mean* is the arithmetic average. The *median* is the middle value: half of the people are above it. For a skewed variable such as salary the two differ, and the median describes the typical person better. *Skewness* measures the lack of symmetry: 0 is symmetric, a positive value means a long tail of high values, and values beyond ± 1 are clearly skewed. *Kurtosis* (excess kurtosis) measures how heavy the tails are; 0 is like a normal distribution.

For the whole company the table reads: Age M = 37.17 (SD = 8.81), Tenure_years M = 5.53 (SD = 4.01, median 4.70, skewness 1.14), Annual_salary M = 45242.77 (SD = 14591.35, median 42310.00, skewness 1.63, kurtosis 4.38), Engagement M = 4.20 (SD = 0.58) and Absence_days M = 3.88 (SD = 1.98). SaNaML adds its own reading: *2 of the 5 rows show clear skewness ($|skewness| > 1$): report medians as well, and read the rank-based tests next to the tests of means.*

What it means for our example

The average salary (45,242.77) is about 2,900 above the median (42,310.00) because a few very high salaries (the maximum is 125,020.00) pull the mean up. Tenure is skewed in the same way: many people are new and a few have worked for twenty years or more. Age and engagement are roughly symmetric. The salary unit is not stated in the file; we simply call the numbers salary units.

2.2 Describing groups

Use **Split by** to see the statistics for each group (Figure 6). Choose **Department**.

Descriptive statistics by Department

What it means: 8 of the 25 rows show clear skewness ($|\text{skewness}| > 1$): report medians as well, and read the rank-based tests next to the tests of means.

Descriptive statistics

Variable	Department	n	Missing	Mean	SD	Median	Min	Max	Ske
Age	Finance	179	0	37.39	9.11	37.00	21.00	63.00	0.3
Age	HR	113	0	36.30	8.79	37.00	21.00	56.00	-0
Age	IT	195	0	37.36	9.06	37.00	21.00	63.00	0.2
Age	Operations	363	0	37.43	8.60	37.00	21.00	63.00	0.2
Age	Sales	350	0	36.96	8.75	36.00	21.00	62.00	0.2
Tenure_years	Finance	179	0	5.71	4.33	5.00	0.10	25.10	1.4
Tenure_years	HR	113	0	5.45	4.45	4.40	0.10	24.60	1.3
Tenure_years	IT	195	0	5.84	4.11	5.00	0.10	19.30	0.9
Tenure_years	Operations	363	0	5.33	3.78	4.50	0.10	21.10	1.1
Tenure_years	Sales	350	0	5.51	3.87	4.80	0.10	18.70	0.9
Annual_salary	Finance	179	0	47920.95	14857.81	44530.00	26850.00	124920.00	2.2
Annual_salary	HR	113	0	44747.96	13458.11	43350.00	25500.00	102110.00	1.5
Annual_salary	IT	195	0	52486.00	17533.10	47880.00	28710.00	125020.00	1.6
Annual_salary	Operations	363	0	41304.52	12491.95	38480.00	22460.00	85620.00	1.1
Annual_salary	Sales	350	0	44081.83	13320.54	41240.00	25510.00	98000.00	1.3
Engagement	Finance	179	0	4.20	0.62	4.25	2.49	5.00	-0
Engagement	HR	113	0	4.24	0.53	4.16	2.89	5.00	-0
Engagement	IT	195	0	4.19	0.55	4.26	2.39	5.00	-0
Engagement	Operations	363	0	4.21	0.61	4.27	2.09	5.00	-0
Engagement	Sales	350	0	4.18	0.57	4.18	2.17	5.00	-0
Absence_days	Finance	179	0	4.05	1.98	4.00	0.00	10.00	0.3
Absence_days	HR	113	0	3.67	1.93	4.00	0.00	10.00	0.6
Absence_days	IT	195	0	4.02	1.80	4.00	0.00	10.00	0.3
Absence_days	Operations	363	0	3.87	2.00	4.00	0.00	10.00	0.4
Absence_days	Sales	350	0	3.80	2.06	4.00	0.00	11.00	0.6

Skewness and kurtosis are bias-corrected (G1, G2; kurtosis is excess kurtosis); standard errors in brackets. Median = weighted average at $(n + 1)p$.

How to report it (APA style): Age: Department = Finance, $M = 37.39$, $SD = 9.11$, $n = 179$; Department = HR, $M = 36.30$, $SD = 8.79$, $n = 113$; Department = IT, $M = 37.36$, $SD = 9.06$, $n = 195$; Department = Operations, $M = 37.43$, $SD = 8.60$, $n = 363$; Department = Sales, $M = 36.96$, $SD = 8.75$, $n = 350$. Tenure_years: Department = Finance, $M = 5.71$, $SD = 4.33$, $n = 179$; Department = HR, $M = 5.45$, $SD = 4.45$, $n = 113$; Department = IT, $M = 5.84$, $SD = 4.11$, $n = 195$; Department = Operations, $M = 5.33$, $SD = 3.78$, $n = 363$; Department = Sales, $M = 5.51$, $SD = 3.87$, $n = 350$. Annual_salary: Department = Finance, $M = 47920.95$, $SD = 14857.81$, $n = 179$; Department = HR, $M = 44747.96$, $SD = 13458.11$, $n = 113$; Department = IT, $M = 52486.00$, $SD = 17533.10$, $n = 195$; Department = Operations, $M = 41304.52$, $SD = 12491.95$, $n = 363$; Department = Sales, $M = 44081.83$, $SD = 13320.54$, $n = 350$. Engagement: Department = Finance, $M = 4.20$, $SD = 0.62$, $n = 179$; Department = HR, $M = 4.24$, $SD = 0.53$, $n = 113$; Department = IT, $M = 4.19$, $SD = 0.55$, $n = 195$; Department = Operations, $M = 4.21$, $SD = 0.61$, $n = 363$; Department = Sales, $M = 4.18$, $SD = 0.57$, $n = 350$. Absence_days: Department = Finance, $M = 4.05$, $SD = 1.98$, $n = 179$; Department = HR, $M = 3.67$, $SD = 1.93$, $n = 113$; Department = IT, $M = 4.02$, $SD = 1.80$, $n = 195$; Department = Operations, $M = 3.87$, $SD = 2.00$, $n = 363$; Department = Sales, $M = 3.80$, $SD = 2.06$, $n = 350$.

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Figure 6. Descriptive statistics by Department (the first rows of the table are Age and Tenure_years).

Table 3

Annual salary by department (simulated data)

Department	n	Mean	SD	Median
Finance	179	47920.95	14857.81	44530.00
HR	113	44747.96	13458.11	43350.00
IT	195	52486.00	17533.10	47880.00
Operations	363	41304.52	12491.95	38480.00

Department	n	Mean	SD	Median
Sales	350	44081.83	13320.54	41240.00

Pay differs by department: IT has the highest median (47,880.00) and Operations the lowest (38,480.00). By education the gap is larger still: Doctorate (n = 46) averages 55,714.13 and Master (n = 292) 55,121.34, against 41,919.87 for Bachelor (n = 595) and 40,040.15 for Diploma (n = 267). By gender, women (n = 563) average 44,213.37 and men (n = 637) 46,152.57. That difference is not yet a statement about pay equity: women and men may differ in level, tenure, department and education. Chapter 4 deals with that.

Engagement, by contrast, hardly differs between departments (means from 4.18 in Sales to 4.24 in HR) and absence differs very little (means from 3.67 in HR to 4.05 in Finance).

Note: small groups

With a second split, groups quickly become small. The Doctorate row has only 46 people. Before you put a table like this in a report, apply the minimum group size you decided on in section 1.7.

2.3 Charts

The **Charts** screen draws six kinds of picture: a histogram, a box plot, means with confidence intervals, a scatter plot with a fitted line, a correlation heatmap and a bar chart of percentages. Choose the **Chart type**, then the variables. Every chart comes with a short description, a summary table, and a sentence and caption for the report. Charts show patterns, not proof: an apparent difference needs a test.

- **Histogram** (Figure 7) shows how the values are spread; the line marks the mean and the dashed line the median.
- **Box plot** (Figure 8) shows the median, the middle half of the values (the box) and unusual values beyond 1.5 times the interquartile range (the circles). Choose **Groups** to split it.
- **Scatter plot** (Figure 9) shows two variables together with the straight-line fit.
- **Bar chart** of percentages and **means with confidence intervals** are used for categories and for group averages.

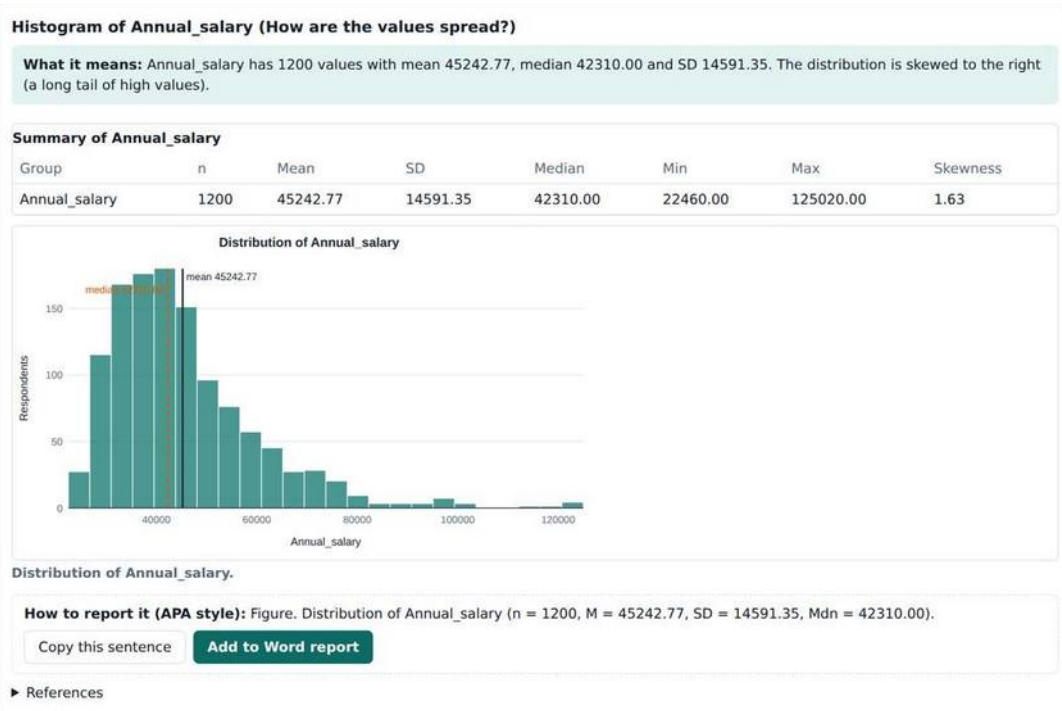


Figure 7. Histogram of Annual_salary: a long tail to the right.

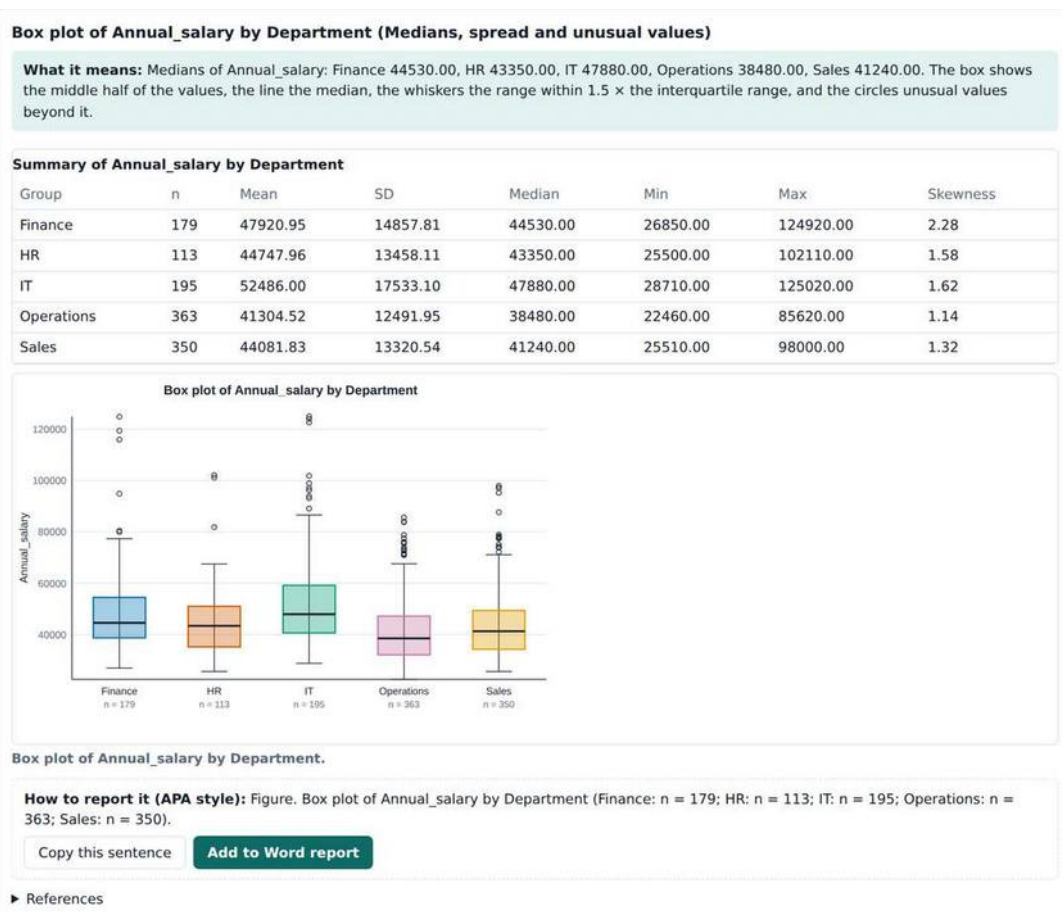


Figure 8. Box plot of Annual_salary by Department. The circles are employees with unusually high salaries.

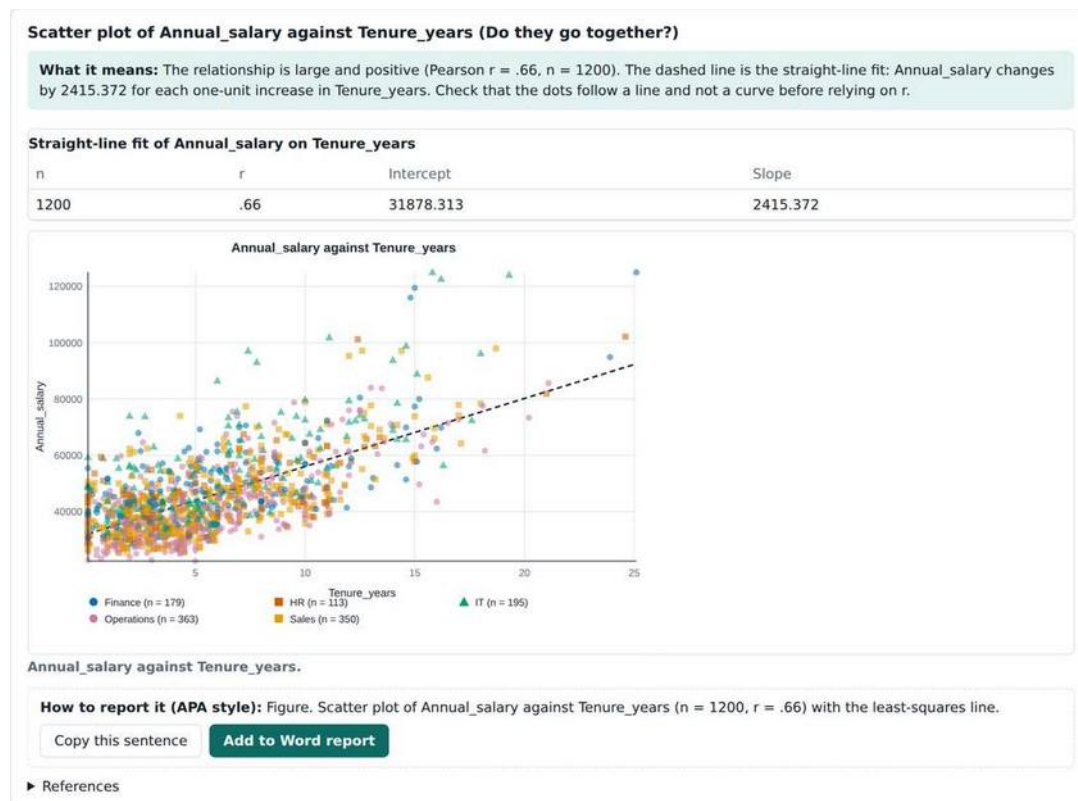


Figure 9. Annual_salary against Tenure_years, with the fitted line and the departments marked.

What it means for our example

The histogram shows the right skew in numbers (skewness 1.63). The box plot shows that IT and Finance have the longest upper tails. The scatter plot shows that salary rises with tenure: $r = .66$, with a fitted slope of 2415.372 salary units per additional year (intercept 31878.313). The fitted line is only a summary; chapter 4 asks how much of this is tenure itself and how much is level and department.

2.4 Salary is skewed: when to transform

Pay is a classic skewed variable: most people earn a modest amount and few earn a lot, and differences between people are better described as percentages than as amounts (a rise of 2,000 means more to someone earning 25,000 than to someone earning 100,000). Taking the natural logarithm of salary makes the distribution more symmetric and turns differences into approximate percentages.

Make the logarithm inside SaNaML (section 1.8): on the Data screen, step 4, choose **Transform a scale variable**, the variable Annual_salary and the transformation **Natural logarithm, $\ln(x)$** , then select **Add the new variable**. The new variable is called Annual_salary_ln. Compare the two histograms.

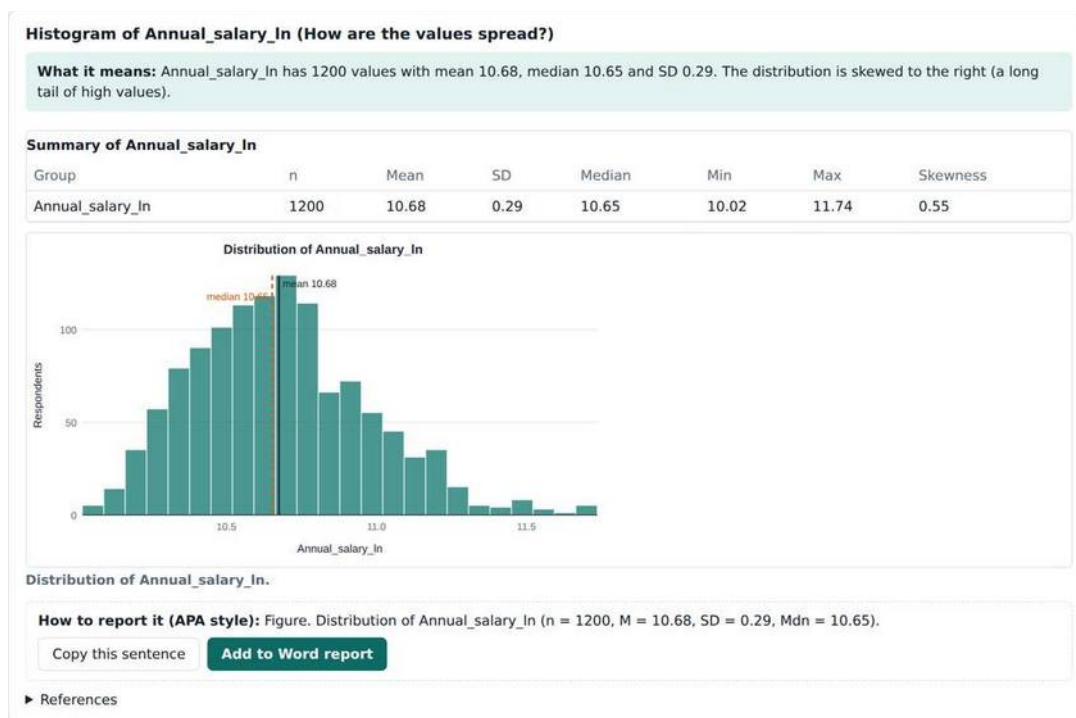


Figure 10. Histogram of Annual_salary_In. The tail is much shorter than in the histogram of Annual_salary.

Table 4

Annual salary and its logarithm

Variable	Mean	SD	Median	Skewness	Kurtosis
Annual_salary	45242.77	14591.35	42310.00	1.63	4.38
Annual_salary_In	10.68	0.29	10.65	0.55	0.32

The logarithm brings the skewness down from 1.63 to 0.55 and the kurtosis from 4.38 to 0.32, and mean and median now nearly agree (10.68 and 10.65). It does not remove the skew completely.

Concept: differences on the log scale are percentages

If log salary differs by a small amount b between two groups, the ratio of their salaries is about e^b . For $b = 0.044$ the ratio is $e^{0.044} = 1.045$: the first group earns about 4.5% more. For small b the percentage is close to $100 \times b$. When you quote a percentage from a coefficient in this manual, we calculated it by hand from the coefficient SaNaML printed.

Tip: when to transform

Transform a variable when its skew or its spread of errors makes the standard methods unreliable, and when the transformed scale has a sensible meaning. Percentage differences in pay are a sensible meaning. Do not transform a variable only to get a significant result.

Exercise

1. Which department has the highest median tenure, and what is it? (Answer: IT and Finance both have a median tenure of 5.00 years.)
1. Draw the histogram of Tenure_years. How skewed is it? (Answer: skewness 1.14, with mean 5.53 and median 4.70.)
1. Split Annual_salary by Education. Which two education groups have the highest means? (Answer: Doctorate 55714.13 and Master 55121.34.)

3. Survey scales: the engagement survey

Most HR questions about attitudes are answered with a questionnaire. A single question is a noisy measure, so questionnaires use several items for one idea and average them. This chapter uses the engagement survey `practice_engagement.csv`: 600 respondents answered nine statements on a scale from 1 to 7. Three items measure *Vigor* (energy at work), three measure *Dedication* (involvement and pride) and three measure *Absorption* (being immersed in the work), the three dimensions of work engagement described by Schaufeli et al. (2002). The responses are simulated; no real person answered them.

3.1 Open the file

Click **Data** → **Choose a file...** and open `practice_engagement.csv`. SaNaML reports 600 rows, 14 variables. `RespondentID`, `Department`, `Gender`, `Tenure_band` and `Works_remotely` are Categorical; the nine items are Scale. Note that `Tenure_band` lists its groups in alphabetical order ("`<2 years`", "`>10 years`", "`2-5 years`", "`6-10 years`"), not in the natural order.

`Vigor2` has 20 missing answers and `Absorption3` has 21; the other items are complete.

3.2 Data quality screening

Survey answers can be careless: the same answer to every item, answers at random, answers given too fast to have been read. Screening takes a minute and is good practice before scoring a questionnaire.

1. On the Data screen click **Check data quality →**.
2. Under *Screening settings (What to check)* SaNaML has ticked the nine items. The file has no completion time, so leave **Completion time in seconds** at "none". Leave the rule at **2 or more indices (Recommended)**.
3. Click **Screen the data**.

Data quality screening (Careless responding, outliers and missing data)

What it means: Careless responding: 3 indices were computed on 9 items. 1 respondent was flagged by 2 or more indices and is suggested for exclusion (0.2% of 600); 24 more raised a single flag, which on its own is not enough to exclude anyone. Outliers: 0 values with $|z| > 3.29$ across 0 variables. Missing data: 0.8% of the answers are missing (at most 3.5% for one item); 559 respondents (93%) answered every item. Little's test does not reject that the data are missing completely at random ($\chi^2(16) = 21.60$, $p = .157$).

Note Careless responding. 1 respondents (0.2%) are suggested for exclusion. Look at their answers before deciding, and report the rule and the number excluded.

Rule: Use several careless-response indices and exclude only respondents flagged by more than one; report the indices, cut-offs and the number excluded (Curran (2016); Meade & Craig (2012)).

OK Outliers. No value is beyond $|z| = 3.29$.

Rule: Univariate outliers: $|z| > 3.29$ ($p < .001$); multivariate outliers: Mahalanobis D^2 with $p < .001$ (Tabachnick & Fidell (2019)).

OK Amount of missing data. At most 3.5% missing for one item.

Rule: Under 10% missing per variable is usually ignorable if the missingness is random; more needs a principled method (FIML or multiple imputation) (Hair et al. (2019)).

OK Missing completely at random. Little's test: $\chi^2(16) = 21.60$, $p = .157$. A non-significant result does not prove MCAR, but it gives no reason to doubt it.

Rule: Little's test $p \geq .05$: no evidence against missing completely at random (Little (1988)).

Careless-response indices	
Index	Flag when
Longstring (Same answer many times in a row)	5 or more identical answers in a row (of 9 items)
Intra-individual response variability (No variation at all)	SD of the answers = 0
Mahalanobis distance (Unusual answer pattern)	D^2 with $p < .001$ (χ^2 , 9 df; complete answers)
Person-total correlation (Answers unlike everyone else's)	Not used: the item means are too similar (SD 0.03 < 0.25) for the index to be informative
Even-odd consistency (Inconsistent within scales)	Not used: needs at least 4 scales with two or more items (found 3); with fewer scales the
Response time (Too fast to have read the items)	Not used: no completion time chosen
A respondent flagged by at least 2 indices is suggested for exclusion; no single index is decisive. Longstring follows the item order of the file. Reverse-coded items (none) were reversed for the person-total and even-odd indices. Scales: Vigor (3), Dedication (3), Absorption (3).	

Figure 11. The start of the data quality report.

Figure 11 shows the result. SaNaML computed three indices on nine items: 1 respondent (0.2% of 600) was flagged by two or more indices and is suggested for exclusion, and 24 more raised a single flag, which on its own is not enough to exclude anyone. No value is an outlier ($|z| > 3.29$), 0.8% of the answers are missing (at most 3.5% for one item), 559 respondents (93%) answered every item, and Little's test does not reject that the data are missing completely at random, $\chi^2(16) = 21.60$, $p = .157$.

Table 5

Careless-response indices for the engagement survey

Index	Flag when	Flagged
Longstring	5 or more identical answers in a row (of 9 items)	25 (4.2%)
Intra-individual response variability	SD of the answers = 0	1 (0.2%)
Mahalanobis distance	D^2 with $p < .001$	0 (0.0%)
Person-total correlation	Not used: item means too similar	-
Even-odd consistency	Not used: needs at least 4 scales (found 3)	-
Response time	Not used: no time chosen	-

The even-odd index compares the two halves of each scale. The program uses it only when the questionnaire has at least four scales with two or more items, because with fewer scales the correlation between the halves is too unstable; the engagement survey has three scales of three items, so the program states that the index is not used. The table shows "Not used" for each index it could not compute, with the reason.

Note: read the indices, do not just count them

The suggested respondent is R364, who gave the same answer to all nine items (longest run 9, response variability 0.00; Mahalanobis $D^2 = 7.9$, $p = .540$): a genuine straight-liner. The other 24 respondents with one flag all have a longstring of 5 or more identical answers (up to 6 of the 9 items). A run of five identical answers can occur in honest responding, so use the flags to look at the answers, not as a verdict. Respondents in the lists are labelled by the ID column of the file (here RespondentID), not by a row number.

In this manual we **keep all 600 respondents**, so that your numbers match ours. In a real project you would look at R364's answers and those of the 24 single-flag respondents, decide, and report the rule and the number excluded (the report sentence is given under the result). Exclusion is done under *Your decision*, with **Exclude the ... ticked respondents**; it can be undone.

3.3 Reliability of the scales

A scale is *reliable* if its items hang together. The usual measure is Cronbach's alpha.

Concept: Cronbach's alpha

Alpha is the average relationship between the items of a scale, raised by the number of items. It runs from 0 to 1. Because it grows with the number of items, a scale of nine items can have a high alpha even if its items measure different things, so alpha does not prove that a scale measures one idea. The *corrected item-total correlation* shows how strongly each item goes with the rest of the scale, and *alpha if deleted* shows what would happen if the item were dropped.

Cut-off values: reliability

Cronbach's α of .70 or higher is usually acceptable for research scales (Nunnally, 1978). A corrected item-total correlation below .30, or a negative one, suggests that the item does not fit or needs reverse-coding.

Step 3 of the Data screen builds construct scores:

1. In **Name of the construct** type Vigor.
2. In **Quick pick: items starting with** type Vigor and press Tab. SaNaML ticks Vigor1 to Vigor3 and sets the lowest and highest answer to 1 and 7. (If an item were worded in the opposite direction, you would tick its **R** box to reverse it. None of these items is.)
3. Read α and the item table.
4. Click **Create the construct score**. Repeat for Dedication and Absorption.
5. For the total, name the construct Work_engagement and tick all nine items (Figure 12).

3 • Build construct scores (optional)

A construct score is the average of the items that measure one idea (for example PE1-PE4 → PE). SaNaML first checks the items hang together (Cronbach's α). If you already have latent variable scores from other software, open that file instead; it needs no construct building.

Name of the construct

Letters, numbers and underscores; must not be used by another variable yet.

Quick pick: items starting with

Lowest answer Highest answer

Used to reverse-code items: on a 1–7 scale, a reversed 2 becomes 6.

Items (tick to include; "R" = reverse-coded)

- ☒ Vigor1 ☐ R
- ☒ Vigor2 ☐ R
- ☒ Vigor3 ☐ R
- ☒ Dedication1 ☐ R
- ☒ Dedication2 ☐ R
- ☒ Dedication3 ☐ R
- ☒ Absorption1 ☐ R
- ☒ Absorption2 ☐ R
- ☒ Absorption3 ☐ R

$\alpha = .88$ Cronbach's α on 559 complete responses. Cronbach's α of .70 or higher is usually acceptable for research scales (Nunnally (1978)).

Item	Corrected item-total r	α if deleted
Vigor1	.68	.86
Vigor2	.64	.86
Vigor3	.63	.86
Dedication1	.66	.86
Dedication2	.64	.86
Dedication3	.60	.87
Absorption1	.57	.87
Absorption2	.57	.87
Absorption3	.60	.87

An item-total correlation below .30 (or a negative one) suggests the item does not fit, or that it needs reverse-coding.

[Create the construct score](#)

Figure 12. Reliability of all nine items together.

Table 6

Reliability of the engagement scales (complete responses only)

Scale	Items	α	Complete responses	Corrected item-total r
Vigor	3	.83	580	.71, .69, .65
Dedication	3	.84	600	.72, .72, .66
Absorption	3	.79	579	.61, .62, .65
All nine items	9	.88	559	.57 to .68

What it means for our example

All three scales are reliable (α from .79 to .84) and no item fits badly (every item-total correlation is above .56). Dropping an item would lower alpha in every scale, so keep all items. The nine items together give $\alpha = .88$, but see the concept box: this reflects the number of items, and the three dimensions are related but not identical (section 3.4). Report the alpha of each scale, and use the total only if your theory says engagement is one overall idea.

3.4 Scale scores

Clicking **Create the construct score** adds the *mean of the items* as a new variable. A person gets a score only if they answered every item of the scale. That is why the scores have fewer valid values than the items: Vigor 580, Absorption 579 and Work_engagement 559 (41 people missing at least one item).

Describe the scores with **Analyses** → **Descriptive statistics**: Vigor M = 3.93 (SD = 1.32), Dedication M = 3.95 (SD = 1.33), Absorption M = 3.90 (SD = 1.30) and Work_engagement M = 3.93 (SD = 1.10); no score is clearly skewed. The scales correlate positively (Figure 13): Vigor with Dedication $r = .57$, Vigor with Absorption $r = .54$ and Dedication with Absorption $r = .47$ (all $p < .001$ after Holm adjustment).

Pearson product-moment correlation (Pearson r)

What it means: 3 of 3 correlations are significant after Holm adjustment. The strongest is Vigor with Dedication ($r = .57$, large, higher goes with higher).

Info Strength. Read the size of a correlation from its absolute value.

Rule: Correlation-type effect sizes (r , rank-biserial, ϕ): about .10 small, .30 medium, .50 large (Cohen (1988)).

Info Method. Pearson's r measures straight-line relationships. With strongly skewed data or many tied ratings, Spearman's ρ is a safer choice.

Correlation matrix

	1	2	3
1. Vigor	—		
2. Dedication	.57***	—	
3. Absorption	.54***	.47***	—

* $p < .05$, ** $p < .01$, *** $p < .001$ (Holm-adjusted). Each pair uses every respondent with both values.

Every pair

Pair	r	95% CI	n	p	p (Holm)
Vigor - Dedication	.566	[.51, .62]	580	< .001	< .001
Vigor - Absorption	.544	[.48, .60]	559	< .001	< .001
Dedication - Absorption	.471	[.41, .53]	579	< .001	< .001

How to report it (APA style): Pearson r correlations (Holm-adjusted p -values) showed: Vigor and Dedication, $r(578) = .57$, $p < .001$, 95% CI [.51, .62]; Vigor and Absorption, $r(557) = .54$, $p < .001$, 95% CI [.48, .60]; Dedication and Absorption, $r(577) = .47$, $p < .001$, 95% CI [.41, .53].

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Figure 13. Pearson correlations of the three engagement scales.

Tip: unreliable scores need care

A score based on three items is a rough measure of a person. Group means (department, gender) are much more trustworthy than a single person's score.

3.5 Comparing departments: ANOVA and Kruskal-Wallis

Does engagement differ between departments? With five groups, use **Analyses** → **One-way ANOVA (Compare three or more groups)**. Choose **Score to compare** Work_engagement and **Groups** Department (Figure 14).

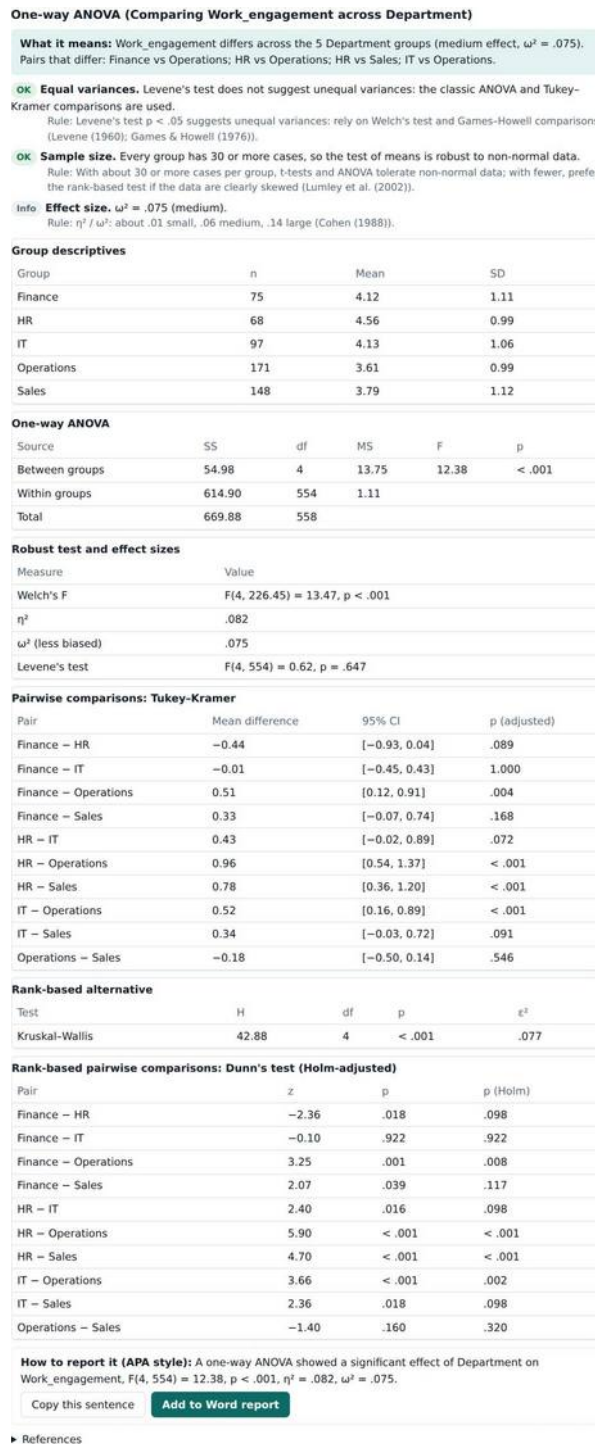


Figure 14. One-way ANOVA of Work_engagement by Department, with the Kruskal-Wallis test.

Concept: one-way ANOVA and its rank-based partner

Analysis of variance (ANOVA) asks whether the group means differ by more than chance would produce. A significant F says that at least one group differs, not which one; *post hoc* comparisons (Tukey-Kramer, which corrects for the number of pairs) tell you which pairs differ. The *Kruskal-Wallis* test does the same job on ranks and does not assume a normal distribution; *Dunn's* test is its pairwise follow-up. SaNaML shows both, reads Levene's test of equal variances for you and tells you which one to use. The effect size ω^2 says how much of the variance the groups explain.

Cut-off values: ANOVA

Levene's test $p < .05$: use Welch's test and Games–Howell comparisons (Levene, 1960; Games & Howell, 1976). Effect size η^2 and ω^2 : about .01 small, .06 medium, .14 large (Cohen, 1988).

Table 7

Work engagement by department (simulated survey)

Department	n	Mean	SD
Finance	75	4.12	1.11
HR	68	4.56	0.99
IT	97	4.13	1.06
Operations	171	3.61	0.99
Sales	148	3.79	1.12

What it means for our example

Work_engagement differs across the five departments, $F(4, 554) = 12.38$, $p < .001$, $\eta^2 = .082$, $\omega^2 = .075$ (a medium effect). Levene's test does not suggest unequal variances ($p = .647$), and Welch's test agrees, $F(4, 226.45) = 13.47$, $p < .001$. The Kruskal–Wallis test gives the same answer, $H(4) = 42.88$, $p < .001$, $\epsilon^2 = .077$. HR is highest (4.56) and Operations lowest (3.61). The difference between HR and Operations is 0.96 [0.54, 1.37] (Tukey–Kramer $p < .001$); Finance and IT are each about 0.5 above Operations ($p = .004$ and $p < .001$), and HR is 0.78 above Sales ($p < .001$). The rank-based pairwise tests (Holm-adjusted) agree on all of these.

The same test on the three scales gives the same picture (Table 8). The department differences are largest for Dedication and smallest for Absorption.

Table 8

ANOVA and Kruskal–Wallis test of the engagement scales by department

Scale	ANOVA	ω^2	Kruskal–Wallis $H(4)$	p
Vigor	$F(4, 575) = 9.18$, $p < .001$	.053	33.46	$< .001$
Dedication	$F(4, 595) = 10.84$, $p < .001$	.062	39.24	$< .001$
Absorption	$F(4, 574) = 6.76$, $p < .001$	.038	25.14	$< .001$
Work_engagement	$F(4, 554) = 12.38$, $p < .001$	.075	42.88	$< .001$

The **Charts** screen draws the group means with their 95% confidence intervals (Figure 15): Finance 4.12 [3.87, 4.37], HR 4.56 [4.32, 4.80], IT 4.13 [3.92, 4.34], Operations 3.61 [3.46, 3.76] and Sales 3.79 [3.60, 3.97].

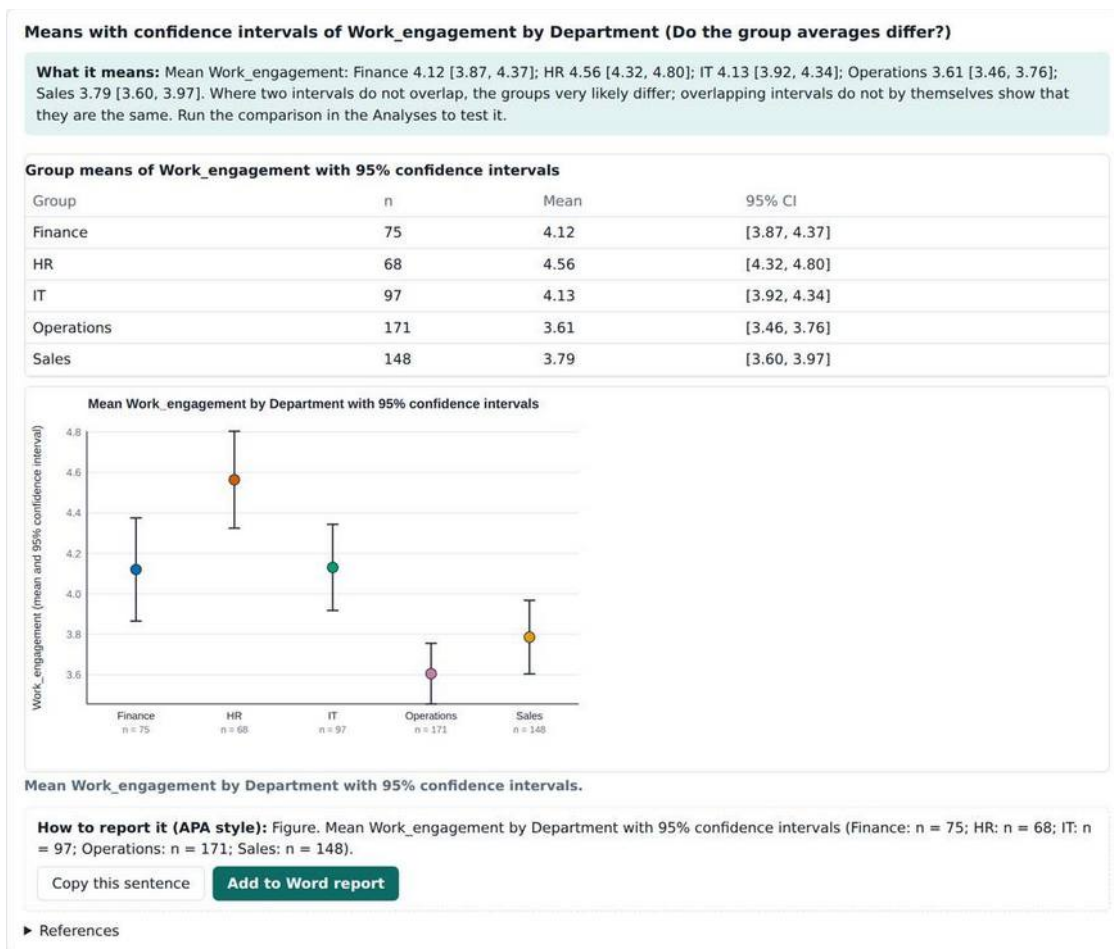


Figure 15. Mean Work_engagement by Department with 95% confidence intervals.

How to report it

"A one-way ANOVA showed a significant effect of Department on Work_engagement, $F(4, 554) = 12.38$, $p < .001$, $\eta^2 = .082$, $\omega^2 = .075$. Tukey-Kramer comparisons showed that HR ($M = 4.56$, $SD = 0.99$) scored higher than Operations ($M = 3.61$, $SD = 0.99$) and Sales ($M = 3.79$, $SD = 1.12$). The Kruskal-Wallis test agreed, $H(4) = 42.88$, $p < .001$."

Note: a department difference is not a manager difference

A difference between departments may come from the work, the people, the managers or the way people were chosen to answer. With survey data you cannot tell which. A department with 68 respondents is also a small group: check the minimum group size before you publish department-level results.

3.6 Comparing gender and remote work

For two groups use **Analyses** → **Independent-samples, paired-samples and one-sample t-tests**, with **Score to compare** Work_engagement and **Groups** Gender. SaNaML reports Welch's t-test first and the Mann-Whitney U test as the rank-based alternative.

What it means for our example

Women ($M = 3.92$, $SD = 1.09$, $n = 256$) and men ($M = 3.94$, $SD = 1.10$, $n = 303$) do not differ, $t(543.87) = -0.26$, $p = .798$, $d = -0.02$, 95% CI $[-0.19, 0.14]$; the Mann-Whitney test agrees ($U = 37820.5$, $p = .613$). The difference is female minus male, which is why the sign is negative.

Respondents who work remotely score a little higher (Yes: $M = 4.02$, $SD = 1.08$, $n = 202$; No: $M = 3.88$, $SD = 1.10$, $n = 357$), but not significantly, $t(426.12) = -1.45$, $p = .148$, $d = -0.13$, 95% CI $[-0.30, 0.05]$; Mann-Whitney $p = .214$. A confidence interval from -0.30 to 0.05 is compatible with no effect and with a small effect: the data are not informative enough to say more.

Tenure_band shows how the two tests can disagree near the boundary: the ANOVA is just significant, $F(3, 555) = 2.80$, $p = .039$, $\omega^2 = .010$ (negligible), while the Kruskal-Wallis test is not, $H(3) = 6.83$, $p = .078$. Do not make much of such a result.

Exercise

1. Compare Dedication by Works_remotely and write the result in one sentence. Then compare your sentence with the one SaNaML writes under *How to report it*.

1. Which pair of departments differs most on Vigor, and by how much? (Answer: HR and Operations, 0.97 , 95% CI $[0.47, 1.48]$, Tukey-Kramer $p < .001$.)

1. Why is the Work_engagement score available for only 559 respondents when the file has 600? (Answer: a score needs all nine items; 41 respondents skipped at least one.)

4. Pay equity: the gender pay gap

Pay equity is one of the most common and most sensitive questions in people analytics. This chapter asks whether women and men in the simulated company earn the same, first in raw terms and then after allowing for the job they do. It also explains, as plainly as we can, what such an analysis can and cannot show. We use `practice_hr.csv` with the variable `Annual_salary_In` made in section 2.4; the company, its employees and its pay are simulated.

Note: before you start

Pay data are among the most sensitive data in an organisation. Work on an anonymised copy, apply the minimum group size of section 1.7, and decide before the analysis who will see the results and what they will be used for. SaNaML has no pay-equity module and no Oaxaca-Blinder decomposition (Oaxaca, 1973; Blinder, 1973); what follows uses the general tools.

4.1 Two questions, not one

- **The raw gap.** Do women and men earn different amounts on average? This describes the organisation as it is.
- **The adjusted gap.** Do women and men earn different amounts when they do comparable work, that is, at the same level, with the same tenure, in the same department and with the same education? This asks whether pay differs for people who look alike on the measured characteristics.

Both are legitimate and they can differ. A large raw gap with a small adjusted gap means that women and men are in different jobs; a small raw gap with a large adjusted gap means that people in the same jobs are paid differently. The two point to different remedies.

4.2 The raw gap: t-test and Mann-Whitney test

1. Click **Analyses** → **Independent-samples, paired-samples and one-sample t-tests**.
2. Choose **Score to compare** `Annual_salary` and **Groups** `Gender`. Leave **Two-tailed test**.

Figure 16 shows the result. Welch's t-test is reported first and the Mann-Whitney U test is the rank-based alternative, which does not depend on the skewed shape of salary.

Independent-samples t-test (Comparing Annual_salary between female and male)

What it means: male scored higher than female on Annual_salary (difference 1939.20 points). The difference is statistically significant and its size is negligible ($|d| = 0.13$).

Info Which t-test. Welch's t-test is reported first because it is accurate whether or not the variances are equal.

Rule: Compare two means with Welch's t-test by default; it stays accurate when the variances differ (Delacre et al. (2017)).

OK Equal variances. Levene's test does not suggest unequal variances; Student's and Welch's tests agree closely.

Rule: Levene's test $p < .05$ suggests unequal variances: rely on Welch's test and Games-Howell comparisons (Levene (1960); Games & Howell (1976)).

OK Sample size. Every group has 30 or more cases, so the test of means is robust to non-normal data.

Rule: With about 30 or more cases per group, t-tests and ANOVA tolerate non-normal data; with fewer, prefer the rank-based test if the data are clearly skewed (Lumley et al. (2002)).

Info Effect size. $|d| = 0.13$ is negligible. The sign of d follows the order of the groups: $d = -0.13$ means male scored higher than female.

Rule: Cohen's d : about 0.2 small, 0.5 medium, 0.8 large (Cohen (1988)).

Group descriptives

Group	n	Mean	SD	Median	Skewness
female	563	44213.37	14237.12	41420.00	1.70
male	637	46152.57	14849.22	42860.00	1.57

t-tests (two-sided)

Test	t	df	p	Mean difference	95% CI
Welch's t (recommended)	-2.31	1190.08	.021	-1939.20	[-3587.92, -290.48]
Student's t (equal variances)	-2.30	1198	.022	-1939.20	[-3592.20, -286.20]

Effect size

Measure	Value	95% CI	Size
Cohen's d	-0.13	[-0.25, -0.02]	negligible
Hedges' g	-0.13		negligible

The CI of d inverts the noncentral t distribution (Steiger & Fouladi, 1997).

Rank-based alternative

Test	U	z	p	Rank-biserial r	Method
Mann-Whitney U	164623.0	-2.45	.014	-.08	normal approximation

Equal variances

Test	F	df	p
Levene (mean)	1.33	1, 1198	.249
Brown-Forsythe (median)	0.56	1, 1198	.456

How to report it (APA style): An independent-samples Welch's t-test showed that Annual_salary differed between female ($M = 44213.37$, $SD = 14237.12$) and male ($M = 46152.57$, $SD = 14849.22$), $t(1190.08) = -2.31$, $p = .021$, $d = -0.13$, 95% CI [-0.25, -0.02].

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References

Figure 16. Independent-samples t-test of Annual_salary by Gender.

Table 9

Raw salary gap between women and men

Group	n	Mean	SD	Median
female	563	44213.37	14237.12	41420.00
male	637	46152.57	14849.22	42860.00

What it means for our example

Men earn more on average: the difference (female minus male) is -1939.20 , 95% CI $[-3587.92, -290.48]$, Welch's $t(1190.08) = -2.31$, $p = .021$, $d = -0.13$, 95% CI $[-0.25, -0.02]$. The Mann-Whitney test agrees, $U = 164623.0$, $z = -2.45$, $p = .014$, with a rank-biserial correlation of $-.08$. In relative terms, men's mean salary is 4.4% above women's ($46152.57 \div 44213.37$, calculated by hand). The effect size is negligible by Cohen's convention, but a salary gap of a few percent is not negligible to the person who earns less: statistical "size" and practical importance are different questions.

The effect-size check of the program reads: " $|d| = 0.13$ is negligible. The sign of d follows the order of the groups: $d = -0.13$ means male scored higher than female." The size is judged on the absolute value of d ; the sign only tells you which group is higher.

On the log scale the picture is the same: Welch's $t(1181.44) = -2.54$, $p = .011$, with a difference of -0.04 in log salary, 95% CI $[-0.08, -0.01]$. The box plot shows heavily overlapping distributions: knowing that someone is a woman tells you very little about her salary.

How to report it

"An independent-samples Welch's t-test showed that Annual_salary differed between female ($M = 44213.37$, $SD = 14237.12$) and male ($M = 46152.57$, $SD = 14849.22$), $t(1190.08) = -2.31$, $p = .021$, $d = -0.13$, 95% CI $[-0.25, -0.02]$."

4.3 Do women and men do different jobs?

If the raw gap came from women and men working in different places, we would see it in the composition. Use **Analyses** → **Chi-square, Fisher's exact and McNemar tests** → **Chi-square test of independence**.

- **Department by Gender:** $\chi^2(4, N = 1200) = 3.68$, $p = .451$, Cramér's $V = .06$. The share of women ranges from 44% (Finance and Operations) to 53% (HR).
- **Job_level by Gender** (set Job_level to Categorical on the Data screen first): $\chi^2(4, N = 1200) = 1.61$, $p = .807$, Cramér's $V = .04$. Women are 46% of level 1, 47% of levels 2 and 3, 42% of level 4 and 10 of the 17 people at level 5.

What it means for our example

Neither department nor level differs significantly by gender. Women and men are distributed over departments and levels in much the same way, so the raw gap is not explained by women being concentrated in lower levels or lower-paying departments. (Level 5 has only 17 people, so do not read its 59% too closely.) Remember to set Job_level back to Scale if you want it as a single straight-line predictor.

4.4 The adjusted gap: regression of log salary

A multiple regression estimates the gender difference while holding the other predictors constant. We use log salary as the outcome, so the coefficients are approximate percentage differences (section 2.4).

1. Click **Analyses** → **Correlation and regression analysis** → **Multiple linear regression**.
2. **Outcome (scale)**: Annual_salary_In.
3. **Block 1 predictors**: Job_level, Tenure_years, Department, Education.
4. **Block 2 predictors**: Gender.

Putting gender alone in the second block is a *hierarchical regression*: SaNaML reports the model without gender, then the model with gender, and the change in R^2 (ΔR^2) that gender adds (Figure 17).

Multiple linear regression (Predicting Annual_salary_In)

What it means: The predictors together explain 83.8% of the variance in Annual_salary_In ($R^2 = .84$, $F = 5.19$, large). Significant predictors: Department: HR (vs Finance) ($\beta = -.09$), Department: IT (vs Finance) ($\beta = .09$), Department: Operations (vs Finance) ($\beta = -.24$), Department: Sales (vs Finance) ($\beta = -.16$), Job_level ($\beta = .68$), Education: Diploma (vs Bachelor) ($\beta = -.07$), Education: Doctorate (vs Bachelor) ($\beta = .07$), Education: Master (vs Bachelor) ($\beta = .08$), Tenure_years ($\beta = .18$), Gender: male (vs female) ($\beta = .08$). Block 2 adds $\Delta R^2 = .006$ ($p < .001$).

- OK Sample size.** 1200 respondents for 10 predictor terms (needed: 130).
Rule: At least 50 + 8k respondents for k predictors to test the whole model (Green (1991)).
- OK Collinearity.** Largest VIF = 3.18.
Rule: VIF below 5: no concern; 5–10: moderate; above 10: serious collinearity (Hair et al. (2019); Kutner et al. (2005)).
- OK Normal residuals.** Shapiro-Wilk $W = 0.998$, $p = .116$.
Rule: Shapiro-Wilk $p \geq .05$ for the residuals; with about 300 or more respondents the coefficient tests are robust anyway (Shapiro & Wilk (1965)).
- OK Constant error variance.** Breusch-Pagan (Koenker) $\chi^2(10) = 17.57$, $p = .063$.
Rule: Breusch-Pagan (Koenker) $p \geq .05$: no evidence that the error spread changes (Breusch & Pagan (1979); Koenker (1981)).
- OK Influential cases.** Largest Cook's distance = 0.020; 1 standardised residual beyond ± 3 .
Rule: No Cook's distance above 1 (Cook & Weisberg (1982)).
- Info Independent errors.** Durbin-Watson = 1.98 (values near 2 mean no order effect; it matters only when the row order means something, such as time).
- Info Effect size.** $f^2 = 5.19$ (large).
Rule: f^2 : .02 small, .15 medium, .35 large (Cohen (1988)).

Model summary							
Model	Predictors	R^2	Adjusted R^2	F	df	p	
1	Department, Job_level, Education, Tenure_years	.833	.831	657.72	9, 1190	<	
2	Department, Job_level, Education, Tenure_years, Gender	.838	.837	616.53	10, 1189	<	

Coefficients (model 2)							
Predictor	B	SE	β	t	p	95% CI of B	VIF
(Intercept)	10.214	0.012		817.32	< .001	[10.190, 10.239]	
Department: HR (vs Finance)	−0.087	0.014	−.088	−6.18	< .001	[−0.115, −0.060]	1.49
Department: IT (vs Finance)	0.070	0.012	.089	5.75	< .001	[0.046, 0.094]	1.75
Department: Operations (vs Finance)	−0.154	0.011	−.243	−14.31	< .001	[−0.175, −0.133]	2.12
Department: Sales (vs Finance)	−0.100	0.011	−.157	−9.30	< .001	[−0.122, −0.079]	2.10
Job_level	0.212	0.006	.681	32.76	< .001	[0.200, 0.225]	3.18
Education: Diploma (vs Bachelor)	−0.048	0.009	−.069	−5.57	< .001	[−0.066, −0.031]	1.14
Education: Doctorate (vs Bachelor)	0.105	0.019	.069	5.49	< .001	[0.067, 0.142]	1.17
Education: Master (vs Bachelor)	0.052	0.011	.077	4.90	< .001	[0.031, 0.073]	1.80
Tenure_years	0.013	0.001	.182	10.08	< .001	[0.011, 0.016]	2.40
Gender: male (vs female)	0.044	0.007	.076	6.48	< .001	[0.031, 0.058]	1.01

n = 1200. Categorical predictors are dummy-coded against their first group. β = standardised coefficient.

How to report it (APA style): A multiple linear regression (two blocks) of Annual_salary_In on Department, Job_level, Education, Tenure_years, Gender was significant, $F(10, 1189) = 616.53$, $p < .001$, $R^2 = .84$, adjusted $R^2 = .84$. Department: HR (vs Finance) predicted Annual_salary_In, $\beta = -.09$, $t(1189) = -6.18$, $p < .001$; Department: IT (vs Finance) predicted Annual_salary_In, $\beta = .09$, $t(1189) = 5.75$, $p < .001$; Department: Operations (vs Finance) predicted Annual_salary_In, $\beta = -.24$, $t(1189) = -14.31$, $p < .001$; Department: Sales (vs Finance) predicted Annual_salary_In, $\beta = -.16$, $t(1189) = -9.30$, $p < .001$; Job_level predicted Annual_salary_In, $\beta = .68$, $t(1189) = 32.76$, $p < .001$; Education: Diploma (vs Bachelor) predicted Annual_salary_In, $\beta = -.07$, $t(1189) = -5.57$, $p < .001$; Education: Doctorate (vs Bachelor) predicted Annual_salary_In, $\beta = .07$, $t(1189) = 5.49$, $p < .001$; Education: Master (vs Bachelor) predicted Annual_salary_In, $\beta = .08$, $t(1189) = 4.90$, $p < .001$; Tenure_years predicted Annual_salary_In, $\beta = .18$, $t(1189) = 10.08$, $p < .001$; Gender: male (vs female) predicted Annual_salary_In, $\beta = .08$, $t(1189) = 6.48$, $p < .001$.

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Figure 17. Regression of Annual_salary_In: model summary and coefficients (block 2 adds Gender).

Concept: what a regression coefficient means

A coefficient B is the average change in the outcome for one unit more of the predictor, holding the other predictors fixed. For a categorical predictor, the coefficient compares a group with the *reference group* (the first category alphabetically): "Gender: male (vs female)" is men minus women. Because the outcome is log salary, B multiplied by 100 is roughly the percentage difference. R^2 is the share of the variance explained.

Table 10

Regression of *Annual_salary_In* ($n = 1200$). Percentages were calculated by hand as $e^B - 1$.

Predictor	B	95% CI of B	p	Approx. %
Job_level (per level)	0.212	[0.200, 0.225]	< .001	+23.6
Tenure_years (per year)	0.013	[0.011, 0.016]	< .001	+1.3
Department: HR (vs Finance)	-0.087	[-0.115, -0.060]	< .001	-8.3
Department: IT (vs Finance)	0.070	[0.046, 0.094]	< .001	+7.3
Department: Operations (vs Finance)	-0.154	[-0.175, -0.133]	< .001	-14.3
Department: Sales (vs Finance)	-0.100	[-0.122, -0.079]	< .001	-9.5
Education: Diploma (vs Bachelor)	-0.048	[-0.066, -0.031]	< .001	-4.7
Education: Master (vs Bachelor)	0.052	[0.031, 0.073]	< .001	+5.3
Education: Doctorate (vs Bachelor)	0.105	[0.067, 0.142]	< .001	+11.1
Gender: male (vs female)	0.044	[0.031, 0.058]	< .001	+4.5

What it means for our example

Job level, tenure, department and education together explain 83.3% of the variance in log salary (model 1: $R^2 = .833$, $F(9, 1190) = 657.72$, $p < .001$). Adding gender raises R^2 to .838 (model 2: $F(10, 1189) = 616.53$, $\Delta R^2 = .006$, $p < .001$). The coefficient for gender is $B = 0.044$ ($SE = 0.007$), $t(1189) = 6.48$, $p < .001$, 95% CI [0.031, 0.058]: **men earn about 4.5% more than women at the same level, with the same tenure, department and education**, with a 95% confidence interval from about 3.1% to 6.0%. This is the gap that was planted in the data (about 4%). Each level adds about 24%, each year of tenure about 1.3%, and Operations pays about 14% less than Finance at the same level.

The checks under the result are all acceptable: no coefficient is hampered by collinearity (largest VIF = 3.18), the residuals are close to normal (Shapiro-Wilk $W = 0.998$, $p = .116$), the error variance is constant (Breusch-Pagan $p = .063$) and no case is influential (largest Cook's distance 0.020). That the log scale works well is part of the reason for using it; the same model on raw salary (not shown) gives a male coefficient of 1935.101 salary units, 95% CI [1233.356, 2636.846], $t = 5.41$.

How to report it

"A hierarchical multiple linear regression of log annual salary on job level, tenure, department and education (block 1, $R^2 = .83$) and gender (block 2) showed that men earned 4.5% more than women with the same level, tenure, department and education, $B = 0.044$, 95% CI [0.031, 0.058], $t(1189) = 6.48$, $p < .001$, $\Delta R^2 = .006$."

4.5 Is the result robust?

A single regression is one set of choices. Try the reasonable alternatives and see whether the gender coefficient moves.

- **Job level as groups.** Set Job_level to Categorical on the Data screen and repeat the regression with all predictors in one block. Each level gets its own coefficient: 0.212 (level 2), 0.431 (level 3), 0.626 (level 4) and 0.841 (level 5), all against level 1. The steps are close to equal, so treating level as a straight line was reasonable. The gender coefficient is unchanged, $B = 0.044$, $t(1186) = 6.47$, 95% CI [0.031, 0.058]. Set Job_level back to Scale afterwards.
- **More controls.** Adding Performance_rating and Age leaves the gender coefficient at $B = 0.044$, 95% CI [0.031, 0.058]; neither Age ($p = .440$) nor Performance_rating ($p = .615$) predicts salary once level is in the model.
- **Fewer controls.** Leaving Job_level out gives a gender coefficient of 0.037, 95% CI [0.019, 0.056], and R^2 falls from .84 to .69. The gap is similar, but the model is much poorer, because level is by far the strongest predictor of pay.

Note: "adjusted" is not the same as "fair"

Whether to control for a variable is a judgement about causes, not a statistical question. If women are held back from higher levels, then level is *part of* the gap, and controlling for it hides that part. Conversely, if level reflects real differences in responsibility, controlling for it is right. Report the gap with and without the contested controls, and say which you consider appropriate and why.

4.6 Does the gap differ by department?

A factorial ANOVA of Annual_salary_In by Gender and Department (**Analyses** → **Factorial ANOVA, ANCOVA and MANOVA**; Figure 18) asks whether the gap differs between departments. It does not adjust for level or tenure; it asks only whether the raw log-salary gap is the same everywhere.

Factorial ANOVA (Annual_salary_In by Department, Gender)

What it means: Significant effects on Annual_salary_In: Department (partial $\eta^2 = .078$, medium); Gender (partial $\eta^2 = .004$, negligible).

OK Equal variances. Levene's test (median) across the 10 cells: $F(9, 1190) = 1.32$, $p = .220$.
Rule: Levene's test $p < .05$ suggests unequal variances: rely on Welch's test and Games-Howell comparisons (Levene (1960); Games & Howell (1976)).

OK Cell sizes. The smallest cell has 53 respondents.

Info Sums of squares. Type III sums of squares with sum-to-zero coding: each effect is tested after all others, which suits unequal group sizes.

Info Effect size. Partial η^2 for every effect.
Rule: η^2 / ω^2 : about .01 small, .06 medium, .14 large (Cohen (1988)).

Cell descriptives			
Cell	n	Mean	SD
Finance - female	79	10.72	0.24
Finance - male	100	10.75	0.29
HR - female	60	10.66	0.26
HR - male	53	10.69	0.30
IT - female	93	10.80	0.30
IT - male	102	10.84	0.29
Operations - female	161	10.57	0.30
Operations - male	202	10.60	0.27
Sales - female	170	10.62	0.27
Sales - male	180	10.69	0.28

Factorial ANOVA (Type III sums of squares)						
Source	SS	df	MS	F	p	Partial η^2
Department	7.86	4	1.96	25.22	< .001	.078
Gender	0.41	1	0.41	5.25	.022	.004
Department x Gender	0.07	4	0.02	0.22	.925	.001
Error	92.68	1190	0.08			
Total (corrected)	101.20	1199				

$R^2 = .084$, adjusted $R^2 = .077$, $n = 1200$.

Estimated marginal means: Department				
Level	n	EMM	SE	95% CI
Finance	179	10.74	0.021	[10.70, 10.78]
HR	113	10.67	0.026	[10.62, 10.72]
IT	195	10.82	0.020	[10.78, 10.86]
Operations	363	10.59	0.015	[10.56, 10.61]
Sales	350	10.65	0.015	[10.62, 10.68]

Averaged equally over the levels of the other factor(s).

Pairwise comparisons of Department (Holm-adjusted)					
Pair	Difference	SE	t	df	p (Holm)
Finance - HR	0.07	0.034	1.98	1190	.096
Finance - IT	-0.08	0.029	-2.88	1190	.016
Finance - Operations	0.15	0.026	5.91	1190	< .001
Finance - Sales	0.08	0.026	3.28	1190	.006
HR - IT	-0.15	0.033	-4.55	1190	< .001
HR - Operations	0.08	0.030	2.81	1190	.016
HR - Sales	0.02	0.030	0.59	1190	.555
IT - Operations	0.24	0.025	9.46	1190	< .001
IT - Sales	0.17	0.025	6.74	1190	< .001
Operations - Sales	-0.07	0.021	-3.20	1190	.007

Estimated marginal means: Gender				
Level	n	EMM	SE	95% CI
female	563	10.67	0.013	[10.65, 10.70]
male	637	10.71	0.012	[10.69, 10.74]

Averaged equally over the levels of the other factor(s).

Pairwise comparisons of Gender (Holm-adjusted)					
Pair	Difference	SE	t	df	p (Holm)
female - male	-0.04	0.018	-2.29	1190	.022

How to report it (APA style): A factorial ANOVA on Annual_salary_In showed a significant effect of Department, $F(4, 1190) = 25.22$, $p < .001$, partial $\eta^2 = .078$; a significant effect of Gender, $F(1, 1190) = 5.25$, $p = .022$, partial $\eta^2 = .004$; no significant effect of Department x Gender, $F(4, 1190) = 0.22$, $p = .925$, partial $\eta^2 = .001$.

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Figure 18. Factorial ANOVA of Annual_salary_In by Department and Gender.

What it means for our example

There is a main effect of Department, $F(4, 1190) = 25.22$, $p < .001$, partial $\eta^2 = .078$, and a small main effect of Gender, $F(1, 1190) = 5.25$, $p = .022$, partial $\eta^2 = .004$, but no interaction, $F(4, 1190) = 0.22$, $p = .925$: the data give no sign that the gap is larger in one department than another.

Every cell has at least 53 people. A non-significant interaction does not prove that the gap is the same everywhere; with five departments the test has limited power to find a difference in one of them.

4.7 What this analysis cannot show

- **Cause.** A gap of 4.5% that remains after the controls does not tell you why. It may reflect discrimination, differences in negotiation, in performance that the file does not capture, in working hours or in the history of each person's pay. The regression shows that a difference exists; it cannot name the reason.
- **Unmeasured differences.** Anything not in the model (the quality of work, the starting salary, the labour market at the time of hiring) is invisible. Adding all available variables does not make the model complete.
- **Individuals.** The result is an average. It does not say that any particular woman is underpaid, and it must not be used to adjust one person's salary without a case-by-case review.
- **Small groups.** In a company with few women at senior levels, the confidence interval at each level is wide. Report the number of people behind each estimate.
- **Other forms of pay.** Bonuses, allowances and benefits are separate questions; so are part-time work and pay per hour.
- **Other characteristics.** SaNaML does not provide tools to compare, for example, the proportion of each group selected or rated highly in the way equal-opportunity audits do. The next section shows the one comparison of that kind that the general tools can make.

4.8 Promotion

Promotion can be compared with a chi-square test: **Chi-square test of independence**, Gender by Promoted_last_3y.

What it means for our example

65 of 563 women (12%) and 69 of 637 men (11%) were promoted in the last three years, $\chi^2(1, N = 1200) = 0.15$, $p = .695$, Cramér's $V = .01$; Fisher's exact test agrees, $p = .714$. There is no sign of a difference in the promotion rate. This is a comparison of overall rates. It does not account for level or tenure, and it is not a substitute for a formal adverse-impact analysis, which SaNaML does not provide.

Exercise

1. Repeat the regression with Gender in block 1 alongside the other predictors (one block). What is the male coefficient? (Answer: $B = 0.044$, the same, because a single block gives the same coefficient as the last block.)
1. What is the planted gap? Does the 95% interval for the coefficient contain it? (Answer: the planted gap is about 4%; the interval 3.1% to 6.0% contains it.)
1. Compare Annual_salary by Education (One-way ANOVA). Which pairs differ most? Run it and compare the groups' means with those in section 2.2.
1. Why is it wrong to conclude from this chapter that "the company discriminates", or that "it does not"?

5. Training evaluation

A company trains its employees, measures a skills test before and after, and wants to know whether the training worked. This chapter shows the usual first answer (a paired test), what else the data allow (does more training go with more gain?) and what the design cannot prove. Data: practice_hr.csv, columns Skills_test_before, Skills_test_after and Training_hours, all simulated.

5.1 The data

Descriptive statistics of the three variables: Skills_test_before M = 58.13 (SD = 11.07), Skills_test_after M = 63.74 (SD = 12.74) and Training_hours M = 23.91 (SD = 13.52, median 21.00, range 1.00 to 96.00, skewness 1.01).

5.2 Did scores change? Paired t-test and Wilcoxon test

The same people were measured twice, so use the paired test, not the independent one.

1. Click **Analyses** → **Independent-samples, paired-samples and one-sample t-tests** → **Paired-samples t-test (The same people measured twice)**.
2. **First measurement:** Skills_test_before. **Second measurement:** Skills_test_after. The change is second minus first.

Paired-samples t-test (Comparing Skills_test_before and Skills_test_after, same respondents)

What it means: Skills_test_after is higher than Skills_test_before by 5.61 points on average. The change is statistically significant and large ($d_z = 0.91$).

OK Sample size. Every group has 30 or more cases, so the test of means is robust to non-normal data.
Rule: With about 30 or more cases per group, t-tests and ANOVA tolerate non-normal data; with fewer, prefer the rank-based test if the data are clearly skewed (Lumley et al. (2002)).

Info Effect size. $d_z = 0.91$ is large (mean change divided by the SD of the changes).
Rule: Cohen's d: about 0.2 small, 0.5 medium, 0.8 large (Cohen (1988)).

Descriptives (complete pairs)

Measure	n	Mean	SD	Median
Skills_test_before	1200	58.13	11.07	58.10
Skills_test_after	1200	63.74	12.74	63.90
Difference (Skills_test_after – Skills_test_before)	1200	5.61	6.19	5.60

Paired t-test (two-sided)

t	df	p	Mean difference	95% CI	d_z	CI of d_z
31.37	1199	< .001	5.61	[5.26, 5.96]	0.91	[0.84, 0.97]

Rank-based alternative

Test	V	z	p	Matched rank-biserial r	Method
Wilcoxon signed-rank	638716.0	23.98	< .001	.80	normal approximation

10 pairs with no change were left out of the signed-rank test.

How to report it (APA style): A paired-samples t-test showed that Skills_test_after ($M = 63.74$, $SD = 12.74$) differed from Skills_test_before ($M = 58.13$, $SD = 11.07$), $t(1199) = 31.37$, $p < .001$, $d_z = 0.91$, 95% CI [0.84, 0.97].

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Figure 19. Paired-samples t-test of the skills test before and after training.

Concept: paired test and d_z

A paired test works on each person's *difference* (after minus before). It asks whether the mean difference is zero. Because it compares each person with themselves, it removes the differences between people, which are large here ($SD = 11$ points), and focuses on the change ($SD = 6.19$). The effect size d_z is the mean change divided by the standard deviation of the changes. The Wilcoxon signed-rank test is the rank-based alternative.

What it means for our example

Scores rose by 5.61 points on average (SD of the change = 6.19), $t(1199) = 31.37$, $p < .001$, 95% CI of the mean change [5.26, 5.96], $d_z = 0.91$, 95% CI [0.84, 0.97] (a large effect). The Wilcoxon signed-rank test agrees, $V = 638716.0$, $z = 23.98$, $p < .001$, matched rank-biserial $r = .80$; 10 pairs with no change were left out of that test.

How to report it

"A paired-samples t-test showed that Skills_test_after (M = 63.74, SD = 12.74) differed from Skills_test_before (M = 58.13, SD = 11.07), $t(1199) = 31.37$, $p < .001$, $d_z = 0.91$, 95% CI [0.84, 0.97]."

5.3 Does more training go with more gain?

If the training works, more hours of training should go with a larger gain. First make the gain: on the Data screen, step 4, choose **Combine two scale variables**, Variable A Skills_test_after, **Difference, A - B**, Variable B Skills_test_before, and name the variable Skills_gain (section 1.8). The paired t-test of section 5.2 found an average gain of 5.61 points. Now ask who gains more.

Gain by department. Skills_gain has means of 5.96 (Finance), 6.02 (HR), 5.29 (IT), 5.39 (Operations) and 5.71 (Sales). A one-way ANOVA finds no difference between departments, $F(4, 1195) = 0.54$, $p = .707$, $\omega^2 = -.002$ (Welch's $F(4, 444.84) = 0.52$, $p = .723$). The training worked about equally everywhere.

Gain and hours. **Bivariate and partial correlation** with Training_hours and Skills_gain gives $r(1198) = .15$, 95% CI [.09, .20], $p < .001$ (Figure 20). A regression of Skills_gain on Training_hours (Figure 21) gives $B = 0.067$ (SE = 0.013), $t(1198) = 5.12$, $p < .001$, 95% CI [0.041, 0.093], $R^2 = .021$: every additional ten hours of training goes with 0.67 more points of gain (10×0.067 , calculated by hand), but hours explain only 2.1% of the variance in the gain.

Pearson product-moment correlation (Pearson r)

What it means: 1 of 1 correlations are significant after Holm adjustment. The strongest is Training_hours with Skills_gain ($r = .15$, small, higher goes with higher).

Info Strength. Read the size of a correlation from its absolute value.

Rule: Correlation-type effect sizes (r , rank-biserial, ϕ): about .10 small, .30 medium, .50 large (Cohen (1988)).

Info Method. Pearson's r measures straight-line relationships. With strongly skewed data or many tied ratings, Spearman's ρ is a safer choice.

Correlation matrix

	1	2
1. Training_hours	—	
2. Skills_gain	.15***	—

* $p < .05$, ** $p < .01$, *** $p < .001$ (Holm-adjusted). Each pair uses every respondent with both values.

Every pair

Pair	r	95% CI	n	p	p (Holm)
Training_hours - Skills_gain	.146	[.09, .20]	1200	< .001	< .001

How to report it (APA style): Pearson r correlations (Holm-adjusted p -values) showed: Training_hours and Skills_gain, $r(1198) = .15$, $p < .001$, 95% CI [.09, .20].

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Figure 20. Correlation of Training_hours with Skills_gain.

Multiple linear regression (Predicting Skills_gain)

What it means: The predictors together explain 2.1% of the variance in Skills_gain ($R^2 = .02$, $f^2 = 0.02$, small). Significant predictors: Training_hours ($\beta = .15$).

OK Sample size. 1200 respondents for 1 predictor term (needed: 58).

Rule: At least $50 + 8k$ respondents for k predictors to test the whole model (Green (1991)).

OK Collinearity. Only one predictor: no collinearity to check.

Rule: VIF below 5: no concern; 5–10: moderate; above 10: serious collinearity (Hair et al. (2019); Kutner et al. (2005)).

OK Normal residuals. Shapiro-Wilk $W = 0.999$, $p = .559$.

Rule: Shapiro-Wilk $p \geq .05$ for the residuals; with about 300 or more respondents the coefficient tests are robust anyway (Shapiro & Wilk (1965)).

OK Constant error variance. Breusch-Pagan (Koenker) $\chi^2(1) = 0.53$, $p = .466$.

Rule: Breusch-Pagan (Koenker) $p \geq .05$: no evidence that the error spread changes (Breusch & Pagan (1979); Koenker (1981)).

OK Influential cases. Largest Cook's distance = 0.026; 3 standardised residuals beyond ± 3 .

Rule: No Cook's distance above 1 (Cook & Weisberg (1982)).

Info Independent errors. Durbin-Watson = 2.07 (values near 2 mean no order effect; it matters only when the row order means something, such as time).

Info Effect size. $f^2 = 0.02$ (small).

Rule: f^2 : .02 small, .15 medium, .35 large (Cohen (1988)).

Model summary

Model	Predictors	R^2	Adjusted R^2	F	df	p	ΔR^2	p (ΔR^2)
1	Training_hours	.021	.021	26.23	1, 1198	< .001		

Coefficients (model 1)

Predictor	B	SE	β	t	p	95% CI of B	VIF
(Intercept)	4.007	0.360		11.14	< .001	[3.301, 4.712]	
Training_hours	0.067	0.013	.146	5.12	< .001	[0.041, 0.093]	

$n = 1200$. Categorical predictors are dummy-coded against their first group. β = standardised coefficient.

How to report it (APA style): A multiple linear regression of Skills_gain on Training_hours was significant, $F(1, 1198) = 26.23$, $p < .001$, $R^2 = .02$, adjusted $R^2 = .02$. Training_hours predicted Skills_gain, $\beta = .15$, $t(1198) = 5.12$, $p < .001$.

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Figure 21. Regression of Skills_gain on Training_hours.

Concept: gain scores and the starting level

A gain score subtracts the starting level, but people who start high have less room to rise, and a high or low first score partly reflects chance, which tends to move back towards the average on the second test (*regression to the mean*; Barnett et al., 2005). The usual remedy is to include the before score as a predictor of the after score, as in section 5.4. Here the two approaches agree because the before score has a coefficient close to 1 (1.006), so the gain score does not depend on the starting level.

Two other ways to answer the same question use the original columns.

Correlations. With Training_hours, Skills_test_before and Skills_test_after, hours are unrelated to the before score ($r = .01$, $p = .761$) and slightly related to the after score ($r = .08$, 95% CI [.02, .13], Holm-adjusted $p = .013$); before and after correlate strongly ($r = .87$, 95% CI [.86, .89]).

Partial correlation. Tick Training_hours and Skills_test_after, and under **Control for** tick Skills_test_before (Figure 22). The correlation between hours and the after score, controlling for the before score, is the correlation between hours and *the part of the after score not predicted by the before score*, which is the change. It is $r(1197) = .15$, 95% CI [.09, .20], $p < .001$.

Partial correlation (Pearson r, controlling for Skills_test_before)

What it means: 1 of 1 partial correlations are significant after Holm adjustment. The strongest is Training_hours with Skills_test_after ($r = .15$, small, higher goes with higher).

Info Strength. Read the size of a correlation from its absolute value.

Rule: Correlation-type effect sizes (r , rank-biserial, ϕ): about .10 small, .30 medium, .50 large (Cohen (1988)).

Info Method. Pearson's r measures straight-line relationships. With strongly skewed data or many tied ratings, Spearman's ρ is a safer choice.

Correlation matrix

	1	2
1. Training_hours	—	
2. Skills_test_after	.15***	—

* $p < .05$, ** $p < .01$, *** $p < .001$ (Holm-adjusted). Complete cases only.

Every pair

Pair	r	95% CI	n	p	p (Holm)
Training_hours - Skills_test_after	.146	[.09, .20]	1200	< .001	< .001

How to report it (APA style): Pearson r partial correlations (Holm-adjusted p -values) showed: Training_hours and Skills_test_after, $r(1197) = .15$, $p < .001$, 95% CI [.09, .20].

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Figure 22. Partial correlation of Training_hours with Skills_test_after, controlling for Skills_test_before.

The hours alone hide the effect, because most of the variation in the after score is the person's starting level. Controlling for the starting level reveals it.

5.4 Regression

A hierarchical regression says the same in the units of the test (Figure 23). Choose **Multiple linear regression**, **Outcome** Skills_test_after, **Block 1** Skills_test_before, **Block 2** Training_hours.

Multiple linear regression (Predicting Skills_test_after)

What it means: The predictors together explain 76.9% of the variance in Skills_test_after ($R^2 = .77$, $f^2 = 3.32$, large). Significant predictors: Skills_test_before ($\beta = .87$), Training_hours ($\beta = .07$). Block 2 adds $\Delta R^2 = .005$ ($p < .001$).

OK Sample size. 1200 respondents for 2 predictor terms (needed: 66).

Rule: At least $50 + 8k$ respondents for k predictors to test the whole model (Green (1991)).

OK Collinearity. Largest VIF = 1.00.

Rule: VIF below 5: no concern; 5–10: moderate; above 10: serious collinearity (Hair et al. (2019); Kutner et al. (2005)).

OK Normal residuals. Shapiro-Wilk $W = 0.999$, $p = .568$.

Rule: Shapiro-Wilk $p \geq .05$ for the residuals; with about 300 or more respondents the coefficient tests are robust anyway (Shapiro & Wilk (1965)).

OK Constant error variance. Breusch-Pagan (Koenker) $\chi^2(2) = 1.03$, $p = .599$.

Rule: Breusch-Pagan (Koenker) $p \geq .05$: no evidence that the error spread changes (Breusch & Pagan (1979); Koenker (1981)).

OK Influential cases. Largest Cook's distance = 0.019; 3 standardised residuals beyond ± 3 .

Rule: No Cook's distance above 1 (Cook & Weisberg (1982)).

Info Independent errors. Durbin-Watson = 2.07 (values near 2 mean no order effect; it matters only when the row order means something, such as time).

Info Effect size. $f^2 = 3.32$ (large).

Rule: f^2 : .02 small, .15 medium, .35 large (Cohen (1988)).

Model summary

Model	Predictors	R^2	Adjusted R^2	F	df	p	ΔR^2	p (ΔR^2)
1	Skills_test_before	.764	.764	3872.34	1, 1198	< .001		
2	Skills_test_before, Training_hours	.769	.768	1989.95	2, 1197	< .001	.005	< .001

Coefficients (model 2)

Predictor	B	SE	β	t	p	95% CI of B	VIF
(Intercept)	3.681	0.995		3.70	< .001	[1.729, 5.633]	
Skills_test_before	1.006	0.016	.873	62.83	< .001	[0.974, 1.037]	1.00
Training_hours	0.067	0.013	.071	5.12	< .001	[0.041, 0.093]	1.00

$n = 1200$. Categorical predictors are dummy-coded against their first group. β = standardised coefficient.

How to report it (APA style): A multiple linear regression (two blocks) of Skills_test_after on Skills_test_before, Training_hours was significant, $F(2, 1197) = 1989.95$, $p < .001$, $R^2 = .77$, adjusted $R^2 = .77$. Skills_test_before predicted Skills_test_after, $\beta = .87$, $t(1197) = 62.83$, $p < .001$; Training_hours predicted Skills_test_after, $\beta = .07$, $t(1197) = 5.12$, $p < .001$.

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Figure 23. Regression of Skills_test_after on Skills_test_before (block 1) and Training_hours (block 2).

What it means for our example

The before score alone explains 76.4% of the variance in the after score ($R^2 = .764$, $F(1, 1198) = 3872.34$). Training_hours adds $\Delta R^2 = .005$ ($p < .001$), $R^2 = .769$. The coefficient for Training_hours is $B = 0.067$ ($SE = 0.013$), $t(1197) = 5.12$, $p < .001$, 95% CI [0.041, 0.093]: with the same before score, every additional ten hours of training goes with 0.67 more points on the after test (10×0.067 , calculated by hand). The coefficient of the before score is 1.006: someone who scored ten points higher before scores about ten points higher after. The scatter plot of hours against the after score shows how weak the raw relation is ($r = .08$, slope 0.074); the effect is only visible once the starting level is controlled.

5.5 What the design cannot show

- **No control group.** A rise from before to after may reflect the training, but also practice on the same test, the time of the year, other development at work, or *regression to the mean*: people chosen for training because of low scores tend to score higher on any retest (Barnett et al., 2005). The paired test cannot tell these apart. A comparison group of similar employees who were not trained is needed.
- **Hours are not randomly assigned.** People who chose more training may be more motivated, and motivated people improve more for other reasons. The dose-response result ($r = .15$) is consistent with an effect of training but does not prove it.
- **Skills are not performance.** A better test score does not show better work. Link training to outcomes that matter (performance, errors, promotion) if you can, and say when you cannot.
- **Statistical and practical significance.** With 1,200 people almost any difference is "significant". Judge the size: 5.61 points on a test with SD 11 to 13 is a large effect, 0.67 points per ten hours is a small one.

5.6 Planning the next evaluation: power analysis

Before the next evaluation, work out how many people you need. Click **Power analysis** and choose **Paired-samples t-test (Same people twice)** (Figure 24). Use a cautious effect size: a new evaluation may find a smaller effect than a first one, so take half of the d_z found here, 0.45. With **A priori power analysis**, $\alpha = .05$ (two-sided) and power .80, SaNaML reports $N = 41$: *An a priori power analysis for a paired-samples t-test indicated that 41 participants are needed to detect $d_z = 0.45$ with $\alpha = .05$ (two-sided) and power = .80 (achieved power .803).*

☒ **Paired-samples t-test (Same people twice)**
 Before vs after, or two ratings per person.

☐ **One-sample t-test (One score against a fixed value)**
 Compares one mean with a fixed value.

☐ **One-way ANOVA, fixed effects (Compare three or more groups)**
 One-way ANOVA, effect size f.

☐ **Chi-square test (Goodness of fit or independence)**
 Goodness of fit or independence; effect size w.

☐ **Pearson correlation test (Is a correlation different from zero?)**
 Test whether a correlation differs from zero.

2 · What do you want to find?

☒ **A priori power analysis (The sample size I need)**
 Most common in proposals.

☐ **Post hoc power analysis (The power of a given sample size)**
 For a fixed number of respondents.

☐ **Sensitivity power analysis (The smallest effect I can detect)**
 Useful when the sample size is already fixed.

Effect size d_z
 0.45
Conventions: small 0.2, medium 0.5, large 0.8 (Cohen, 1988). Better: take it from earlier studies.

Significance level α
 0.05

Power ($1 - \beta$)
 0.8
Plan for a power of .80 at $\alpha = .05$ unless the field expects otherwise.

Tails
 Two-sided (usual)

N = 41

How to report it: An a priori power analysis for a paired-samples t-test indicated that 41 participants are needed to detect $d_z = 0.45$ with $\alpha = .05$ (two-sided) and power = .80 (achieved power .803).

► References

Figure 24. A priori power analysis for a paired-samples t-test.

Two other questions can be asked of the same screen. If only one class of 40 people is available, **Sensitivity power analysis** shows the smallest effect it can detect with power .80: $d_z = 0.454$. And **Post hoc power analysis** with $N = 100$ shows that a study of 100 people has a power of only .508 to detect $d_z = 0.20$: it would miss a small effect about half of the time. Power analyses are not added to the Word report; copy their sentence.

Exercise

1. What is the mean gain on the skills test, and its 95% interval? (Answer: 5.61, 95% CI [5.26, 5.96].)
1. Run the partial correlation of Training_hours and Skills_test_after controlling for Skills_test_before. (Answer: $r = .15$.)
1. Change the effect size in the power analysis to $d_z = 0.30$. How many people are needed? Run it and report the sentence.
1. List two reasons why the 5.61-point gain could be smaller than it looks.

6. Engagement and job satisfaction: what goes with them?

Which working conditions go with higher engagement and higher job satisfaction? This chapter uses correlations to see the whole picture, multiple regression to see each condition's share, a hierarchical regression to separate background from working conditions, and a repeated-measures analysis to follow engagement through the year. Data: practice_hr.csv (simulated). Engagement and Job_satisfaction are survey scores from 1 to 5.

6.1 The correlation matrix

Click **Analyses** → **Correlation and regression analysis** → **Bivariate and partial correlation**. Tick Engagement, Job_satisfaction, Manager_support, Work_life_balance, Overtime_hours_month, Performance_rating, Tenure_years and Training_hours. Keep Pearson's r.

Concept: correlation

A correlation r measures how closely two variables follow a straight line, from -1 to $+1$. By convention .10 is small, .30 medium and .50 large (Cohen, 1988). With many pairs, some will look significant by chance; SaNaML therefore adjusts the p-values for the number of pairs (the Holm method) and uses every person with both values for each pair. *Partial* correlation removes the effect of a third variable.

What it means for our example

8 of the 28 correlations are significant after Holm adjustment. Engagement goes with Job_satisfaction ($r = .66$, 95% CI [.63, .69]), with Manager_support ($r = .40$, [.35, .45]) and with Work_life_balance ($r = .29$, [.24, .34]). Overtime_hours_month goes with lower Work_life_balance ($r = -.33$, [-.38, -.28]) and, weakly, with lower Engagement ($r = -.11$, Holm-adjusted $p = .003$). Job_satisfaction also goes with Manager_support ($r = .26$), Work_life_balance ($r = .25$) and Tenure_years ($r = .14$). Performance_rating and Training_hours are unrelated to everything else ($|r| \leq .06$). Manager support and work-life balance are unrelated to each other ($r = -.05$).

The **Charts** screen draws the same matrix as a heatmap (Figure 25): red for positive and blue for negative correlations.



Figure 25. Correlation heatmap of the same eight variables.

6.2 Multiple regression of engagement

A multiple regression estimates each condition's association with engagement while holding the others constant. Choose **Multiple linear regression**, **Outcome** Engagement, and tick Manager_support, Work_life_balance, Overtime_hours_month and Performance_rating as **Block 1 predictors** (Figure 26).

Multiple linear regression (Predicting Engagement)

What it means: The predictors together explain 26.3% of the variance in Engagement ($R^2 = .26$, $f^2 = 0.36$, large). Significant predictors: Manager_support ($\beta = .42$), Work_life_balance ($\beta = .31$).

OK Sample size. 1160 respondents for 4 predictor terms (needed: 82).

Rule: At least $50 + 8k$ respondents for k predictors to test the whole model (Green (1991)).

OK Collinearity. Largest VIF = 1.12.

Rule: VIF below 5: no concern; 5–10: moderate; above 10: serious collinearity (Hair et al. (2019); Kutner et al. (2005)).

OK Normal residuals. Shapiro-Wilk $W = 0.991$, $p < .001$; with this many respondents the coefficient tests are robust.

Rule: Shapiro-Wilk $p \geq .05$ for the residuals; with about 300 or more respondents the coefficient tests are robust anyway (Shapiro & Wilk (1965)).

Note Constant error variance. Breusch-Pagan (Koenker) $\chi^2(4) = 36.49$, $p < .001$.

Rule: Breusch-Pagan (Koenker) $p \geq .05$: no evidence that the error spread changes (Breusch & Pagan (1979); Koenker (1981)).

OK Influential cases. Largest Cook's distance = 0.034; 3 standardised residuals beyond ± 3 .

Rule: No Cook's distance above 1 (Cook & Weisberg (1982)).

Info Independent errors. Durbin-Watson = 2.02 (values near 2 mean no order effect; it matters only when the row order means something, such as time).

Info Effect size. $f^2 = 0.36$ (large).

Rule: f^2 : .02 small, .15 medium, .35 large (Cohen (1988)).

Model summary

Model	Predictors	R^2	Adjusted
1	Performance_rating, Overtime_hours_month, Manager_support, Work_life_balance	.263	.260

Coefficients (model 1)

Predictor	B	SE	β	t	p	95% CI of B	VIF
(Intercept)	2.440	0.114		21.31	< .001	[2.215, 2.664]	
Performance_rating	0.016	0.017	.023	0.90	.369	[-0.018, 0.050]	1.00
Overtime_hours_month	-0.002	0.002	-.022	-0.81	.418	[-0.007, 0.003]	1.12
Manager_support	0.277	0.017	.418	16.53	< .001	[0.244, 0.310]	1.00
Work_life_balance	0.235	0.020	.310	11.59	< .001	[0.195, 0.275]	1.12

$n = 1160$ (40 respondents with missing values left out). Categorical predictors are dummy-coded against their first group. β = standardised coefficient.

How to report it (APA style): A multiple linear regression of Engagement on Performance_rating, Overtime_hours_month, Manager_support, Work_life_balance was significant, $F(4, 1155) = 102.84$, $p < .001$, $R^2 = .26$, adjusted $R^2 = .26$. Manager_support predicted Engagement, $\beta = .42$, $t(1155) = 16.53$, $p < .001$; Work_life_balance predicted Engagement, $\beta = .31$, $t(1155) = 11.59$, $p < .001$.

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Figure 26. Multiple regression of Engagement on four working conditions.

Table 11

Regression of Engagement ($n = 1160$; 40 people with missing values left out)

Predictor	B	β	t	p
Manager_support	0.277	.42	16.53	< .001
Work_life_balance	0.235	.31	11.59	< .001
Overtime_hours_month	-0.002	-.02	-0.81	.418
Performance_rating	0.016	.02	0.90	.369

What it means for our example

The four predictors explain 26.3% of the variance in Engagement ($R^2 = .263$, adjusted $R^2 = .260$, $F(4, 1155) = 102.84$, $p < .001$). A one-point rise in manager support goes with 0.28 points higher engagement, and a one-point rise in work-life balance with 0.24 points, each with the other conditions held constant. Overtime and performance add nothing once the other two are in the model. Overtime was weakly correlated with engagement ($r = -.11$), but it is related to work-life balance ($r = -.33$), which carries the association. Whether overtime *causes* lower engagement through work-life balance is a question of mediation, which SaNaML does not test; and survey data measured at one time cannot settle the order of cause and effect.

SaNaML's checks all pass except one, flagged *Note*: the error variance is not constant (Breusch-Pagan $\chi^2(4) = 36.49$, $p < .001$). Engagement cannot exceed 5, so its spread shrinks near the top of the scale. The residuals are not perfectly normal (Shapiro-Wilk $p < .001$), but with 1,160 people the coefficient tests are robust, as SaNaML says. VIF is at most 1.12.

Cut-off values: regression

Sample size at least $50 + 8k$ for k predictors (Green, 1991). VIF below 5: no concern (Hair et al., 2019). Breusch-Pagan $p \geq .05$: no evidence that the error spread changes (Breusch & Pagan, 1979; Koenker, 1981).

6.3 Hierarchical regression: background first, working conditions second

Does the background (level, gender, age, tenure) explain engagement, and what do the working conditions add? Put Job_level, Gender, Age and Tenure_years in block 1 and the four conditions in block 2 (Figure 27).

Multiple linear regression (Predicting Engagement)

What it means: The predictors together explain 26.9% of the variance in Engagement ($R^2 = .27$, $F^2 = 0.37$, large). Significant predictors: Tenure_years ($\beta = -.09$), Manager_support ($\beta = .42$), Work_life_balance ($\beta = .31$). Block 2 adds $\Delta R^2 = .261$ ($p < .001$).

OK Sample size. 1160 respondents for 8 predictor terms (needed: 114).

Rule: At least 50 + 8k respondents for k predictors to test the whole model (Green (1991)).

OK Collinearity. Largest VIF = 1.89.

Rule: VIF below 5: no concern; 5–10: moderate; above 10: serious collinearity (Hair et al. (2019); Kutner et al. (2005)).

OK Normal residuals. Shapiro-Wilk $W = 0.991$, $p < .001$; with this many respondents the coefficient tests are robust.

Rule: Shapiro-Wilk $p \geq .05$ for the residuals; with about 300 or more respondents the coefficient tests are robust anyway (Shapiro & Wilk (1965)).

Note Constant error variance. Breusch-Pagan (Koenker) $\chi^2(8) = 36.56$, $p < .001$.

Rule: Breusch-Pagan (Koenker) $p \geq .05$: no evidence that the error spread changes (Breusch & Pagan (1979); Koenker (1981)).

OK Influential cases. Largest Cook's distance = 0.023; 2 standardised residuals beyond ± 3 .

Rule: No Cook's distance above 1 (Cook & Weisberg (1982)).

Info Independent errors. Durbin-Watson = 2.02 (values near 2 mean no order effect; it matters only when the row order means something, such as time).

Info Effect size. $F^2 = 0.37$ (large).

Rule: F^2 : .02 small, .15 medium, .35 large (Cohen (1988)).

Model summary

Model	Predictors
1	Job_level, Gender, Age, Tenure_years
2	Job_level, Gender, Age, Tenure_years, Performance_rating, Overtime_hours_month, Manager_support

Coefficients (model 2)

Predictor	B	SE	β	t	p	95% CI of B	VIF
(Intercept)	2.429	0.133		18.24	< .001	[2.168, 2.691]	
Job_level	0.036	0.021	.057	1.68	.093	[-0.006, 0.077]	1.81
Gender: male (vs female)	-0.036	0.029	-.031	-1.22	.222	[-0.093, 0.022]	1.00
Age	0.001	0.002	.017	0.65	.516	[-0.002, 0.004]	1.08
Tenure_years	-0.014	0.005	-.094	-2.71	.007	[-0.023, -0.004]	1.89
Performance_rating	0.014	0.017	.020	0.80	.423	[-0.020, 0.048]	1.02
Overtime_hours_month	-0.002	0.002	-.022	-0.83	.409	[-0.007, 0.003]	1.12
Manager_support	0.276	0.017	.417	16.50	< .001	[0.243, 0.309]	1.01
Work_life_balance	0.235	0.020	.310	11.60	< .001	[0.195, 0.274]	1.12

n = 1160 (40 respondents with missing values left out). Categorical predictors are dummy-coded against their first group. β = standardised coefficient.

How to report it (APA style): A multiple linear regression (two blocks) of Engagement on Job_level, Gender, Age, Tenure_years, Performance_rating, Overtime_hours_month, Manager_support, Work_life_balance was significant, $F(8, 1151) = 52.82$, $p < .001$, $R^2 = .27$, adjusted $R^2 = .26$. Tenure_years predicted Engagement, $\beta = -.09$, $t(1151) = -2.71$, $p = .007$; Manager_support predicted Engagement, $\beta = .42$, $t(1151) = 16.50$, $p < .001$; Work_life_balance predicted Engagement, $\beta = .31$, $t(1151) = 11.60$, $p < .001$.

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Figure 27. Hierarchical regression of Engagement: background in block 1, working conditions in block 2.

What it means for our example

The background explains almost nothing (model 1: $R^2 = .007$, $F(4, 1155) = 2.15$, $p = .073$). Adding the working conditions raises R^2 to .269 ($\Delta R^2 = .261$, $p < .001$). In the full model only Tenure_years ($B = -0.014$, $\beta = -.09$, $p = .007$), Manager_support ($\beta = .42$) and Work_life_balance ($\beta = .31$) are significant; gender is not ($B = -0.036$, $p = .222$). Engagement depends much more on how people are managed and how they live than on who they are.

How to report it

"A hierarchical multiple linear regression of Engagement on background variables (block 1) and working conditions (block 2) showed that the working conditions added significantly, $\Delta R^2 = .261$, $p < .001$, final model $R^2 = .27$, $F(8, 1151) = 52.82$, $p < .001$. Manager support ($\beta = .42$, $p < .001$) and work-life balance ($\beta = .31$, $p < .001$) predicted engagement."

6.4 Job satisfaction

Repeat the regression with Job_satisfaction as the outcome and Manager_support, Work_life_balance, Overtime_hours_month, Performance_rating and Engagement as predictors ($n = 1111$; 89 people with missing values are left out). The model explains 43.5% of the variance ($R^2 = .435$, $F(5, 1105) = 170.08$, $p < .001$). Engagement dominates ($B = 0.656$, $\beta = .63$, $t = 23.70$, $p < .001$) and Work_life_balance adds a little ($B = 0.067$, $\beta = .08$, $p = .001$); Manager_support, which correlated with satisfaction at $r = .26$, has no coefficient of its own once engagement is in the model ($B = 0.011$, $p = .524$). The pattern fits the idea that support works through engagement, but this is an interpretation that the analysis cannot test.

Note: two scores that are nearly one thing

Engagement and satisfaction correlate at $r = .66$, and both are answers from the same person at the same time. Predicting one from the other is partly asking the same question twice. Do not read the large coefficient as "engagement drives satisfaction".

6.5 Engagement over the year: repeated-measures ANOVA

The file has engagement measured in four quarters. The same people were measured four times, so use **Analyses** → **One-way repeated-measures ANOVA**, and tick Engagement_Q1 to Engagement_Q4 in the order they were taken (Figure 28).

One-way repeated-measures ANOVA (Comparing Engagement_Q1, Engagement_Q2, Engagement_Q3, Engagement_Q4, same respondents)

What it means: The scores differ across Engagement_Q1, Engagement_Q2, Engagement_Q3, Engagement_Q4 (negligible effect, generalised $\eta^2 = .002$). Pairs that differ: Engagement_Q1 vs Engagement_Q2; Engagement_Q1 vs Engagement_Q3; Engagement_Q1 vs Engagement_Q4; Engagement_Q2 vs Engagement_Q4.

OK Sphericity. Mauchly's test is not significant: no correction needed.

Rule: If Mauchly's test is significant, correct the degrees of freedom: Greenhouse-Geisser when $\epsilon < .75$, Huynh-Feldt when $\epsilon \geq .75$ (Girden (1992)).

OK Sample size. 1200 complete respondents: the ANOVA is robust to non-normal data.

Rule: With about 30 or more cases per group, t-tests and ANOVA tolerate non-normal data; with fewer, prefer the rank-based test if the data are clearly skewed (Lumley et al. (2002)).

Info Effect size. Generalised $\eta^2 = .002$ (negligible).

Rule: η^2 / ω^2 : about .01 small, .06 medium, .14 large (Cohen (1988)).

Descriptives (complete cases)

Measure	n	Mean	SD
Engagement_Q1	1200	4.19	0.61
Engagement_Q2	1200	4.22	0.61
Engagement_Q3	1200	4.24	0.61
Engagement_Q4	1200	4.25	0.61

Repeated-measures ANOVA

Correction	df	F	p	ϵ
None (sphericity assumed)	3, 3597	15.79	< .001	1
Greenhouse-Geisser	2.99, 3584.78	15.79	< .001	.997
Huynh-Feldt	3.00, 3594.73	15.79	< .001	.999

Partial $\eta^2 = .013$; generalised $\eta^2 = .002$ (Bakeman, 2005).

Mauchly's test of sphericity

W	χ^2	df	p
0.995	6.13	5	.294

Pairwise paired t-tests (Holm-adjusted)

Pair	Mean difference	t	df	p	p (Holm)	d_z
Engagement_Q1 – Engagement_Q2	–0.04	–3.84	1199	< .001	< .001	–0.11
Engagement_Q1 – Engagement_Q3	–0.05	–5.23	1199	< .001	< .001	–0.15
Engagement_Q1 – Engagement_Q4	–0.07	–6.59	1199	< .001	< .001	–0.19
Engagement_Q2 – Engagement_Q3	–0.01	–1.40	1199	.161	.321	–0.04
Engagement_Q2 – Engagement_Q4	–0.03	–2.75	1199	.006	.018	–0.08
Engagement_Q3 – Engagement_Q4	–0.01	–1.33	1199	.184	.321	–0.04

Rank-based alternative

Test	χ^2	df	p	Kendall's W
Friedman	47.19	3	< .001	.013

Rank-based pairwise comparisons: Wilcoxon (Holm-adjusted)

Pair	V	p	p (Holm)	r
Engagement_Q1 – Engagement_Q2	281352.5	< .001	< .001	–.13
Engagement_Q1 – Engagement_Q3	260728.0	< .001	< .001	–.18
Engagement_Q1 – Engagement_Q4	243144.5	< .001	< .001	–.23
Engagement_Q2 – Engagement_Q3	300085.0	.331	.662	–.03
Engagement_Q2 – Engagement_Q4	283528.0	.005	.015	–.10
Engagement_Q3 – Engagement_Q4	299619.5	.335	.662	–.03

How to report it (APA style): A one-way repeated-measures ANOVA showed a significant difference across Engagement_Q1, Engagement_Q2, Engagement_Q3, Engagement_Q4, $F(3, 3597) = 15.79$, $p < .001$, partial $\eta^2 = .013$, generalised $\eta^2 = .002$.

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Figure 28. Repeated-measures ANOVA of Engagement_Q1 to Engagement_Q4.

Cut-off values: sphericity

If Mauchly's test is significant, correct the degrees of freedom: Greenhouse-Geisser when $\epsilon < .75$, Huynh-Feldt when $\epsilon \geq .75$ (Girden, 1992). The ANOVA assumes the differences between all pairs of time points have equal variance (sphericity).

What it means for our example

Mean engagement rose slightly each quarter: 4.19, 4.22, 4.24 and 4.25 (SD = 0.61 in each). Mauchly's test is not significant ($W = 0.995$, $\chi^2(5) = 6.13$, $p = .294$), so no correction is needed, $F(3, 3597) = 15.79$, $p < .001$, partial $\eta^2 = .013$, generalised $\eta^2 = .002$ (negligible). The Friedman test agrees, $\chi^2(3) = 47.19$, $p < .001$, Kendall's $W = .013$. The difference between the first and the last quarter is 0.07 points (Holm-adjusted $p < .001$, $d_z = -0.19$). The change is statistically clear and practically tiny: with 1,200 people SaNaML can detect a change of a few hundredths of a point, but whether 0.07 points on a five-point scale matters to the company is a question for the company.

Exercise

1. Which working condition has the largest standardised coefficient for Engagement? (Answer: Manager_support, $\beta = .42$.)
1. Run the regression of Engagement on Tenure_years alone. Is the sign the same as in the hierarchical model? (Run it and compare with $B = -0.014$.)
1. Draw a scatter plot of Engagement against Manager_support. What is r and the slope? (Answer: $r = .40$, slope 0.266, $n = 1160$.)
1. Why is the repeated-measures ANOVA better than four separate tests of the quarters?

7. Absence and department differences

This chapter uses non-parametric tests, which do not assume a bell-shaped distribution, and the chi-square test for categories. The questions are whether absence and leaving differ between departments and what goes with absence.

7.1 Absence by department

Absence_days is a count from 0 to 11 with a median of 4.00 in every department (Figure 29 shows the report). Choose **One-way ANOVA (Compare three or more groups)**, **Score to compare** Absence_days and **Groups** Department.

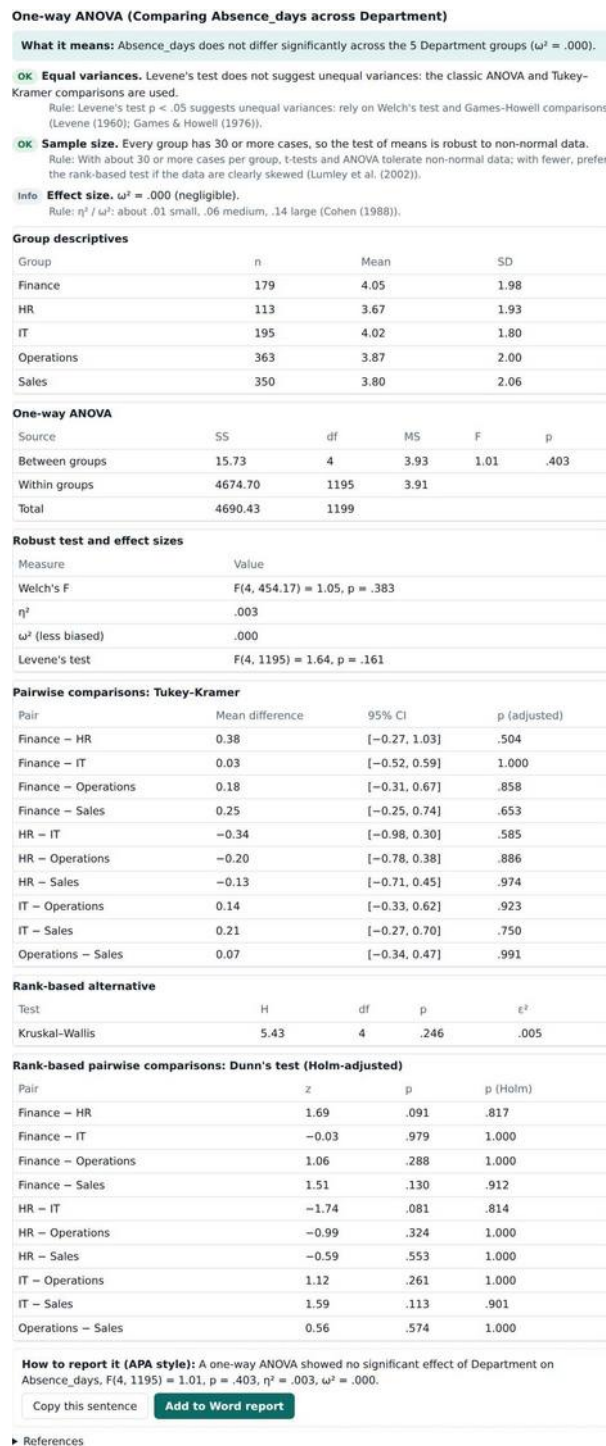


Figure 29. One-way ANOVA and Kruskal-Wallis test of Absence_days by Department.

What it means for our example

Mean absence is 4.05 days in Finance, 3.67 in HR, 4.02 in IT, 3.87 in Operations and 3.80 in Sales. The departments do not differ: $F(4, 1195) = 1.01, p = .403, \eta^2 = .003, \omega^2 = .000$. Levene's test does not suggest unequal variances ($p = .161$), and the Kruskal-Wallis test, which does not depend on the distribution of a count, agrees: $H(4) = 5.43, p = .246, \epsilon^2 = .005$. None of the ten pairs differs after correction. The same is true of Engagement in this file: $F(4, 1195) = 0.28, p = .891$ and Kruskal-Wallis $p = .838$ (means from 4.18 to 4.24).

Note that a non-significant result is a finding. "No evidence of a difference between departments" is useful to know. Do not describe it as "no difference" without the confidence intervals; the Tukey-Kramer intervals in the report show the range of differences that remain plausible (for example Finance minus HR, 0.38 days, 95% CI [-0.27, 1.03]).

7.2 What goes with absence?

Absence is a count with many ties, so use Spearman's rank correlation: **Bivariate and partial correlation**, choose **Spearman's rank-order correlation**, and tick Absence_days, Engagement, Job_satisfaction, Commute_km, Overtime_hours_month and Tenure_years.

What it means for our example

Absence is lower among more engaged employees ($p = -.18$, 95% CI [-.24, -.13], Holm-adjusted $p < .001$) and, less strongly, among more satisfied ones ($p = -.10$, [-.16, -.05], $p = .006$). It is slightly higher with a longer commute ($p = .08$, $p = .044$ after adjustment). Overtime ($p = .07$, adjusted $p = .114$) and tenure ($p = -.00$) are not related to absence. Absence does not differ by gender (3.86 against 3.90 days, Welch $p = .729$) or between people who stayed and people who left (3.88 days in both groups, $p = .957$).

These are weak associations: engagement explains about 3% of the variation in absence ($p^2 \approx .03$, calculated by hand). Absence is a poor tool for judging an individual.

7.3 Leaving by department: the chi-square test

Is leaving linked to department? Both variables are categories, so use **Chi-square, Fisher's exact and McNemar tests** → **Chi-square test of independence: Categorical variable** Department, **Second categorical variable** Left_company (Figure 30).

Chi-square test of independence (Department by Left_company)

What it means: There is no significant relationship between Department and Left_company (Cramér's V = .08).

OK Expected counts. All expected counts are adequate (smallest 16.57).

Rule: Chi-square: no expected count below 1 and no more than 20% below 5; otherwise use the exact test (Cochran (1954)).

Info Effect size. Cramér's V = .08 (negligible).

Rule: Cramér's V: small, medium and large are .10, .30 and .50 divided by $\sqrt{(\text{smaller table dimension} - 1)}$ (Cohen (1988)).

Counts (row %)

Department	No	Yes	Total
Finance	155 (87%)	24 (13%)	179
HR	97 (86%)	16 (14%)	113
IT	173 (89%)	22 (11%)	195
Operations	294 (81%)	69 (19%)	363
Sales	305 (87%)	45 (13%)	350

Tests

Test	Value	df	p
Pearson chi-square	8.42	4	.077
Likelihood ratio	8.20	4	.085

Adjusted standardised residuals

Department	No	Yes
Finance	0.52	-0.52
HR	0.16	-0.16
IT	1.46	-1.46
Operations	-2.80 *	2.80 *
Sales	1.14	-1.14

* beyond ± 1.96 : this cell holds clearly more (positive) or fewer (negative) cases than expected.

Effect size

Measure	Value	Size
Cramér's V	.084	negligible

How to report it (APA style): A chi-square test of independence showed no significant association between Department and Left_company, $\chi^2(4, N = 1200) = 8.42, p = .077$, Cramér's V = .08.

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Figure 30. Chi-square test of Department by Left_company.

Concept: the chi-square test of independence

The test compares the counts in a table with the counts expected if the two variables were unrelated. The *adjusted standardised residual* of each cell says how far its count is from expectation, in units that behave like z-scores: beyond ± 1.96 is notable. Cramér's V is the effect size, from 0 to 1. The chi-square test needs expected counts of at least 5 in nearly all cells; otherwise use Fisher's exact test.

Table 12*Left_company by Department*

Department	Stayed	Left	% left
Finance	155	24	13%
HR	97	16	14%
IT	173	22	11%
Operations	294	69	19%
Sales	305	45	13%

What it means for our example

Overall 176 of the 1,200 employees (14.7%) left. The test does not reject independence: $\chi^2(4, N = 1200) = 8.42$, $p = .077$, Cramér's $V = .08$ (negligible). One cell is flagged: Operations has more leavers than expected (adjusted residual 2.80, 19% left against 11% to 14% elsewhere). Because the overall test is not significant, treat that as a lead to examine, not as a finding: with ten cells, one will fall outside ± 1.96 by chance fairly often. If Operations is a concern, test it as a pre-planned question on new data.

The same test on two other variables gives no significant relation either. Promotion and leaving: 15% of those not promoted left against 11% of those promoted, $\chi^2(1, N = 1200) = 1.45$, $p = .228$, Cramér's $V = .03$ (Fisher's exact test $p = .247$). Gender and leaving: 17% of women and 13% of men left, $\chi^2(1, N = 1200) = 2.91$, $p = .088$, Cramér's $V = .05$ (Fisher $p = .102$).

The variables made in section 1.8 can be used here. Tenure_group (three equal-sized groups) against Left_company (Figure 31): 18% of the low-tenure third left (71 of 401), 17% of the middle third (68 of 407) and 9% of the high-tenure third (37 of 392), $\chi^2(2, N = 1200) = 12.87$, $p = .002$, Cramér's $V = .10$ (small). The high-tenure third has fewer leavers than expected (adjusted residual -3.57 for leaving) and the low-tenure third more (2.11).

Chi-square test of independence (Tenure_group by Left_company)

What it means: Tenure_group and Left_company are related (Cramér's V = .10, small). The cells with large adjusted residuals (beyond ± 1.96) show where the pattern comes from.

OK Expected counts. All expected counts are adequate (smallest 57.49).

Rule: Chi-square: no expected count below 1 and no more than 20% below 5; otherwise use the exact test (Cochran (1954)).

Info Effect size. Cramér's V = .10 (small).

Rule: Cramér's V: small, medium and large are .10, .30 and .50 divided by $\sqrt{(\text{smaller table dimension} - 1)}$ (Cohen (1988)).

Counts (row %)			
Tenure_group	No	Yes	Total
Low third	330 (82%)	71 (18%)	401
Middle third	339 (83%)	68 (17%)	407
High third	355 (91%)	37 (9%)	392

Tests			
Test	Value	df	p
Pearson chi-square	12.87	2	.002
Likelihood ratio	13.71	2	.001

Adjusted standardised residuals		
Tenure_group	No	Yes
Low third	-2.11 *	2.11 *
Middle third	-1.43	1.43
High third	3.57 *	-3.57 *

* beyond ± 1.96 : this cell holds clearly more (positive) or fewer (negative) cases than expected.

Effect size		
Measure	Value	Size
Cramér's V	.104	small

How to report it (APA style): A chi-square test of independence showed a significant association between Tenure_group and Left_company, $\chi^2(2, N = 1200) = 12.87, p = .002$, Cramér's V = .10.

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Figure 31. Chi-square test of Tenure_group by Left_company.

Merging Finance and HR into one category called Corporate (Dept_merged) changes the department test: $\chi^2(3, N = 1200) = 8.39, p = .039$, Cramér's V = .08 (negligible), with the same flagged cell, Operations (adjusted residual 2.80).

Note: merging categories after looking at the data

With all five departments the test gave $p = .077$; with Finance and HR merged it gives $p = .039$. No employee changed; only the number of categories did. If you choose a merge, a cut point or a grouping after seeing which version gives a significant result, the p-value is no longer trustworthy. Decide such groupings from the meaning of the groups (here, perhaps, whether Finance and HR share a manager) before you look at the outcome, and report every version you tried.

7.4 Leavers and stayers compared

Descriptive statistics split by Left_company gives a first look at how leavers differ, which chapter 8 turns into a prediction.

Table 13

Leavers and stayers (means)

Variable	Stayed (n = 1024)	Left (n = 176)
Tenure_years	5.71	4.48
Annual_salary	45776.58	42136.93
Overtime_hours_month	8.76	11.67
Engagement	4.25	3.91

Leavers had shorter tenure, lower pay, more overtime and lower engagement. Because these variables are correlated with each other and with the outcome, a regression is needed to separate them; that is what chapter 8 does.

Exercise

1. Run the chi-square test of Gender by Left_company. Is the difference significant? (Answer: no, $p = .088$; 17% of women and 13% of men left.)
1. Repeat the Department test on Job_satisfaction. Compare the Kruskal-Wallis and the ANOVA results.
1. In the chi-square table, why should you not conclude that Operations has a leaving problem? (Answer: the overall test is not significant and one flagged cell out of ten is not unusual.)

8. Predicting attrition

Which employees are likely to leave in the next year, and why? This chapter first fits a logistic regression, then uses the **Machine learning** screen to compare prediction models on employees they have not seen, explain what drives the predictions, check that the predicted probabilities can be trusted, and look at the trade-off between catching leavers and raising false alarms. The outcome is `Left_company` ("Yes" for 176 of the 1,200 employees, 14.7%) in `practice_hr.csv`; the data are simulated.

Note: read this before you build a model of people

A prediction of leaving describes a group of employees with similar characteristics. It is not a statement about a particular person, it is wrong for many people, and it can harm the people it is wrong about. Section 8.9 sets out how such predictions may be used. SaNaML's output is meant for understanding and planning (where is attention needed, and why), not for ranking or selecting individual employees.

8.1 What is being predicted

`Left_company` records whether the person left within the year. It does not record when, voluntarily or not, or why. Three consequences:

- People who left in January and in December are treated alike (no time-to-event analysis; SaNaML has no survival analysis).
- Dismissals, retirements and resignations are mixed. Real projects usually predict voluntary leaving only, using the reasons held in HR records.
- The predictors must be known **before** the outcome. `Engagement_Q1` to `Engagement_Q4` measure engagement through the same year; someone who left in the second quarter has no engagement for the third and fourth. We therefore leave them out and use the annual Engagement score, which we treat as measured at the start of the year. In real data, check the dates of every variable.

8.2 Logistic regression first

Logistic regression is the standard model for a yes/no outcome. Choose **Analyses** → **Correlation and regression analysis** → **Binary logistic regression**, **Outcome** `Left_company`, the event "Yes", and as predictors the fifteen work-related variables: `Department`, `Job_level`, `Tenure_years`, `Annual_salary`, `Performance_rating`, `Training_hours`, `Overtime_hours_month`, `Commute_km`, `Remote_days_week`, `Manager_support`, `Work_life_balance`, `Engagement`, `Job_satisfaction`, `Absence_days` and `Promoted_last_3y` (Figure 32). We deliberately leave out `Gender`, `Age` and `Education`, the employee number and the variables of chapters 4 to 6 that duplicate others (section 8.3 explains why).

Binary logistic regression (Predicting Left_company = "Yes")

What it means: The model predicts "Yes" better than chance (Nagelkerke $R^2 = .16$, AUC = .75, acceptable). Tenure_years: each unit divides the odds of "Yes" by 1.08; Overtime_hours_month: each unit multiplies the odds of "Yes" by 1.05; Commute_km: each unit multiplies the odds of "Yes" by 1.03; Engagement: each unit divides the odds of "Yes" by 2.61.

Check: Events per variable. EPV = 8.9.

Rule: At least 10 respondents in the smaller outcome group per coefficient (EPV ≥ 10) (Peduzzi et al. (1996)).

OK Calibration. Hosmer-Lemeshow $\chi^2(8) = 12.33$, $p = .137$.

Rule: Hosmer-Lemeshow $p \geq .05$: no evidence of poor fit (Hosmer et al. (2013)).

OK Discrimination. AUC = .750 (acceptable).

Rule: AUC .7-.8 acceptable, .8-.9 excellent, $\geq .9$ outstanding (Hosmer et al. (2013)).

Model fit

χ^2	df	p	Nagelkerke R^2	McFadden R^2	Tjur R^2	AUC	95% CI of AUC
104.71	18	< .001	.163	.116	.108	.750	[.71, .79]

Coefficients

Predictor	B	SE	Wald	p	Odds ratio	95% CI of OR
(Intercept)	1.969	1.016	3.76	.053	7.163	[0.978, 52.445]
Department: HR (vs Finance)	0.198	0.394	0.25	.615	1.219	[0.563, 2.640]
Department: IT (vs Finance)	-0.280	0.361	0.60	.438	0.756	[0.373, 1.533]
Department: Operations (vs Finance)	0.454	0.340	1.79	.181	1.575	[0.809, 3.066]
Department: Sales (vs Finance)	0.142	0.311	0.21	.647	1.153	[0.627, 2.120]
Job_level	-0.187	0.225	0.69	.406	0.829	[0.534, 1.289]
Tenure_years	-0.077	0.034	5.14	.023	0.926	[0.867, 0.990]
Annual_salary	0.000	0.000	0.23	.630	1.000	[1.000, 1.000]
Performance_rating	-0.125	0.109	1.31	.252	0.882	[0.712, 1.093]
Training_hours	0.009	0.007	1.85	.174	1.009	[0.996, 1.022]
Overtime_hours_month	0.046	0.013	12.09	< .001	1.047	[1.020, 1.075]
Commute_km	0.034	0.009	14.83	< .001	1.035	[1.017, 1.053]
Remote_days_week	0.037	0.105	0.12	.727	1.037	[0.844, 1.275]
Manager_support	0.096	0.117	0.67	.411	1.101	[0.876, 1.384]
Work_life_balance	-0.114	0.135	0.71	.398	0.892	[0.685, 1.162]
Engagement	-0.960	0.227	17.95	< .001	0.383	[0.246, 0.597]
Job_satisfaction	0.003	0.207	0.00	.990	1.003	[0.669, 1.503]
Absence_days	-0.075	0.047	2.57	.109	0.928	[0.847, 1.017]
Promoted_last_3y: Yes (vs No)	-0.215	0.332	0.42	.517	0.807	[0.421, 1.545]

"Yes" = 1, "No" = 0. Categorical predictors are dummy-coded against their first group.

Classification at a 0.5 cut-off

Observed	Predicted No	Predicted Yes	% correct
No	903	14	98.5
Yes	147	13	8.1
Overall			85.1

Guessing the larger group for everyone would be 85.1% correct.

How to report it (APA style): A binary logistic regression predicting Left_company = "Yes" (vs "No") from Department, Job_level, Tenure_years, Annual_salary, Performance_rating, Training_hours, Overtime_hours_month, Commute_km, Remote_days_week, Manager_support, Work_life_balance, Engagement, Job_satisfaction, Absence_days, Promoted_last_3y was significant, $\chi^2(18, N = 1077) = 104.71$, $p < .001$, Nagelkerke $R^2 = .16$. Tenure_years: OR = 0.93, 95% CI [0.87, 0.99], $p = .023$; Overtime_hours_month: OR = 1.05, 95% CI [1.02, 1.07], $p < .001$; Commute_km: OR = 1.03, 95% CI [1.02, 1.05], $p < .001$; Engagement: OR = 0.38, 95% CI [0.25, 0.60], $p < .001$.

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Figure 32. Binary logistic regression of Left_company on fifteen predictors.

Concept: odds ratio

The *odds* of an event are its probability divided by one minus its probability: a probability of 15% is odds of 0.15 / 0.85, about 0.18. An *odds ratio* (OR) is the factor by which the odds change when the predictor rises by one unit. OR = 1 means no relationship; OR above 1 means higher odds of leaving; OR below 1, lower odds. For small probabilities the odds ratio is close to the ratio of probabilities, but the two are not the same.

Cut-off values: logistic regression

At least 10 people in the smaller outcome group per coefficient (EPV ≥ 10 ; Peduzzi et al., 1996). AUC .70 acceptable, .80 excellent, .90 outstanding (Hosmer et al., 2013).

What it means for our example

The model uses 1,077 employees (123 with missing values are left out) and is significant, $\chi^2(18) = 104.71$, $p < .001$, Nagelkerke $R^2 = .16$, AUC = .750, 95% CI [.71, .79] (acceptable). Four predictors stand out:

- Engagement: OR = 0.383, 95% CI [0.246, 0.597], $p < .001$. Each extra point of engagement divides the odds of leaving by 2.61.
- Overtime_hours_month: OR = 1.047, [1.020, 1.075], $p < .001$: each extra hour of overtime per month raises the odds by about 5%.
- Commute_km: OR = 1.035, [1.017, 1.053], $p < .001$: each extra kilometre raises the odds by about 3.5%.
- Tenure_years: OR = 0.926, [0.867, 0.990], $p = .023$: each extra year lowers the odds by about 7%.

Promotion (OR = 0.807, $p = .517$), salary, level, performance, training, manager support, work-life balance, satisfaction and absence are not significant once the others are in the model. The salary coefficient is shown as 1.000 because one salary unit is a very small step; with salary in thousands it would be readable. The new-variable tools of section 1.8 have no division by a constant; standardising Annual_salary (z-score) would give an odds ratio per standard deviation instead.

The *Check* flag on the sample size is worth noting: EPV = 8.9, a little under 10, because there are 18 coefficients for 160 leavers with complete data. A model with only the four strongest predictors (Engagement, Overtime_hours_month, Tenure_years, Promoted_last_3y) has EPV = 44.0 and gives nearly the same discrimination (AUC = .707, 95% CI [.67, .75], Nagelkerke $R^2 = .12$; Engagement OR = 0.390, Overtime OR = 1.051, Tenure OR = 0.906).

The table at the end of the result classifies each person as a predicted leaver when the predicted probability exceeds 0.5. Of the 160 employees with complete data who left, the model flagged 13 (8.1%); of the 917 who stayed, it flagged 14 (so 1.5% were wrongly flagged). Overall it classified 85.1% correctly, which is exactly what you would get by guessing that nobody leaves (85.1%). The model ranks people reasonably (AUC .75) but, with only 15% leavers, almost nobody reaches a predicted probability of 0.5. Sections 8.5 and 8.8 return to this.

How to report it

"A binary logistic regression predicting leaving from fifteen work-related variables was significant, $\chi^2(18, N = 1077) = 104.71$, $p < .001$, Nagelkerke $R^2 = .16$, AUC = .75. Engagement (OR = 0.38, 95% CI [0.25, 0.60]), overtime (OR = 1.05, 95% CI [1.02, 1.07]), commuting distance (OR = 1.03, 95% CI [1.02, 1.05]) and tenure (OR = 0.93, 95% CI [0.87, 0.99]) were significant predictors."

8.3 Machine learning on unseen employees

A regression is judged by how well it fits the data it was estimated on, which flatters it. The **Machine learning** screen asks a harder question: how well does a model predict employees it has never seen? It splits the employees into parts, trains on most, tests on the rest, and repeats, so every employee is predicted by a model that did not see them (*cross-validation*). Every step that learns from the data, including filling in missing values and choosing the settings of the model (*tuning*), is repeated inside each training set so that nothing leaks from the test employees (*nested cross-validation*).

Concept: cross-validation

The employees are split into k parts (*folds*). Each model is trained on $k - 1$ parts and tested on the remaining one, k times over. Repeating the whole split several times averages out the luck of one particular split (Kuhn & Johnson, 2013).

Step 1 • Prediction task. Choose **Outcome to predict** Left_company. SaNaML recognises a yes/no outcome and asks which answer is the event; keep "Yes". It ticks every variable as a predictor except the identifier EmployeeID, which it leaves unticked (Figure 33). Untick the variables we do not want, so that the fifteen predictors listed above remain. The reasons for what we unticked:

- **EmployeeID** is a label. Every employee has a different one, so a model could "learn" individuals. SaNaML leaves it unticked by default, and always leaves it out of the model. If you tick it, the check *Respondent number* appears: "EmployeeID looks like a respondent identifier (every value different). A model could learn the order in which data were collected, or memorise individual respondents, rather than anything general about them. It is always left out of the model, whether or not its box is ticked."
- **Gender, Age and Education** are personal characteristics. We do not want a model of leaving to use them (section 8.9). Note that the remaining variables can still stand in for them.
- **Skills_test_before** and **Skills_test_after** belong to the training evaluation.
- **Engagement_Q1 to Engagement_Q4** overlap with Engagement and run through the year of the outcome (section 8.1).
- **Annual_salary_In** is a duplicate of Annual_salary, and **Skills_gain** is made from the two skills tests.

1 • Prediction task (What to predict, from what)

Outcome to predict

Left_company (yes/no: No / Yes) ▼

A score (regression), a variable with two answers (binary classification) or one with 3–10 categories (multiclass classification), for example a construct score such as behavioural intention.

Which answer is the event (coded 1)?

☒ **Yes**
Predict the probability of “Yes”.

☐ **No**
Predict the probability of “No”.

Predictors

Select all Select none

☐ EmployeeID (groups)	☒ Department (groups)	☒ Job_level	☐ Gender (groups)	☐ Age
☐ Education (groups)	☒ Tenure_years	☒ Annual_salary	☒ Performance_rating	☒ Training_hours
☒ Overtime_hours_month	☒ Commute_km	☒ Remote_days_week	☒ Manager_support	☒ Work_life_balance
☒ Engagement	☒ Job_satisfaction	☒ Absence_days	☒ Promoted_last_3y (groups)	☐ Skills_test_before
☐ Skills_test_after	☐ Engagement_Q1	☐ Engagement_Q2	☐ Engagement_Q3	☐ Engagement_Q4
☐ Annual_salary_ln	☐ Skills_gain			

Info **Unbalanced answers.** The answers are unbalanced (15% vs 85%). Read AUC and balanced accuracy rather than plain accuracy.

OK **Sample size.** 1200 respondents.

Info **Leakage protection.** Filling in missing values, standardising and tuning are all learned from the training rows of each fold only, so the test rows never influence the model (nested cross-validation).

Figure 33. 1 • Prediction task: outcome, event, the fifteen predictors and the checks.

Under the predictors SaNaML shows three checks (a fourth, *Respondent number*, appears if you tick EmployeeID). *Unbalanced answers (Info)*: "The answers are unbalanced (15% vs 85%). Read AUC and balanced accuracy rather than plain accuracy." *Sample size (OK)*: 1200 respondents. *Leakage protection (Info)*: the filling in of missing values, standardising and tuning are learned from the training rows of each fold only.

Step 2 • Validation plan. The recommended plan for 1,200 employees is *Repeated 10-fold cross-validation, 3 repeats* (30 train/test splits). The main measure for a yes/no outcome is AUC. The random seed (2026) makes every split reproducible.

Step 3 • Models. Seven models are compared with the always-included null model: logistic regression, elastic net (a penalised logistic regression), a decision tree (CART), a random forest, XGBoost and LightGBM, and optionally a stacked ensemble (we left it unticked). Click **Run the comparison**. With 216 steps on one background worker, the run took about 16 minutes on our two-core computer; a faster computer, or the **Quick look** plan, takes less. **Stop** cancels the run.

Tip: use the Quick look to try things out

The **5-fold cross-validation, no repeats (Quick look)** plan is fast but noisy. Use it to try settings; use the recommended plan for the results you report.

8.4 Results • Model comparison

Machine learning model comparison (Predicting Left_company = "Yes")

What it means: Elastic net predicted Left_company = "Yes" best on respondents it had not seen (AUC = .711, 95% CI [.652, .769]; Balanced accuracy = .511). It was not significantly better than logistic regression (p = .934); the relationships appear largely linear and additive, which supports a linear model for this outcome. The simplest model that is not significantly worse than the best is logistic regression; prefer it when interpretability matters.

OK Sample size. 1200 respondents.

Info Leakage protection. Filling in missing values, standardising and tuning are all learned from the training rows of each fold only, so the test rows never influence the model (nested cross-validation).

Rule: Tune and evaluate in separate (nested) cross-validation loops; tuning on the test rows makes performance look better than it is (Varma & Simon (2006); Cawley & Talbot (2010)).

OK Validation. 30 train/test splits; the same splits for every model, so the comparison is paired.

Rule: With survey-sized samples, repeat k-fold cross-validation several times to stabilise the estimate (Kuhn & Johnson (2013)).

Info Comparison test. Differences between models are tested with the corrected resampled t-test, which allows for the overlap between training sets.

Rule: Compare models on the same folds with the corrected resampled t-test; the ordinary t-test is too liberal because the training sets overlap (Nadeau & Bengio (2003); Bouckaert & Frank (2004)).

OK Discrimination. Best AUC = .711.

Rule: AUC .7–.8 acceptable, .8–.9 excellent, ≥ .9 outstanding (Hosmer et al. (2013)).

Out-of-sample performance (nested cross-validation)								
Model	AUC M (SD)	95% CI	Accuracy	Balanced accuracy	F1	MCC	Brier score	Δ vs best (p)
Elastic net	.711 (.075)	[.652, .769]	.850	.511	.053	.056	0.116	best
Logistic regression	.710 (.076)	[.651, .769]	.850	.525	.108	.111	0.117	–.001 (.934)
XGBoost	.693 (.078)	[.632, .754]	.848	.506	.037	.030	0.118	–.018 (.097)
LightGBM	.688 (.074)	[.630, .746]	.847	.502	.024	.009	0.119	–.023 (.045)
Random forest	.678 (.065)	[.627, .728]	.851	.510	.050	.054	0.119	–.033 (.038)
Decision tree (CART)	.645 (.075)	[.587, .704]	.847	.527	.117	.105	0.126	–.065 (.009)
Null model	.500 (.000)	[.500, .500]	.853	.500	.000	.000	0.125	–.211 (< .001)

AUC M (SD) over 30 test folds; 95% CI with the Nadeau-Bengio corrected variance. Δ vs best: mean per-fold difference from the best model, corrected resampled t-test p.

Models and tuning	
Model	What is tuned

Figure 34. The model comparison report: what it means, the checks and the out-of-sample table.

Figure 34 shows the report. All numbers are measured on employees the models did not see.

Table 14

Out-of-sample prediction of Left_company = "Yes" (10-fold cross-validation, 3 repeats)

Model	AUC M (SD)	95% CI	Accuracy	Balanced accuracy	Δ vs best (p)
Elastic net	.711 (.075)	[.652, .769]	.850	.511	best
Logistic regression	.710 (.076)	[.651, .769]	.850	.525	–.001 (.934)
XGBoost	.693 (.078)	[.632, .754]	.848	.506	–.018 (.097)
LightGBM	.688 (.074)	[.630, .746]	.847	.502	–.023 (.045)
Random forest	.678 (.065)	[.627, .728]	.851	.510	–.033 (.038)
Decision tree (CART)	.645 (.075)	[.587, .704]	.847	.527	–.065 (.009)
Null model	.500 (.000)	[.500, .500]	.853	.500	–.211 (< .001)

Concept: AUC

The *area under the ROC curve* is the probability that a randomly chosen leaver gets a higher predicted risk than a randomly chosen stayer. .50 is guessing and 1.00 is perfect. It measures how well the model *ranks* people, and it does not depend on a cut-off, which makes it suitable for an outcome that is rare.

What it means for our example

The elastic net predicts best (AUC = .711, 95% CI [.652, .769]), but it is not significantly better than plain logistic regression ($\Delta\text{AUC} = .001$, corrected resampled $t(29) = 0.08$, $p = .934$). The flexible models are no better and some are significantly worse (random forest $p = .038$, LightGBM $p = .045$, decision tree $p = .009$). SaNaML therefore names logistic regression as the simplest model that is not significantly worse than the best; its summary says that the relationships "appear largely linear and additive, which supports a linear model for this outcome". With an effect structure that is nearly linear, the extra flexibility of boosting and forests buys nothing. Only the null model, which has AUC .500 by definition, is clearly worse.

Notice the accuracy column. The null model, which says "nobody leaves" for everyone, is 85.3% accurate, more than any of the real models (.847 to .851). Balanced accuracy is close to .50 for every model. These numbers do not say that the models are useless (the AUCs show that they rank employees much better than chance); they show that accuracy and balanced accuracy at a fixed cut-off of 0.5 are the wrong yardsticks when only 15% of employees leave (section 8.5).

How to report it

"Seven prediction models were compared with 10-fold cross-validation repeated 3 times (30 train/test splits), with every model tuned by an inner 5-fold cross-validation inside each training set (nested cross-validation; seed 2026). Elastic net performed best (AUC = .711, 95% CI [.652, .769]), but not significantly higher than logistic regression, $\Delta\text{AUC} = .001$, corrected resampled $t(29) = 0.08$, $p = .934$."

8.5 The imbalance problem

Only 14.7% of the employees leave. This is typical for attrition and for many other HR outcomes (promotions, misconduct, long absences), and it causes three problems.

1. **Accuracy is misleading.** Predicting "stays" for everybody is right 85% of the time. Any model that has to beat this on accuracy must be very good indeed.
2. **A fixed cut-off of 0.5 flags almost nobody.** A probability above 0.5 is rare when the average is 0.15, so sensitivity (the share of leavers flagged) is very low at 0.5 (8.1% for the logistic regression of section 8.2, and 3% for the elastic net in section 8.8), and so are F1 (between .024 and .117 for the real models in the full report) and the balanced accuracy (Table 14).

3. **There are few leavers to learn from.** With 176 leavers, a flexible model has little to work on, which is another reason why the simple models do as well.

What SaNaML does about it, and what it does not:

- It **flags** the imbalance before the run ("The answers are unbalanced (15% vs 85%). Read AUC and balanced accuracy rather than plain accuracy") and uses **AUC** as the main measure for a yes/no outcome.
- It reports balanced accuracy, F1, MCC (the Matthews correlation coefficient, which is high only if both groups are predicted well) and the Brier score (the average squared error of the predicted probabilities; lower is better; the null model's is 0.125 and the elastic net's is 0.116) next to accuracy.
- It checks **calibration**, which does not depend on a cut-off (section 8.7).
- It lets you **try cut-offs** and shows precision, recall and the average precision against the prevalence (section 8.8).
- It does **not** rebalance the data or weight the classes: the models are fitted to the data as they are, and the remedy is to judge them by AUC, precision and recall.

8.6 Results • Explanations: what drives leaving?

Explain this model chooses the model to explain; here the elastic net, which is the best model. The explanations describe what the model learned.

- **SHAP feature importance** (Figure 35) shares each prediction among the predictors. The mean absolute SHAP value shows how far a predictor moves the predictions, in log-odds of leaving; the report adds the direction and a second opinion, the *permutation importance* (how much the AUC falls on the test employees when the predictor is shuffled).
- The **SHAP summary plot** (Figure 36) shows every employee as a dot: right means the predictor raised that person's predicted risk; red means a high value of the predictor.
- **Accumulated local effects (ALE)** (Figure 37) show how the prediction changes across the range of each main driver, with the other predictors kept realistic.

SHAP feature importance (What drives the predictions of Elastic net)

What it means: In the elastic net model, the strongest drivers of the prediction are Engagement (higher Engagement → lower the chance of Left_company = "Yes"), Tenure_years (higher Tenure_years → lower the chance of Left_company = "Yes"), Commute_km (higher Commute_km → higher the chance of Left_company = "Yes"). The ranking by SHAP agrees well with the ranking by held-out permutation importance, so it is not an artefact of one method. Shuffling Remote_days_week, Training_hours, Manager_support, Job_satisfaction did not hurt the predictions at all, so they add nothing beyond the other predictors in this model.

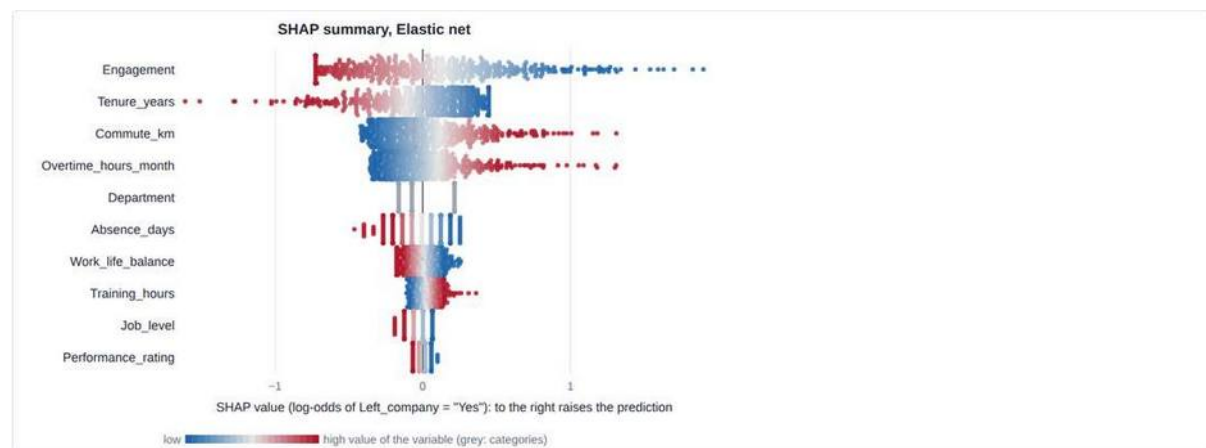
Info Meaning. Importance describes how this model uses the predictors; correlated predictors share credit, and importance is not evidence of a causal effect.

Rule: SHAP values, permutation importance, partial dependence and ALE describe what the model learned, not causal effects (Molnar (2022)).

Importance of each predictor

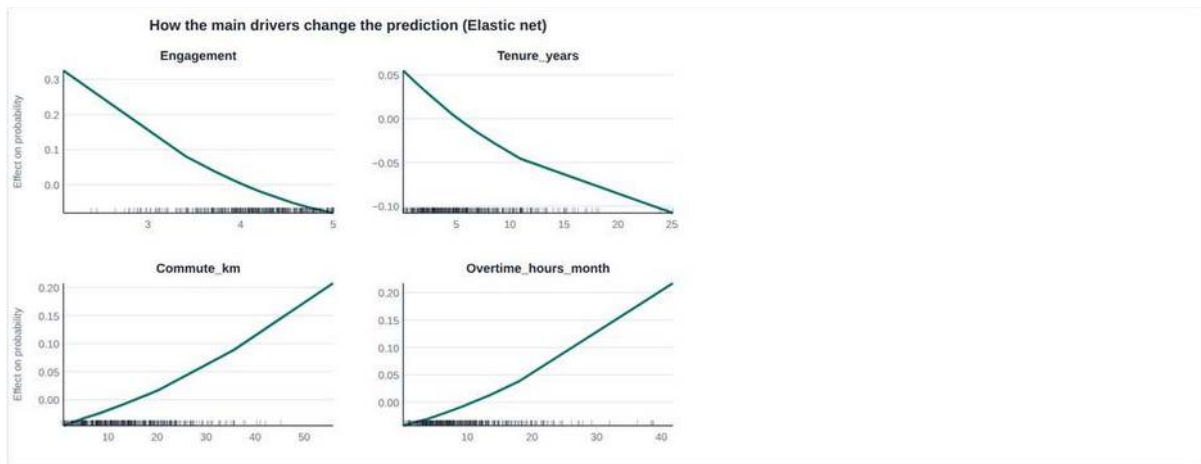
Predictor	Mean SHAP	SHAP rank	Direction	Permutation importance
Engagement	0.425	1	higher Engagement → lower the chance of Left_company = "Yes"	0.086 (0.050)
Tenure_years	0.257	2	higher Tenure_years → lower the chance of Left_company = "Yes"	0.024 (0.036)
Commute_km	0.220	3	higher Commute_km → higher the chance of Left_company = "Yes"	0.030 (0.024)
Overtime_hours_month	0.198	4	higher Overtime_hours_month → higher the chance of Left_company = "Yes"	0.026 (0.032)
Department	0.130	5	depends on the category	0.007 (0.013)
Absence_days	0.102	6	higher Absence_days → lower the chance of Left_company = "Yes"	0.003 (0.012)
Work_life_balance	0.066	7	higher Work_life_balance → lower the chance of Left_company = "Yes"	0.002 (0.009)
Training_hours	0.053	8	higher Training_hours → higher the chance of Left_company = "Yes"	-0.001 (0.008)
Job_level	0.044	9	higher Job_level → lower the chance of Left_company = "Yes"	0.001 (0.006)
Performance_rating	0.030	10	higher Performance_rating → lower the chance of Left_company = "Yes"	0.000 (0.004)
Manager_support	0.023	11	higher Manager_support → higher the chance of Left_company = "Yes"	-0.001 (0.005)
Annual_salary	0.000	12	mixed or non-linear	0.000 (0.003)
Remote_days_week	0.000	13	mixed or non-linear	-0.001 (0.003)
Job_satisfaction	0.000	14	mixed or non-linear	-0.001 (0.003)

Figure 35. SHAP feature importance in the elastic net model (the start of the report).



SHAP summary plot (Elastic net). Each dot is one respondent. Position: how much that predictor moved this respondent's prediction; colour: the respondent's value on the predictor (red high, blue low).

Figure 36. SHAP summary plot: one dot per employee and predictor.



Accumulated local effects of the main drivers (Elastic net). Numeric predictors: accumulated local effects (centred), with a rug of observed values. Categorical predictors: partial dependence (average prediction per category).

Figure 37. Accumulated local effects of the four main drivers.

Table 15

The strongest drivers in the elastic net model

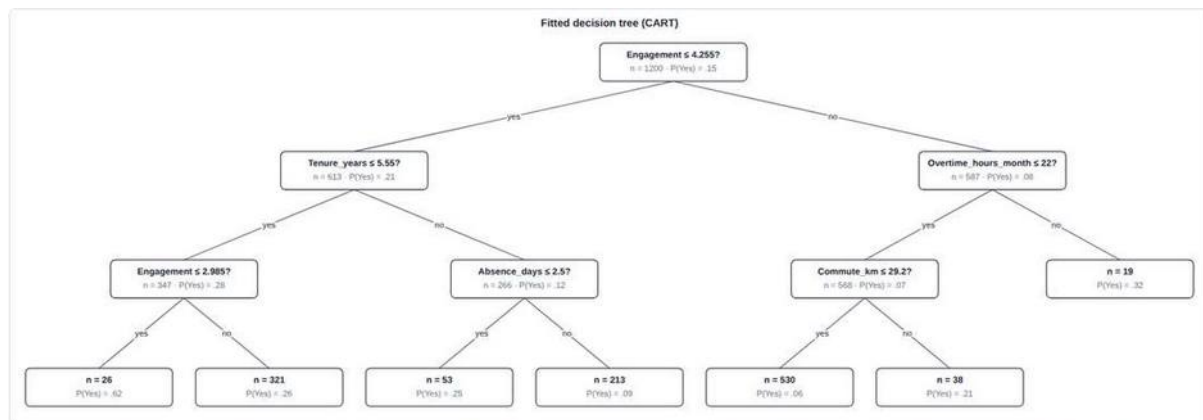
Predictor	Mean abs. SHAP	SHAP rank	Permutation importance M (SD)	Direction
Engagement	0.425	1	0.086 (0.050)	higher engagement, lower chance of leaving
Tenure_years	0.257	2	0.024 (0.036)	longer tenure, lower chance
Commute_km	0.220	3	0.030 (0.024)	longer commute, higher chance
Overtime_hours_month	0.198	4	0.026 (0.032)	more overtime, higher chance
Department	0.130	5	0.007 (0.013)	depends on the category
Absence_days	0.102	6	0.003 (0.012)	weak

What it means for our example

Engagement is by far the strongest driver (mean $|SHAP| = 0.425$), followed by tenure (0.257), commute (0.220) and overtime (0.198). The two methods agree (Spearman $\rho = .88$ between the rankings), and Engagement is ranked first by every one of the six models SaNaML compares in its stability table; Spearman correlations with the other models' rankings range from .68 (random forest) to .93 (logistic regression). Annual_salary, Remote_days_week, Job_satisfaction and Promoted_last_3y add nothing once the others are included (SHAP 0.000). Job_satisfaction is almost the same as Engagement ($r = .66$), so it adds nothing beyond it; that is different from saying that satisfaction does not matter.

The accumulated local effects show nearly straight lines: across its observed range, engagement changes the predicted probability of leaving by about 0.41, tenure by about 0.16, commute by about 0.25 and overtime by about 0.26. At the lowest engagement (2.09) the effect is +0.325; at the highest (5.00) it is -0.081. Short commutes and little overtime lower the predicted probability, and the effects rise sharply for the longest commutes (above 35 km) and heaviest overtime (above 18 hours a month).

Choose **Decision tree (CART)** under **Explain this model** to see one readable tree (Figure 38). The first question is Engagement ≤ 4.255 . Of the 613 employees at or below it, 21% leave; among those whose engagement is at most 2.985 and whose tenure is at most 5.55 years (26 employees), 62% leave. Of the 587 above it, only 8% leave, and among those with at most 22 overtime hours and a commute of at most 29.2 km (530 employees) only 6%.



The fitted decision tree (Decision tree (CART)): follow the questions from the top; each box shows the respondents who reach it and the prediction there. Depth 3, 7 leaves. Left branch = yes.

Figure 38. The fitted decision tree (CART), drawn as a diagram.

Cut-off values: reading explanations

SHAP values, permutation importance and accumulated local effects describe what the model learned, **not causal effects** (Molnar, 2022). Correlated predictors share credit. Trust a driver that ranks high on both SHAP and permutation importance. ALE is preferred over partial dependence when predictors are correlated (Apley & Zhu, 2020).

Note: drivers are not levers

That engagement is the strongest predictor does not mean that raising someone's engagement will stop them from leaving. Engagement may be a symptom (a person already planning to leave answers a survey differently), and unmeasured causes may sit behind both. Use the drivers to decide where to look and what to ask, and test any intervention.

8.7 Results • Calibration

A model is *calibrated* if its predicted probabilities mean what they say: of the employees given a 20% chance, about 20% should leave. This matters more than the AUC if the numbers will be used for planning. Under *Results • Precision and calibration*, SaNaML checks calibration on the out-of-fold predictions (Figure 39, Figure 40).

Calibration (Do the predictions mean what they say?) • Elastic net

What it means: Calibration (Do the predicted probabilities match how often “Yes” actually happens?): for the elastic net, the calibration slope was 1.01 and calibration-in-the-large 0.00; the predictions are about as spread out as they should be. A predicted 30% can be read as roughly a 30% chance.

OK Calibration slope. 1.01, 95% CI [0.78, 1.24] (mean over 3 repeats; range 0.99 to 1.02): the predictions are about as spread out as they should be.

Rule: Calibration slope close to 1 (below 1: predictions too extreme, a sign of overfitting; above 1: too timid) and calibration-in-the-large close to 0 (no systematic over- or under-prediction) (Van Calster et al. (2019)).

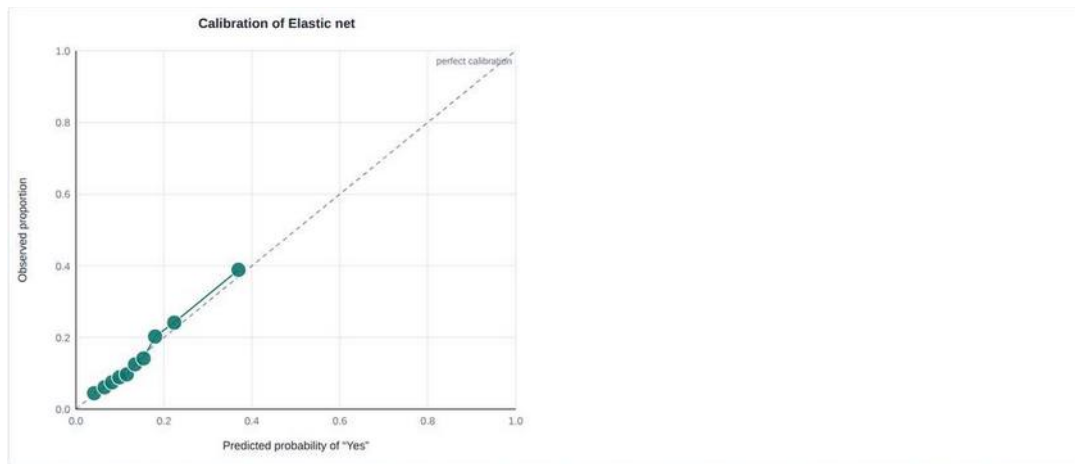
OK Calibration-in-the-large. 0.00 on the log-odds scale, 95% CI [−0.16, 0.17]: no systematic over- or under-prediction (observed – mean predicted proportion = 0.000).

Rule: Calibration slope close to 1 (below 1: predictions too extreme, a sign of overfitting; above 1: too timid) and calibration-in-the-large close to 0 (no systematic over- or under-prediction) (Van Calster et al. (2019)).

Reliability table of Elastic net (ten equal-count bins of the predicted probability, pooled over repeats)

Bin	Mean predicted probability	Observed proportion	Predictions
1	.041	.044	360
2	.065	.061	360
3	.082	.075	360
4	.099	.089	360
5	.116	.097	360
6	.134	.125	360
7	.154	.142	360
8	.180	.203	360
9	.224	.242	360
10	.370	.380	360

Figure 39. Calibration of the elastic net: slope, calibration-in-the-large and the reliability table.



Reliability diagram of Elastic net: points on the dashed diagonal are perfectly calibrated; point size shows the number of predictions.

Figure 40. Reliability diagram: predicted probability against the observed share of leavers.

What it means for our example

The elastic net is well calibrated: the calibration slope is 1.01, 95% CI [0.78, 1.24], and the calibration-in-the-large is 0.00, 95% CI [-0.16, 0.17] (no systematic over- or under-prediction). A predicted 30% can be read as roughly a 30% chance. The ten equal-sized groups in the reliability table follow the diagonal closely: the group with the lowest predicted risk (mean .041) had an observed share of .044, and the group with the highest (mean .370) had .389.

Cut-off values: calibration

Slope close to 1 and calibration-in-the-large close to 0 (Van Calster et al., 2019). A slope well below 1 means the predictions are too extreme (overfitting); well above 1, too cautious.

Calibration says that the probabilities are honest; it does not say that they are sharp. The highest-risk tenth of employees has an average predicted risk of 37%: it is the best the model can do for them, and it also means that most of them stay.

8.8 Results • Cut-offs, precision and recall

A model gives each employee a probability of leaving. To act on it, someone has to turn the probability into yes or no with a *cut-off*. For a yes/no outcome the Machine learning screen has a section for this, *Results • Cut-offs, precision and recall (What happens when probabilities become yes or no?)* (Figure 41). It uses the out-of-sample predictions of the model that was explained (here the elastic net). Suppose that HR can offer a supportive "stay conversation" to a limited number of employees. Whom to invite depends on a trade-off:

- *Recall* (also called sensitivity) is the share of all leavers who are flagged.
- *Precision* is the share of the flagged employees who really would have left.

Flagging more people raises recall and lowers precision. The box *Cut-off (probability, between 0 and 1)* starts at 0.50.

Results · Cut-offs, precision and recall (What happens when probabilities become yes or no?)

A model gives each respondent a probability. To call someone a case you need a cut-off. Try different cut-offs and watch how many cases are found (recall) and how many of the flagged respondents really are cases (precision).

Cut-off (probability, between 0 and 1)

Cut-offs, precision and recall (What happens when the model's probabilities are turned into yes or no, Elastic net)

What it means: At the cut-off 0.50 (everyone with a predicted probability of 0.50 or more is called a case), 1% of respondents are flagged. Of the respondents who really have Left_company = "Yes", 3% are found (sensitivity, recall); of those flagged, 38% really are cases (precision). Left_company = "Yes" occurs in 14.7% of respondents, so a model that found nothing would still be 85.3% accurate: read precision and recall rather than accuracy.

Note Rare outcome. Yes occurs in 14.7% of respondents. Accuracy is misleading here (calling everyone "no" is 85.3% accurate); use precision, recall and the average precision.

Rule: Choose the cut-off from the cost of a missed case against the cost of a false alarm, not by default at 0.5; with a rare outcome (under about 20%) judge the model by precision, recall and average precision rather than accuracy (Saito & Rehmsmeier (2015); Fawcett (2006)).

OK Average precision. 0.30 against 0.15 for a model with no information (the prevalence).

Info Choosing the cut-off. There is no single right cut-off: lower it to find more cases (recall rises, precision usually falls), raise it to be surer of the cases you flag. In this study the best F1 is at 0.15 (F1 = 0.37) and the best balance of sensitivity and specificity at 0.15. Choose from the cost of a missed case against the cost of a false alarm.

Rule: Choose the cut-off from the cost of a missed case against the cost of a false alarm, not by default at 0.5; with a rare outcome (under about 20%) judge the model by precision, recall and average precision rather than accuracy (Saito & Rehmsmeier (2015); Fawcett (2006)).

Info Meaning. These figures come from predictions made for respondents the models had not seen. The cut-off you choose here should be fixed before you apply the model to new people; do not tune it on the same data you then report.

Counts at the cut-off 0.50 (average over the repeats of the cross-validation)

Actual outcome	Predicted: case	Predicted: not a case	Total
Actually Yes	5.3	170.7	176
Actually not Yes	9	1015	1024
Total	14.3	1185.7	1200

Measures at every cut-off

Cut-off	Flagged	Sensitivity (recall)	Specificity	Precision	F1	Accuracy
0.05	92%	0.98	0.08	0.15	0.27	0.22
0.10	64%	0.85	0.39	0.19	0.32	0.46
0.15	37%	0.64	0.68	0.26	0.37	0.67

Figure 41. The cut-offs section at the default cut-off of 0.50: what it means, the checks and the counts.

What it means for our example

At 0.50 the program says: "1% of respondents are flagged. Of the respondents who really have Left_company = "Yes", 3% are found (sensitivity, recall); of those flagged, 38% really are cases (precision)." Of the 176 leavers, on average 5.3 are flagged and 170.7 are missed. The default cut-off is useless for this purpose, as section 8.5 predicted. The program also reports the average precision, 0.30, against 0.15 for a model with no information (the prevalence): the model is better than chance, but far from precise.

Type a lower cut-off, for example 0.20 (Figure 42). The program now says: "19% of respondents are flagged. ... 42% are found (sensitivity, recall); of those flagged, 32% really are cases (precision)." The counts table gives the numbers behind these percentages, averaged over the three repeats of the cross-validation.

Results · Cut-offs, precision and recall (What happens when probabilities become yes or no?)

A model gives each respondent a probability. To call someone a case you need a cut-off. Try different cut-offs and watch how many cases are found (recall) and how many of the flagged respondents really are cases (precision).

Cut-off (probability, between 0 and 1)

Cut-offs, precision and recall (What happens when the model's probabilities are turned into yes or no, Elastic net)

What it means: At the cut-off 0.20 (everyone with a predicted probability of 0.20 or more is called a case), 19% of respondents are flagged. Of the respondents who really have Left_company = "Yes", 42% are found (sensitivity, recall); of those flagged, 32% really are cases (precision). Left_company = "Yes" occurs in 14.7% of respondents, so a model that found nothing would still be 85.3% accurate: read precision and recall rather than accuracy.

Note Rare outcome. Yes occurs in 14.7% of respondents. Accuracy is misleading here (calling everyone "no" is 85.3% accurate); use precision, recall and the average precision.

Rule: Choose the cut-off from the cost of a missed case against the cost of a false alarm, not by default at 0.5; with a rare outcome (under about 20%) judge the model by precision, recall and average precision rather than accuracy (Saito & Rehmsmeier (2015); Fawcett (2006)).

OK Average precision. 0.30 against 0.15 for a model with no information (the prevalence).

Info Choosing the cut-off. There is no single right cut-off: lower it to find more cases (recall rises, precision usually falls), raise it to be surer of the cases you flag. In this study the best F1 is at 0.15 (F1 = 0.37) and the best balance of sensitivity and specificity at 0.15. Choose from the cost of a missed case against the cost of a false alarm.

Rule: Choose the cut-off from the cost of a missed case against the cost of a false alarm, not by default at 0.5; with a rare outcome (under about 20%) judge the model by precision, recall and average precision rather than accuracy (Saito & Rehmsmeier (2015); Fawcett (2006)).

Info Meaning. These figures come from predictions made for respondents the models had not seen. The cut-off you choose here should be fixed before you apply the model to new people; do not tune it on the same data you then report.

Counts at the cut-off 0.20 (average over the repeats of the cross-validation)

Actual outcome	Predicted: case	Predicted: not a case	Total
Actually Yes	74.3	101.7	176
Actually not Yes	158	866	1024
Total	232.3	967.7	1200

Measures at every cut-off

Cut-off	Flagged	Sensitivity (recall)	Specificity	Precision	F1	Accuracy
0.05	92%	0.98	0.08	0.15	0.27	0.22
0.10	64%	0.85	0.39	0.19	0.32	0.46
0.15	37%	0.64	0.68	0.26	0.37	0.67

Figure 42. The same section at a cut-off of 0.20.

Table 16

Counts at the cut-off 0.20 (average over the repeats of the cross-validation).

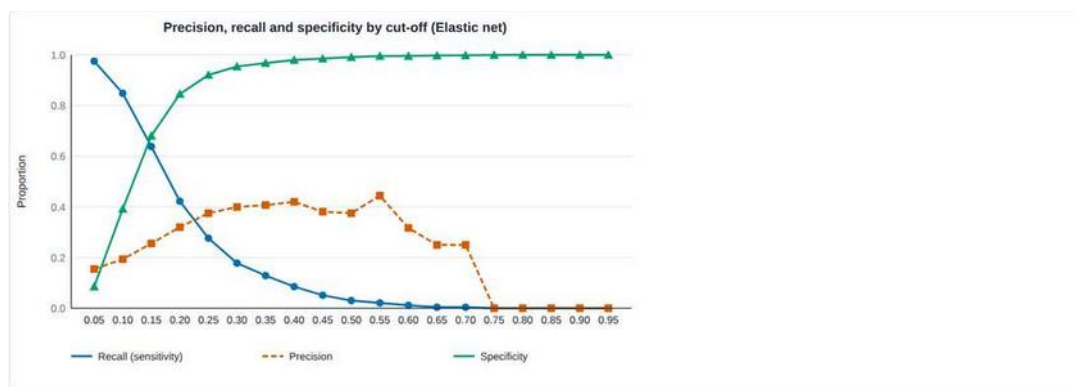
Actual outcome	Predicted: case	Predicted: not a case	Total
Actually Yes	74.3	101.7	176
Actually not Yes	158	866	1024
Total	232.3	967.7	1200

Table Table 17 is part of the program's table *Measures at every cut-off*, and the chart shows the same measures (Figure 43).

Table 17

Measures at some cut-offs (from the program; elastic net, out-of-sample).

Cut-off	Flagged	Sensitivity (recall)	Specificity	Precision	F1	Accuracy
0.10	64%	0.85	0.39	0.19	0.32	0.46
0.15	37%	0.64	0.68	0.26	0.37	0.67
0.20	19%	0.42	0.85	0.32	0.36	0.78
0.30	7%	0.18	0.95	0.40	0.25	0.84
0.50	1%	0.03	0.99	0.38	0.06	0.85



Precision, recall and specificity of Elastic net at every cut-off. Where precision is undefined (nobody flagged at a very high cut-off) it is drawn as 0.

Figure 43. Precision, recall and specificity of the elastic net at every cut-off.

What it means for our example

At the cut-off 0.20 about 232 employees (19%) are flagged, which reaches 42% of the leavers (74 of 176), but 158 of the 232 flagged (68%, a hand calculation from the counts) would have stayed anyway. Lowering the cut-off to 0.10 reaches 85% of the leavers but flags 64% of all employees, of whom only 19% would leave: nearly the same as the base rate of 14.7%, so the flag tells you little. Raising it to 0.30 gives a more accurate list (precision 0.40) that finds only 18% of the leavers. The program notes that the best F1 is at 0.15 ($F1 = 0.37$), but "there is no single right cut-off": choose it from the cost of a missed case against the cost of a false alarm.

The program also warns that the cut-off "should be fixed before you apply the model to new people; do not tune it on the same data you then report". The numbers in Table 17 are honest estimates for a given cut-off, but picking the best-looking cut-off from the same predictions makes the reported precision and recall slightly optimistic.

There is no single right threshold. It depends on the cost of the intervention, the cost of losing someone and, above all, whether being flagged is a benefit or a burden for the employee. If the "intervention" could damage careers, the precision of the list must be much higher than it is here. For comparison, the classification table of the logistic regression at the 0.5 cut-off (section 8.2) gives a precision of 13 of 27 flagged employees (48%) with a recall of only 8.1%: a short, fairly accurate list that misses nearly everybody who leaves.

8.9 Ethical use of attrition predictions

- **Use it for groups and for support.** Findings such as "low engagement, long commutes and heavy overtime go with leaving" support action on workload, commuting arrangements and management. They do not justify naming employees as "flight risks".
- **Never use a prediction to withhold something.** A person flagged as likely to leave must not be denied training, promotion or responsibility because "they will leave anyway". That turns a prediction into a cause.
- **Keep it away from line managers' decisions about individuals.** A list of names with risk scores changes how people are treated, and it can become true because of the way people are treated. (SaNaML's screens do not list individual predictions; do not build such a list from the model without a clear policy.)

- **Think about fairness.** We did not give the model Gender, Age or Education. That is not enough: commute distance, department and tenure may be related to protected characteristics, so a model can still treat groups differently. SaNaML has no fairness tools; if you use a model to select people for anything, compare the rates of flagging and the error rates across groups with other software before using it, and involve the people responsible for equality.
- **Be open about it.** Employees should know that such analyses exist and how the results are used. In many places, automated decisions about people are regulated: check the law that applies to you.
- **Remember the limits.** The model was tested on the same company in the same year. Next year's conditions will differ; recheck calibration on new data before trusting the numbers, and update the model when conditions change.

Exercise

1. Run the logistic regression with only Engagement, Overtime_hours_month, Tenure_years and Promoted_last_3y. What is the AUC and EPV? (Answer: AUC = .707, EPV = 44.0.)
1. In the model comparison, which model has the highest accuracy, and why is this not a good sign? (Answer: the null model, .853; it predicts that nobody leaves.)
1. Which four predictors have the highest SHAP importance? (Answer: Engagement 0.425, Tenure_years 0.257, Commute_km 0.220, Overtime_hours_month 0.198.)
1. Using Table 17, what share of the leavers is found at the cut-off 0.30, and what share of the flagged employees really leave? (Answer: 18% of the leavers; precision 0.40, so 40% of those flagged leave.)
1. Type the cut-off 0.15 in the program. What does it say about the share flagged? (Answer: 37% flagged, sensitivity 0.64, precision 0.26.)
1. Describe in three sentences how you would explain the results of this chapter to a works council, without naming individuals.

9. Employee types: clustering

Are there groups of employees who resemble each other, so that one policy does not have to fit everyone? Cluster analysis looks for such groups. It is **exploratory**: it always returns groups, whether or not real groups exist, and the groups depend on which variables and which method you choose. This chapter uses the **Clustering** screen on `practice_hr.csv` (simulated data; no real employee appears) and is honest about how weak the structure here is.

Concept: what clustering does

K-means clustering places a chosen number of centres in the space of the variables and assigns every employee to the nearest centre, then moves the centres and repeats until nothing changes. Hierarchical clustering instead merges the most similar employees step by step. Both need the variables to be on comparable scales, so SaNaML standardises each variable before clustering. The result is a *description* of the sample, not proof that employee types exist.

9.1 Choosing the variables

Open **Clustering**. Under "Variables that describe the respondents (numeric)" tick the variables that define a type. We choose four: `Tenure_years`, `Overtime_hours_month`, `Engagement` and `Annual_salary_In` (Figure 44). `Annual_salary_In` is the variable made in section 2.4; it is used instead of `Annual_salary` because salary is strongly right-skewed. Leave the method at "k-means clustering" and the number of clusters on "the best silhouette (Recommended)", with the seed at its default.

1 • Settings (What to cluster)

Variables that describe the respondents (numeric)

Select all Select none

☐ Job_level	☐ Age	☒ Tenure_years	☐ Annual_salary	☐ Performance_rating	☐ Training_hours	☒ Overtime_hours_month
☐ Commute_km	☐ Remote_days_week	☐ Manager_support	☐ Work_life_balance	☒ Engagement	☐ Job_satisfaction	☐ Absence_days
☐ Skills_test_before	☐ Skills_test_after	☐ Engagement_Q1	☐ Engagement_Q2	☐ Engagement_Q3	☐ Engagement_Q4	☒ Annual_salary_In
☐ Skills_gain						

Variables are standardised before clustering, so a variable on a 1–7 scale does not outweigh one on a 1–5 scale. Respondents with a missing value on any chosen variable are left out. Use construct or latent variable scores, or items measured the same way.

Method

Compare solutions from to clusters Show

☐ Also compute the gap statistic (slower) Random seed

▼ Compare the clusters on other variables (optional)

Variables to describe the clusters

☐ Department	☐ Job_level	☐ Gender	☒ Age	☐ Education	☐ Annual_salary	☒ Performance_rating
☐ Training_hours	☒ Commute_km	☐ Remote_days_week	☒ Manager_support	☒ Work_life_balance	☒ Job_satisfaction	☒ Absence_days
☐ Promoted_last_3y	☐ Skills_test_before	☐ Skills_test_after	☐ Engagement_Q1	☐ Engagement_Q2	☐ Engagement_Q3	☐ Engagement_Q4
☐ Left_company	☐ Skills_gain					

Find the clusters

Figure 44. The Clustering settings: four variables, k-means, the best silhouette.

Two cautions about the choice of variables:

- **Clusters reflect what you put in.** If you add many pay variables, you will get pay clusters. Choose variables that are meaningful for the question and measured in the same way.
- **Do not include the outcome you want to explain.** We keep Left_company out of the clustering and compare the clusters on it afterwards. If it were inside, the finding that one cluster contains more leavers would be circular.

Respondents with a missing value on any chosen variable are left out; here all 1,200 are used.

9.2 The default solution: two clusters

Select **Find the clusters**. The first result is the solution with the best average silhouette, which is two clusters. The text says: "K-means clustering split the 1200 respondents into 2 clusters (average silhouette = .27: weak structure, possibly artificial). Cluster 1 (64%): low Tenure_years and Annual_salary_In. Cluster 2 (36%): high Tenure_years and Annual_salary_In."

Concept: the average silhouette

For each employee, the silhouette compares the average distance to the others in the same cluster with the average distance to the nearest other cluster. It runs from -1 to 1. The average over all employees is judged with the scale of Kaufman and Rousseeuw (1990): .71 to 1.00 strong structure, .51 to .70 reasonable, .26 to .50 weak (possibly artificial), .25 or less no substantial structure.

What it means

The silhouette of .27 is only just above the .25 line. The two clusters split the employees mainly by tenure and pay, which are strongly related: this is a gradient, not two separate groups. The program says so itself ("weak structure, possibly artificial"), and the analyst should repeat it.

9.3 Comparing solutions and choosing k

Below the first result, SaNaML lists the solutions with two to eight clusters (Table 18).

Table 18

Solutions with 2 to 8 clusters (from the program).

Clusters	Within-cluster sum of squares	Average silhouette	Calinski-Harabasz	Smallest cluster
2 (shown)	3503.9	.270	443.1	35.6%
3	2864.1	.265	404.5	17.0%
4	2410.6	.228	395.2	13.9%
5	2134.9	.219	372.9	9.8%
6	1933.9	.216	353.9	7.8%
7	1805.4	.203	329.8	7.6%
8	1693.2	.208	312.4	6.4%

The within-cluster sum of squares always falls as clusters are added, so one looks for an elbow; the program notes: "silhouette suggests 2, elbow 3." The silhouette chart is shown in Figure 45; the program also draws the sum-of-squares chart.

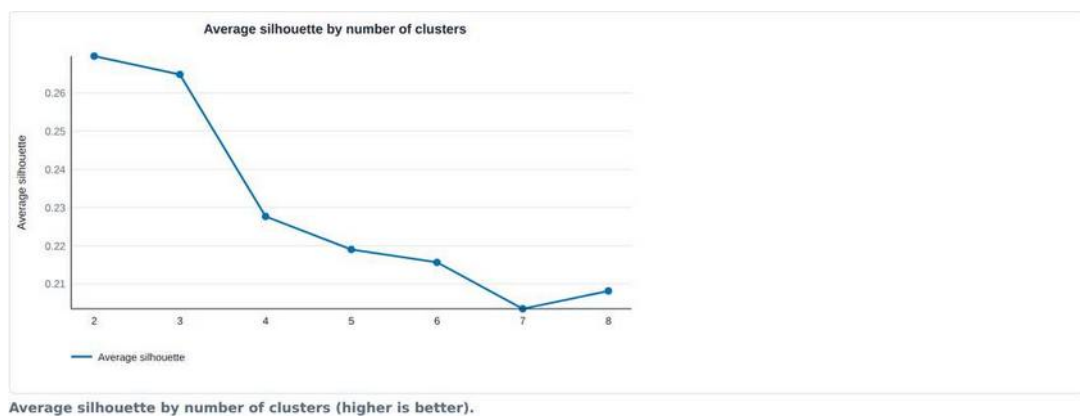


Figure 45. Average silhouette by number of clusters ($k = 3$ solution displayed).

The criteria disagree, as they often do. Two clusters give the best silhouette, but the pattern is a tenure gradient that the manager already knows. Three clusters have a silhouette that is almost the same (.265 against .270) and an elbow. We therefore choose three, which is a decision of the analyst, not of the program: change "Show" from the best silhouette to "3 clusters" and select **Find the clusters** again (Figure 46).

Cluster analysis (Are there groups of similar respondents?)

What it means: K-means clustering split the 1200 respondents into 3 clusters (average silhouette = .26: weak structure, possibly artificial). Cluster 1 (55%): low Tenure_years. Cluster 2 (28%): high Tenure_years and Annual_salary_In. Cluster 3 (17%): high Overtime_hours_month; low Engagement.

Note **Cluster structure.** Average silhouette .26: weak structure, possibly artificial.

Rule: Average silhouette: .71–1.00 strong structure, .51–.70 reasonable, .26–.50 weak (possibly artificial), .25 or less no substantial structure (Kaufman & Rousseeuw (1990)).

OK **Cluster sizes.** The smallest cluster holds 17.0% of respondents (204).

Rule: Every cluster should hold at least 5% of the respondents (and at least 20) to be interpretable and stable (Common practice (as for latent profiles, Spurk et al., 2020)).

OK **Stability.** Mean adjusted Rand index .90 (lowest .75) between the full solution and 20 random 80% subsamples.

Rule: Mean adjusted Rand index between the full solution and 80% subsamples of .80 or more: the clusters are stable (informal guide; the resampling idea is from Hennig, 2007) (Hubert & Arabie (1985); Hennig (2007)).

Info **Number of clusters.** Chosen by you; silhouette suggests 2, elbow 3. Try neighbouring values and prefer the solution that is stable and easy to interpret.

Solutions with 2 to 8 clusters

Clusters	Within-cluster sum of squares	Average silhouette	Calinski-Harabasz	Smallest cluster
2	3503.9	.270	443.1	35.6%
3 (shown)	2864.1	.265	404.5	17.0%
4	2410.6	.228	395.2	13.9%
5	2134.9	.219	372.9	9.8%
6	1933.9	.216	353.9	7.8%
7	1805.4	.203	329.8	7.6%
8	1693.2	.208	312.4	6.4%

Larger silhouette, Calinski-Harabasz and gap are better; the within-cluster sum of squares always falls as clusters are added, so look for an elbow. The criteria can disagree: silhouette suggests 2, elbow 3.

The 3 clusters (largest first)

Cluster	Respondents	Share	Average silhouette
Cluster 1	657	54.8%	.33
Cluster 2	339	28.2%	.20

Figure 46. The three-cluster solution: sizes, silhouettes and stability.

9.4 Reading the three-cluster solution

The program summarises: "K-means clustering split the 1200 respondents into 3 clusters (average silhouette = .26: weak structure, possibly artificial). Cluster 1 (55%): low Tenure_years. Cluster 2 (28%): high Tenure_years and Annual_salary_In. Cluster 3 (17%): high Overtime_hours_month; low Engagement." The checks say the smallest cluster holds 17.0% of respondents (204), and that the mean adjusted Rand index over 20 random 80% subsamples is .90 (lowest .75). The program's informal guide is .80 or more for stable clusters; here the mean passes but the lowest value does not, so the solution is reasonably, not perfectly, stable.

Concept: stability

The program repeats the clustering on 20 random 80% subsamples and compares each result with the full solution using the adjusted Rand index (Hubert & Arabie, 1985; the resampling idea is from Hennig, 2007). A value of 1 means identical groupings; 0 means agreement no better than chance. A stable solution is necessary, though not sufficient, for the clusters to be real.

Table 19 gives the size and silhouette of each cluster, and Table 20 the profile.

Table 19

The three clusters (from the program).

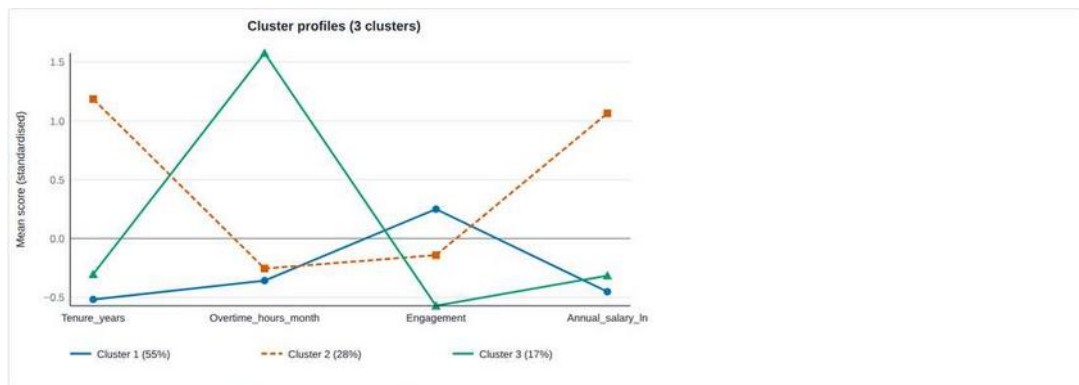
Cluster	Respondents	Share	Average silhouette
Cluster 1	657	54.8%	.33
Cluster 2	339	28.2%	.20
Cluster 3	204	17.0%	.16

Table 20

Mean of each clustering variable by cluster, original units (standardised value in brackets).

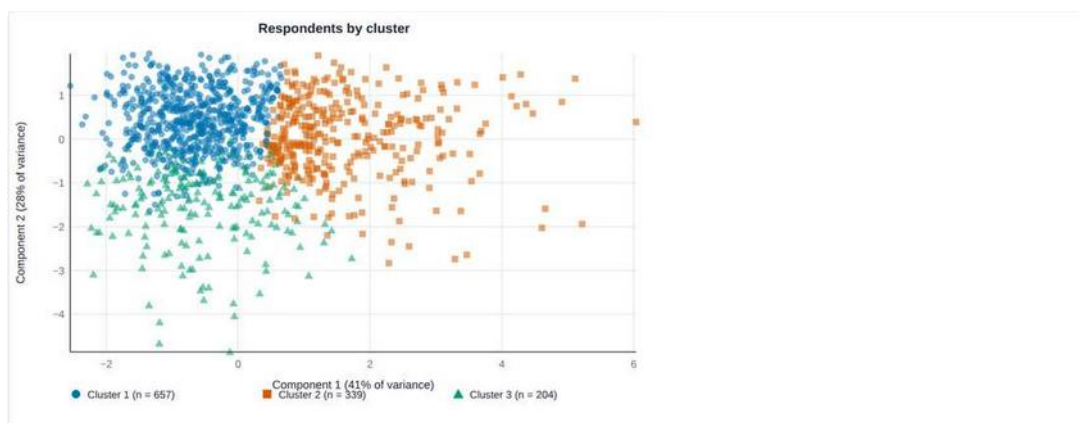
Variable	Cluster 1 (n = 657)	Cluster 2 (n = 339)	Cluster 3 (n = 204)	η^2 (descriptive)
Tenure_years	3.46 (-0.52)	10.28 (1.19)	4.32 (-0.30)	.56
Overtime_hours_month	6.87 (-0.36)	7.54 (-0.25)	19.41 (1.58)	.51
Engagement	4.34 (0.25)	4.11 (-0.14)	3.86 (-0.57)	.10
Annual_salary_In	10.54 (-0.45)	10.98 (1.06)	10.58 (-0.32)	.45

The three profile charts are on the screen: the cluster profiles (Figure 47) and the respondents on the first two principal components (Figure 48).



Cluster profiles: mean standardised scores of the 3 clusters.

Figure 47. Cluster profiles: mean standardised scores of the three clusters.



Respondents on the first two principal components (69% of the variance), coloured by cluster.

Figure 48. Respondents on the first two principal components (69% of the variance), coloured by cluster.

Note: differences are large by construction

The program states: "The clusters were built from these variables, so differences between clusters are large by construction." The η^2 column only shows which variables separate the clusters most; it carries no p-values on purpose. Here tenure and overtime separate them most, engagement least.

Putting the profiles into words, with a descriptive label as a working name only:

- **Cluster 1, "newer employees" (55%).** Short tenure (mean 3.46 years), average overtime (6.87 hours a month), the highest engagement (4.34).
- **Cluster 2, "established and well paid" (28%).** Long tenure (10.28 years) and the highest Annual_salary_ln (10.98).
- **Cluster 3, "overworked" (17%).** Much overtime (19.41 hours a month) and the lowest engagement (3.86).

Tip: be careful with labels

A label such as "overworked" is shorthand for a pattern in the numbers. It must never become a label attached to a person, in a file or in conversation. State the criterion instead: "employees with more than about 19 hours of overtime a month on average in this cluster". Also remember the silhouette of .26: cluster boundaries are fuzzy, and many employees sit near a boundary (the average silhouettes of clusters 2 and 3, .20 and .16, are low).

9.5 Describing the clusters with other variables

The **Compare the clusters on other variables (optional)** section lists variables that were not used to form the clusters. For the numeric ones the program gives mean, SD and Welch's test (Table 21).

Table 21

Other numeric variables by cluster: mean (SD) and Welch's test (from the program).

Variable	Cluster 1	Cluster 2	Cluster 3	Welch F (df)	p	η^2
Age	36.39 (9.37)	39.26 (7.16)	36.20 (8.88)	17.04 (2, 527.2)	< .001	.02
Performance_rating	3.39 (0.84)	3.47 (0.81)	3.35 (0.89)	1.66 (2, 502.4)	.192	.00
Commute_km	14.09 (9.14)	14.21 (9.27)	14.83 (9.40)	0.47 (2, 482.9)	.626	.00
Manager_support	3.42 (0.87)	3.31 (0.89)	3.28 (0.85)	2.93 (2, 485.8)	.055	.00
Work_life_balance	3.51 (0.72)	3.41 (0.72)	2.86 (0.74)	61.37 (2, 504.3)	< .001	.10
Job_satisfaction	3.73 (0.57)	3.80 (0.61)	3.40 (0.68)	24.21 (2, 461.4)	< .001	.05
Absence_days	3.85 (2.00)	3.78 (1.90)	4.15 (2.03)	2.28 (2, 509.9)	.103	.00

The program adds: "These variables were not used to form the clusters, so the tests are meaningful, but the clusters were still chosen from the same sample: confirm in new data before calling them

types." Cluster 3 stands out on work-life balance (2.86 against 3.51 and 3.41) and job satisfaction (3.40 against 3.73 and 3.80). Absence does not differ. Because the clusters were chosen to be different, the p-values here are optimistic and should be treated as descriptions, not as confirmatory tests.

9.6 Keeping the clusters as a variable

Under "2 · Keep the clusters", type a name for the new variable (we use Employee_type) and select **Replace it with these clusters** (Figure 49). The program confirms: Added "Employee_type". Employee_type is now a grouping variable that every other screen can use. Respondents who were left out would get no cluster; here there are none.



2 · Keep the clusters (Use them in other analyses)

Add each respondent's cluster to the data as a grouping variable, then compare the clusters in the Analyses (for example ANOVA or chi-square) or use it in Machine learning.

Name of the new variable **Replace it with these clusters**

Letters, numbers and underscores. Respondents who were left out get no cluster.

Added "Employee_type". **Open the analyses →**

Figure 49. Keeping the clusters as the new variable Employee_type.

Note: the clusters change when the data change

The new column is a fixed record of this run. If you change the variables, the method or the number of clusters and keep the clusters again under the same name, the button replaces the old column. Choose a new name if you want to keep both.

9.7 Do the employee types differ in leaving?

Choose **Analyses** → **Categories** → **Chi-square test of independence** with Employee_type and Left_company (Figure 50). The program reports Pearson chi-square(2, N = 1200) = 32.08, $p < .001$, Cramér's V = .16 (small); all expected counts are adequate (smallest 29.92).

Chi-square test of independence (Employee_type by Left_company)

What it means: Employee_type and Left_company are related (Cramér's V = .16, small). The cells with large adjusted residuals (beyond ± 1.96) show where the pattern comes from.

OK Expected counts. All expected counts are adequate (smallest 29.92).

Rule: Chi-square: no expected count below 1 and no more than 20% below 5; otherwise use the exact test (Cochran (1954)).

Info Effect size. Cramér's V = .16 (small).

Rule: Cramér's V: small, medium and large are .10, .30 and .50 divided by $\sqrt{(\text{smaller table dimension} - 1)}$ (Cohen (1988)).

Counts (row %)

Employee_type	No	Yes	Total
Cluster 1	569 (87%)	88 (13%)	657
Cluster 2	306 (90%)	33 (10%)	339
Cluster 3	149 (73%)	55 (27%)	204

Tests

Test	Value	df	p
Pearson chi-square	32.08	2	< .001
Likelihood ratio	28.82	2	< .001

Adjusted standardised residuals

Employee_type	No	Yes
Cluster 1	1.37	-1.37
Cluster 2	3.03 *	-3.03 *
Cluster 3	-5.45 *	5.45 *

* beyond ± 1.96 : this cell holds clearly more (positive) or fewer (negative) cases than expected.

Effect size

Measure	Value	Size
Cramér's V	.163	small

How to report it (APA style): A chi-square test of independence showed a significant association between Employee_type and Left_company, $\chi^2(2, N = 1200) = 32.08, p < .001$, Cramér's V = .16.

Copy this sentence

Add to Word report

► References

Figure 50. Chi-square test of Employee_type by Left_company.

Table 22

Employee_type by Left_company: counts (row %) and adjusted residuals (from the program).

Employee_type	No	Yes	Total	Residual (No)	Residual (Yes)
Cluster 1	569 (87%)	88 (13%)	657	1.37	-1.37
Cluster 2	306 (90%)	33 (10%)	339	3.03 *	-3.03 *
Cluster 3	149 (73%)	55 (27%)	204	-5.45 *	5.45 *

27% of cluster 3 left, against 13% of cluster 1 and 10% of cluster 2. The adjusted residuals beyond ± 1.96 mark cluster 2 (fewer leavers than expected) and cluster 3 (more).

How to report

A chi-square test of independence showed a significant association between Employee_type and Left_company, $\chi^2(2, N = 1200) = 32.08, p < .001$, Cramér's $V = .16$.

Two ANOVAs complete the picture. Job_satisfaction differs across the types (Welch's $F(2, 461.40) = 24.21, p < .001, \omega^2 = .048$, small); Games–Howell comparisons show cluster 3 below both others (differences 0.33 and 0.40 points, both $p < .001$) and no difference between clusters 1 and 2 ($p = .125$). Absence_days does not differ ($F(2, 1197) = 2.35, p = .096, \omega^2 = .002$).

What it means

The group with heavy overtime and low engagement has about twice the leaving rate of the others, and also reports lower work-life balance and satisfaction. This agrees with the machine learning results of chapter 8 (engagement and overtime among the main drivers). It is an association in one simulated sample, not evidence that overtime causes leaving, and the clusters were chosen using overtime and engagement themselves, so the finding is partly built in.

Exercise

1. Run the clustering with Engagement, Overtime_hours_month, Work_life_balance and Job_satisfaction (all four are survey or workload measures). Compare the silhouette with the .27 found here. (Check your own result: no answer is given, because it is not part of the planted analyses.)

1. In Table 18, how much does the average silhouette fall between 3 and 4 clusters? (Answer: from .265 to .228, a fall of .037; hand calculation.)

1. What share of cluster 2 left the company? (Answer: 33 of 339, about 10%.)

1. Why is the p-value for tenure not reported in Table 20? (Answer: tenure was used to build the clusters, so a test would be circular.)

1. Write two sentences that describe cluster 3 for a managers' meeting without using a label for people.

10. Reporting and the Word report

SaNaML does not write the report for you, but it collects the results you choose into one Word document that contains, for each result, a sentence in APA style, the interpretation, the checks, the tables and, at the end, the rules of thumb and the references. This chapter shows how to build and use that document.

10.1 Adding results

Under every analysis there are two buttons: **Copy this sentence**, which places the APA-style sentence on the clipboard, and **Add to Word report**, which adds the whole result. Add the results you may want to keep as you work; you can remove or reorder them later. In the session of this manual we added four results: the chi-square test of Employee_type by Left_company, the independent-samples t-test of Annual_salary by Gender, the paired t-test of Skills_test_before and Skills_test_after, and the multiple regression for Annual_salary_In.

10.2 The Word report screen

Open **Word report**. The screen explains what it does and lists the results with their interpretation (Figure 51). Each result has three buttons: up, down and **Remove**. Below the list is a **Report title** box (we typed "HR analytics, practice data") and the button **Download Word report (.docx)**.

removed words for each result, removed by the student user and the references, and results that the Add to Word report button under any analysis.

Word report (4 results)

Chi-square test of independence (Employee_type by Left_company)
Employee_type and Left_company are related (Cramér's V = .16, small). The cells with large adjusted residuals (beyond ± 1.96) show where the pattern comes from.

↑ ↓ Remove

Independent-samples t-test (Comparing Annual_salary between female and male)
male scored higher than female on Annual_salary (difference 1939.20 points). The difference is statistically significant and its size is negligible ($|d| = 0.13$).

↑ ↓ Remove

Paired-samples t-test (Comparing Skills_test_before and Skills_test_after, same respondents)
Skills_test_after is higher than Skills_test_before by 5.61 points on average. The change is statistically significant and large ($d_z = 0.91$).

↑ ↓ Remove

Multiple linear regression (Predicting Annual_salary_In)
The predictors together explain 83.8% of the variance in Annual_salary_In ($R^2 = .84$, $F^2 = 5.19$, large). Significant predictors: Department: HR (vs Finance) ($\beta = -.09$), Department: IT (vs Finance) ($\beta = .09$), Department: Operations (vs Finance) ($\beta = -.24$), Department: Sales (vs Finance) ($\beta = -.16$), Job_level ($\beta = .68$), Education: Diploma (vs Bachelor) ($\beta = -.07$), Education: Doctorate (vs Bachelor) ($\beta = .07$), Education: Master (vs Bachelor) ($\beta = .08$), Tenure_years ($\beta = .18$), Gender: male (vs female) ($\beta = .08$). Block 2 adds $\Delta R^2 = .006$ ($p < .001$).

↑ ↓ Remove

Report title

Download Word report (.docx)

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Figure 51. The Word report screen with four results, the report title and the download button.

Note: what the report does not contain

The power analyses of chapter 5 are not added to the Word report in this version: copy their sentence and keep your own record. Charts and the machine learning results do have an [Add to Word report](#) button; we did not add them in our session, so check on your own screen how they appear in the document.

10.3 What is inside the downloaded document

We downloaded the report and opened it. It starts with the title, then a line that reads "Analyses run in SaNaML 0.10.2 on 2026-10-06. All computations were done on the analyst's own computer." Then each result follows with its own heading, the APA-style sentence, the interpretation, the checks, and numbered tables. The document ends with a table of the rules of thumb used and the references. None of the four results we added was a chart, so the document of our run contains no pictures.

Table 23 shows how the pieces of the screen map to the document.

Table 23

Parts of the Word report.

Part	Content
Title and date line	The report title and the version, date and statement that computation was local
Per result	Heading, APA-style sentence, interpretation, checks, numbered tables
Rules	The rule of thumb behind each check (for example, "Cramér's V: small, medium and large are .10, .30 and .50 divided by $\sqrt{(\text{smaller table dimension} - 1)}$ ")
References	The sources cited by the rules and tests used

10.4 Using the report well

- **Edit it.** The document is a draft. Replace the generic variable names by plain words, delete what is not needed, and add the context.
- **Check each sentence against the table.** The APA sentence is generated from the numbers; confirm that it says what you mean. For instance, in our run the t-test result reads "male scored higher than female on Annual_salary (difference 1939.20 points)" and the effect-size check explains: " $|d| = 0.13$ is negligible. The sign of d follows the order of the groups: $d = -0.13$ means male scored higher than female." The reported $d = -0.13$ in the APA sentence has the sign of the group order (female minus male), so read the two together.
- **Report the non-significant results too.** For example, the absence by cluster test ($p = .096$) belongs in the report as a null result.
- **Say what the data are.** State the source, the date, the number of employees, and who was excluded.
- **Share with care.** A report on groups of employees can still identify people in small groups (a department of five). Combine or hide small groups before sharing.

- **Keep the data and the report together.** Download the report at the end of the session: SaNaML keeps the data and results in the browser only until the site data are cleared.

10.5 A checklist for an HR analysis

1. Is the question clear, and is the data good enough for it (chapter 1, responsible people analytics)?
2. Have you looked at the data (counts, missing values, outliers) before any test (chapter 2)?
3. Are the scales reliable before you use their scores (chapter 3)?
4. Does the test fit the design (independent groups, repeated measures, categories; chapters 3 to 7)?
5. Have you looked at the effect size and its interval, not only the p-value?
6. For predictions: was the model tested on unseen employees, is it calibrated, and who is harmed if it errs (chapter 8)?
7. Is the result written in plain words, with its limits, for people who can act on it?

Exercise

1. Add the results of three analyses of this manual to the Word report, give the report a title and download it. Open the file and find the rules table.
1. In the downloaded report, which line states where the computations were done? (Answer: "All computations were done on the analyst's own computer.")
1. Pick one APA sentence and rewrite it for a non-statistician in two sentences.

Appendix A. Codebooks

Both files are simulated; no real person or company appears in them. Missing values are empty cells. The planted effects are listed so that you can check your results.

A.1 practice_hr.csv (1,200 employees)

Table 24

Variables of practice_hr.csv

Variable	Meaning
EmployeeID	Identifier (a label; do not analyse it)
Department	Sales, Operations, Finance, IT, HR
Job_level	1 to 5
Gender	female, male
Age	Years
Education	Diploma, Bachelor, Master, Doctorate
Tenure_years	Years with the company
Annual_salary	Skewed; after controls, women earn about 4% less (planted)
Performance_rating	1 to 5
Training_hours	Hours of training
Overtime_hours_month	Hours of overtime per month
Commute_km	Commuting distance (35 missing)
Remote_days_week	Days worked remotely per week
Manager_support	Survey score, 1 to 5 (40 missing)
Work_life_balance	Survey score, 1 to 5
Engagement	Survey score, 1 to 5
Job_satisfaction	Survey score, 1 to 5 (52 missing)
Absence_days	Count of days absent
Promoted_last_3y	Yes/No (about 11%)
Skills_test_before, Skills_test_after	Skills test around a training programme (paired)
Engagement_Q1 to Engagement_Q4	Engagement in four quarters (repeated measures)
Left_company	Yes/No, left within the year (about 15%)

Added by you in section 1.8 (Data screen, step 4): **Annual_salary_ln**, the natural logarithm of Annual_salary; **Tenure_group**, tertiles of Tenure_years; **Left_yes**, a 0/1 indicator of Left_company = Yes; **Dept_merged**, Department with Finance and HR merged into Corporate. Chapter 9 adds **Employee_type**.

What is planted: pay gap after controls (chapter 4); gain on the skills test and a dose-response with Training_hours (chapter 5); engagement driven by Manager_support and Work_life_balance

(chapter 6); leaving linked to engagement, overtime and tenure (chapter 8); about 15% of employees leave.

A.2 practice_engagement.csv (600 respondents)

Table 25

Variables of practice_engagement.csv

Variable	Meaning
RespondentID	Identifier
Department	Sales, Operations, Finance, IT, HR
Gender	female, male
Tenure_band	<2 years, 2-5 years, 6-10 years, >10 years
Works_remotely	Yes/No
Vigor1 to Vigor3	Energy at work, rated 1 to 7 (Vigor2: 20 missing)
Dedication1 to Dedication3	Involvement and pride, rated 1 to 7
Absorption1 to Absorption3	Immersion in work, rated 1 to 7 (Absorption3: 21 missing)

What is planted: three correlated scales, and differences between departments with Operations lowest (chapter 3).

Appendix B. Which tool answers which HR question?

Table 26

HR questions and the SaNaML tools that answer them (or do not)

HR question	Tool in SaNaML 0.10.2	Chapter
What does the workforce look like?	Analyses → Descriptive statistics; Charts	2
Is a survey scale reliable?	Data → Build construct scores (Cronbach's α)	3
Were survey answers careless?	Data → Check data quality	3
Which variable would be better as a logarithm, as groups or as an indicator?	Data → Compute and recode variables	1, 2, 7
Does engagement differ between departments or groups?	One-way ANOVA with Kruskal-Wallis; t-test with Mann-Whitney	3, 7
Do women and men earn the same?	t-test; regression of log salary (Annual_salary_ln) with gender	4
Are women and men in different jobs?	Chi-square test of Gender by Department or Job_level	4
Did training change test scores?	Paired t-test with Wilcoxon test	5
Does more training go with more gain?	Partial correlation; regression with the baseline score	5
How many people do I need to evaluate a programme?	Power analysis	5
What goes with engagement or satisfaction?	Correlation; multiple and hierarchical regression	6
Does a score change across several occasions?	Repeated-measures ANOVA with Friedman test	6
Is absence or leaving different between departments?	ANOVA / Kruskal-Wallis; chi-square test	7
Which employees are at risk of leaving?	Logistic regression; Machine learning	8
What drives the prediction?	Machine learning → SHAP, accumulated local effects	8
Are the predicted probabilities trustworthy?	Machine learning → Calibration	8
Are there types of employee?	Clustering	9
How long do employees stay? (time to leaving)	Not available (survival analysis)	-
Is a selection procedure fair to all groups?	Not available (adverse-impact tools)	-
Does a factor act through another (mediation, moderation)?	Not available	-
Ordinal or multinomial outcomes	Not available (collapse to yes/no for logistic regression)	-
Hide small groups automatically	Not available (check group sizes yourself)	-

HR question	Tool in SaNaML 0.10.2	Chapter
Choose a cut-off for a rare outcome	Machine learning → Cut-offs, precision and recall	8
Rebalance a rare outcome (weights, resampling)	Not available (read AUC, precision and recall)	-
A new variable from two variables (difference, ratio)	Not available (the new variable tools work on one variable)	-

Appendix C. Glossary

Table 27

Terms used in this manual

Term	Meaning
Adjusted gap	A difference between groups that remains after other measured variables are held constant in a regression.
Alpha (Cronbach's)	A measure of how well the items of a scale hang together, from 0 to 1.
ANOVA	Analysis of variance: a test of whether three or more group means differ.
AUC	Area under the ROC curve: the probability that a randomly chosen leaver gets a higher predicted risk than a randomly chosen stayer. .50 is chance, 1.00 perfect.
Balanced accuracy	The average of sensitivity and specificity; it is not inflated by a large majority group.
Calibration	Whether predicted probabilities match the observed frequencies.
Confidence interval	A range of values compatible with the data at a stated level (usually 95%).
Cross-validation	Testing a model on people it was not trained on, by splitting the data into folds.
d, d _z	Standardised mean differences: 0.2 small, 0.5 medium, 0.8 large (Cohen, 1988).
Dummy coding	Turning a categorical predictor into 0/1 comparisons with a reference group.
Effect size	How large an effect is, as opposed to whether it is statistically significant.
Hierarchical regression	A regression entered in blocks, to see what each block adds (ΔR^2).
Kruskal–Wallis test	A rank-based test for three or more groups.
Leakage	Information in a predictor that would not be available when the prediction is made, or that contains the outcome.
Mann–Whitney U test	A rank-based test for two independent groups.
Odds ratio	The factor by which the odds of the outcome change for one unit more of a predictor.
p-value	The probability of a result at least as extreme as the one observed if there were no effect. It is not the probability that the hypothesis is true.
Partial correlation	A correlation after removing the effect of one or more other variables.
Precision (positive predictive value)	Of those predicted to leave, the share who did leave.
R ²	The share of the variance of the outcome explained by the model.
Reference group	The comparison group for a categorical predictor (the first category alphabetically in SaNaML).
Sensitivity (recall)	Of those who left, the share the model flagged.
Specificity	Of those who stayed, the share the model did not flag.
SHAP	A method that shares each prediction among the predictors; mean absolute SHAP shows how much each predictor moves the predictions.
Wilcoxon signed-rank test	A rank-based test for paired measurements.

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